

Outcome_analysis

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Create the folders

```
dir.create("output")
dir.create("figures")
dir.create("reports")
```

Load necessary libraries

1 Import data and define variables

```
Outcome <- read.csv(
  here("data", "Outcome_data.csv"),
  header = TRUE,
  check.names = FALSE
)

Outcome <- Outcome %>%
  mutate(
    across(c(PID, QCODE), as.character),

    across(c(AGS, YEARS,BLDLOSS, BMI, CATHPERIOD, FUPM, HOSPTIME,OPRTIME,
      POSTUTIFUP, POSTUTI_T), as.numeric),

    across(c(ANASTOMO,BLDCompleak,BLDCompRetn,BLDCompVoid,BLDSPASM,CATH,CENTER,
      CLAVIENCLASS,EPIDURAL,GENDER, GU30, GU90N, GUER30, GUER90, HEMAT,
      HISTOINJ,ILEUS, INCCOMPBLEED, INCCOMPINF, INCCOMPSCAR, NARC,
      NONARC, PAIN, PREOPEPISODE1, PREOPRNPLN, PREOPRNSCR, PREOPTYPEU1,
      PREOPVURGRADE, PROC365, STENT, SURGIND, SURGTECH2, TRT2, UDR7M,
      URCOMPVUR, URCOMPINJ, URCOMPOBST, URCOMPSTEN, POSTUTI, VURPHASE,
      VURSIDE, RECGRADE_MAX), as.factor)
  )

str(Outcome)

'data.frame':   412 obs. of  55 variables:
 $ PID          : chr  "P1" "P2" "P3" "P4" ...
 $ QCODE        : chr  "102" "111" "3" "4" ...
 $ REC011       : int   0 0 0 1 0 1 1 0 1 1 ...
```

```

$ AGS : num 5 6 7 10 7 9 10 8 7 11 ...
$ YEARS : num 2011 2014 2017 2014 2011 ...
$ ANASTOMO : Factor w/ 2 levels "0","1": 1 1 1 1 1 1 1 1 1 1 ...
$ BLDCompleak : Factor w/ 2 levels "0","1": 1 1 1 1 1 1 1 1 1 1 ...
$ BLDCompRetn : Factor w/ 2 levels "0","1": 1 1 2 1 1 2 1 2 2 2 ...
$ BLDCompVoid : Factor w/ 2 levels "0","1": 2 1 1 1 1 1 1 1 1 1 ...
$ BLDLOSS : num 70 300 150 55 55 190 75 100 55 55 ...
$ BLDSPASM : Factor w/ 2 levels "0","1": 1 2 2 1 1 2 1 2 2 1 ...
$ BMI : num 13.5 13.7 12 13.3 15 15.8 15.3 15.3 14.8 15.3 ...
$ CATH : Factor w/ 2 levels "0","1": 2 2 2 2 2 2 2 2 2 2 ...
$ CATHPERIOD : num 2 7 5 5 4 7 4 5 5 7 ...
$ CENTER : Factor w/ 4 levels "1","2","3","5": 1 1 1 1 1 1 1 1 1 1 ...
$ CLAVIENCLASS : Factor w/ 4 levels "1","2","3","4": 2 2 2 2 1 2 2 1 2 3 ...
$ EPIDURAL : Factor w/ 2 levels "0","1": 1 1 1 1 1 1 2 1 1 1 ...
$ FUPM : num 24 24 44 36 120 36 36 37 12 36 ...
$ GENDER : Factor w/ 2 levels "0","1": 1 1 2 2 2 1 1 2 2 1 ...
$ GU30 : Factor w/ 2 levels "0","1": 1 2 1 1 1 1 2 1 2 1 ...
$ GU90N : Factor w/ 3 levels "0","1","2": 1 2 2 2 1 2 2 1 2 1 ...
$ GUER30 : Factor w/ 2 levels "0","1": 1 2 1 1 1 1 2 1 2 1 ...
$ GUER90 : Factor w/ 2 levels "0","1": 2 2 1 2 1 1 2 1 2 1 ...
$ HEMAT : Factor w/ 2 levels "0","1": 1 2 2 2 1 2 2 1 2 1 ...
$ HISTOINJ : Factor w/ 2 levels "0","1": 2 1 1 1 1 1 1 2 1 1 ...
$ HOSPTIME : num 2 10 5 5 4 5 8 5 5 4 ...
$ ILEUS : Factor w/ 2 levels "0","1": 1 1 1 1 1 1 1 1 1 1 ...
$ INCCOMPBLEED : Factor w/ 2 levels "0","1": 2 1 2 1 1 1 1 1 1 1 ...
$ INCCOMPINF : Factor w/ 2 levels "0","1": 1 1 1 1 1 2 1 1 2 1 ...
$ INCCOMPSCAR : Factor w/ 2 levels "0","1": 1 2 1 1 1 1 1 1 1 1 ...
$ NARC : Factor w/ 2 levels "0","1": 1 2 2 1 1 2 1 1 2 1 ...
$ NONARC : Factor w/ 2 levels "0","1": 2 2 2 2 2 2 2 2 2 2 ...
$ OPRTIME : num 76 80 80 83 90 75 90 88 85 80 ...
$ PAIN : Factor w/ 4 levels "0","1","2","3": 2 4 4 2 2 3 3 2 4 2 ...
$ PREOPEPISODE1 : Factor w/ 3 levels "1","2","3": 2 3 3 3 3 3 3 2 3 3 ...
$ PREOPRNPLN : Factor w/ 2 levels "0","1": 1 1 1 1 1 1 1 1 1 1 ...
$ PREOPRNSCR : Factor w/ 2 levels "0","1": 1 2 1 2 1 1 1 1 2 1 ...
$ PREOPTYPEU1 : Factor w/ 2 levels "0","1": 2 2 2 2 2 2 2 2 2 2 ...
$ PREOPVURGRADE : Factor w/ 3 levels "3","4","5": 3 3 3 3 2 3 2 2 3 2 ...
$ PROC365 : Factor w/ 2 levels "0","1": 1 1 1 1 1 1 2 1 1 1 ...
$ STENT : Factor w/ 2 levels "0","1": 1 1 1 1 1 1 1 1 1 1 ...
$ SURGIND : Factor w/ 11 levels "1","2","3","4",...: 3 3 3 3 4 4 4 3 3 3 ...
$ SURGTECH2 : Factor w/ 3 levels "1","2","3": 3 2 2 2 2 2 2 2 2 2 ...
$ TRT2 : Factor w/ 2 levels "0","1": 2 2 2 2 2 2 2 2 2 2 ...
$ UDR7M : Factor w/ 2 levels "0","1": 2 2 2 2 1 2 2 2 2 2 ...
$ URCOMPVCUR : Factor w/ 2 levels "0","1": 1 1 1 1 1 1 1 1 1 1 ...
$ URCOMPINJ : Factor w/ 2 levels "0","1": 1 1 1 1 1 1 1 1 1 1 ...
$ URCOMPOBST : Factor w/ 2 levels "0","1": 1 1 1 1 1 1 1 1 2 1 ...
$ URCOMPSTEN : Factor w/ 2 levels "0","1": 1 1 1 1 1 1 1 1 1 2 ...
$ POSTUTI : Factor w/ 2 levels "0","1": 2 1 1 1 1 1 2 1 1 2 ...
$ POSTUTIFUP : num 1 24 44 36 120 36 1 37 12 1 ...
$ POSTUTI_T : num 1 0 0 0 0 0 1 0 0 1 ...
$ VURPHASE : Factor w/ 3 levels "1","2","3": 1 1 3 3 1 1 1 1 1 1 ...
$ VURSIDE : Factor w/ 2 levels "0","1": 1 2 2 2 1 2 2 1 1 2 ...
$ RECGRADE_MAX : Factor w/ 6 levels "0","1","2","3",...: 2 2 1 3 1 3 3 2 3 3 ...

```

2 Data Imputation using KNN (k-nearest neighbors)

```
# Missingness in the dataset
m_outcome <- colSums(is.na(Outcome)) / nrow(Outcome) * 100
print(m_outcome)
```

PID	QCODE	REC011	AGS	YEARS
0.0000000	0.0000000	33.7378641	0.0000000	0.0000000
ANASTOMO	BLDCOMPLEAK	BLDCOMPRETN	BLDCOMPVOID	BLDLOSS
0.0000000	0.0000000	0.0000000	0.0000000	1.9417476
BLDSPASM	BMI	CATH	CATHPERIOD	CENTER
0.0000000	0.0000000	0.0000000	0.0000000	0.0000000
CLAVIENCLASS	EPIDURAL	FUPM	GENDER	GU30
22.0873786	0.0000000	0.0000000	0.0000000	0.0000000
GU90N	GUER30	GUER90	HEMAT	HISTOINJ
0.0000000	0.0000000	0.0000000	0.0000000	0.0000000
HOSPTIME	ILEUS	INCCOMPBLEED	INCCOMPINF	INCCOMPSCAR
0.0000000	0.0000000	0.0000000	0.0000000	0.0000000
NARC	NONARC	OPRTIME	PAIN	PREOPEPISODE1
0.0000000	0.0000000	0.0000000	0.9708738	2.9126214
PREOPRNPLN	PREOPRNSCR	PREOPTYPEU1	PREOPVURGRADE	PROC365
20.6310680	0.0000000	0.2427184	0.0000000	0.0000000
STENT	SURGIND	SURGTECH2	TRT2	UDR7M
0.0000000	0.0000000	0.0000000	0.0000000	0.0000000
URCOMPVCUR	URCOMPINJ	URCOMPOBST	URCOMPSTEN	POSTUTI
8.7378641	0.0000000	0.0000000	0.0000000	0.0000000
POSTUTIFUP	POSTUTI_T	VURPHASE	VURSIDE	RECGRADE_MAX
0.0000000	0.0000000	0.0000000	0.0000000	33.7378641

```
# KNN imputation
Outcome_knn <- Outcome # create a temporary dataset copy for KNN using scaled variables

# Convert categorical variables
cat_vars <- c(
  "GENDER", "PREOPVURGRADE", "VURPHASE", "VURSIDE",
  "UDR7M", "PREOPRNSCR", "HISTOINJ", "SURGIND",
  "CENTER", "TRT2", "SURGTECH2",
  "PREOPEPISODE1", "PREOPRNPLN", "PREOPTYPEU1"
)
Outcome_knn[cat_vars] <- lapply(Outcome_knn[cat_vars], as.factor)

# Scale continuous variables
continuous_vars <- c("AGS", "BMI", "YEARS")
Outcome_knn[continuous_vars] <- scale(Outcome_knn[continuous_vars])

# KNN imputation without subsetting the dataset
Outcome_knn <- knn(
  Outcome_knn,
  variable = c("PREOPEPISODE1", "PREOPRNPLN", "PREOPTYPEU1"), # Variables to impute
  dist_var = c("AGS", "GENDER", "BMI", # Variables used to compute patient similarity
    "PREOPVURGRADE", "VURPHASE", "VURSIDE", "UDR7M",
```

```

      "PREOPRNSCR", "HISTOINJ", "SURGIND", "YEARS",
      "TRT2"
    ),
    k = 5,
    imp_var = FALSE
  )

```

```

Out <- Outcome # New dataset for analysis including imputed variables
Out$PREOPEPISODE1 <- Outcome_knn$PREOPEPISODE1
Out$PREOPRNPLN <- Outcome_knn$PREOPRNPLN
Out$PREOPTYPEU1 <- Outcome_knn$PREOPTYPEU1

```

3 Recode and create (composite) variables

```

# Recode SURGIND into SURGIND2

```

```

Out <- Out %>%
  mutate(
    SURGIND2 = case_when(
      SURGIND == 4 ~ 1,
      SURGIND == 3 ~ 2,
      SURGIND == 1 ~ 3,
      SURGIND == 2 ~ 4,
      SURGIND %in% c(5, 6, 7, 8, 9, 10, 11) ~ 5,
      TRUE ~ NA_real_
    ),
    SURGIND2 = factor(SURGIND2)
  )

```

```

# Create a new variable LUTD by combining BLDCOMPRETN, BLDCOMPVOID, BLDCOMPLEAK

```

```

Out <- Out %>%
  mutate(
    LUTD = case_when(
      BLDCOMPRETN == 1 |
      BLDCOMPVOID == 1 |
      BLDCOMPLEAK == 1 ~ 1,

      BLDCOMPRETN == 0 &
      BLDCOMPVOID == 0 &
      BLDCOMPLEAK == 0 ~ 0,

      TRUE ~ NA_real_
    ),
    LUTD = factor(LUTD)
  )

```

```

# Create a new variable OBST by combining URCOMPOBST, URCOMPSTEN, ANASTOMO

```

```

Out <- Out %>%

```

```

mutate(
  OBST = case_when(
    URCOMPOBST == 1 |
    URCOMPSTEN == 1 |
    ANASTOMO == 1 ~ 1,

    URCOMPOBST == 0 &
    URCOMPSTEN == 0 &
    ANASTOMO == 0 ~ 0,

    TRUE ~ NA_real_
  ),
  OBST = factor(OBST)
)

# Create a new variable SURG by combining INCCOMPBLEED, INCCOMPINF, INCCOMPSCAR
Out <- Out %>%
  mutate(
    SURG = case_when(
      INCCOMPBLEED == 1 |
      INCCOMPINF == 1 |
      INCCOMPSCAR == 1 ~ 1,

      INCCOMPBLEED == 0 &
      INCCOMPINF == 0 &
      INCCOMPSCAR == 0 ~ 0,

      TRUE ~ NA_real_
    ),
    SURG = factor(SURG)
  )

# Create a new variable GU90 by combining no readmission (0) and planned readmission (2)
Out <- Out %>%
  mutate(
    GU90 = recode(as.character(GU90N),
      "0" = "0",
      "1" = "1",
      "2" = "0"),
    GU90 = factor(GU90)
  )

# Create a categorical variable CATHPERIOD_G for the numerical variable CATHPERIOD
Out <- Out %>%
  mutate(CATHPERIOD_G = case_when(
    CATHPERIOD <= 3 ~ "Early removal 3 days",
    CATHPERIOD > 3 & CATHPERIOD <= 7 ~ "Standard removal 4-7 days",
    CATHPERIOD > 7 ~ "Delayed removal >7 days",
  ))

Out$CATHPERIOD_G <- factor(

```

```

    Out$CATHPERIOD_G,
    levels = c("Early removal 3 days", "Standard removal 4-7 days", "Delayed removal >7 days")
)

# Recode complications

comp_vars <- c(
  "LUTD",
  "OBST",
  "URCOMPINJ",
  "SURG",
  "ILEUS"
)

Out[comp_vars] <- lapply(
  Out[comp_vars],
  function(x) as.numeric(as.character(x))
)

Out$ANY <- as.numeric(
  rowSums(
    Out[, comp_vars],
    na.rm = TRUE
  ) > 0
)

minor_vars <- c(
  "LUTD",
  "SURG",
  "ILEUS"
)

Out$MINOR <- as.numeric(
  rowSums(
    Out[, minor_vars],
    na.rm = TRUE
  ) > 0
)

major_vars <- c(
  "OBST",
  "URCOMPINJ"
)

Out$MAJOR <- as.numeric(
  rowSums(
    Out[, major_vars],
    na.rm = TRUE
  ) > 0
)

```

```

# Convert variables to appropriate data types for analysis
Out <- Out %>%
  mutate(
    across(c(PID, QCODE), as.character),

    across(c(AGS, YEARS, BLDLOSS, BMI, CATHPERIOD, FUPM, HOSPTIME, OPRTIME,
              POSTUTIFUP, POSTUTI_T), as.numeric),

    across(c(ANASTOMO, BLDCompleak, BLDCompRetn,
              BLDCompVoid, BLDSPASM, CATH, CENTER, CLAVIENCLASS, EPIDURAL, GENDER,
              GU30, GU90N, GUER30, GUER90, HEMAT,
              HISTOINJ, ILEUS, INCCOMPBLEED, INCCOMPINF, INCCOMPSCAR, NARC,
              NONARC, PAIN, PREOPEPISODE1, PREOPRNPLN, PREOPRNSCR, PREOPTYPEU1,
              PREOPVURGRADE, PROC365, STENT, SURGIND, SURGTECH2, TRT2, UDR7M,
              URCOMPVUR, URCOMPINJ, URCOMPOBST, URCOMPSTEN, POSTUTI, VURPHASE,
              VURSIDE, RECGRADE_MAX, SURGIND2, LUTD, OBST, SURG, GU90,
              CATHPERIOD_G, ANY, MINOR, MAJOR), as.factor)
  )

str(Out)

'data.frame':  412 obs. of  64 variables:
 $ PID      : chr  "P1" "P2" "P3" "P4" ...
 $ QCODE     : chr  "102" "111" "3" "4" ...
 $ RECO11    : int   0 0 0 1 0 1 1 0 1 1 ...
 $ AGS       : num   5 6 7 10 7 9 10 8 7 11 ...
 $ YEARS     : num  2011 2014 2017 2014 2011 ...
 $ ANASTOMO  : Factor w/ 2 levels "0","1": 1 1 1 1 1 1 1 1 1 1 ...
 $ BLDCompleak : Factor w/ 2 levels "0","1": 1 1 1 1 1 1 1 1 1 1 ...
 $ BLDCompRetn : Factor w/ 2 levels "0","1": 1 1 2 1 1 2 1 2 2 2 ...
 $ BLDCompVoid : Factor w/ 2 levels "0","1": 2 1 1 1 1 1 1 1 1 1 ...
 $ BLDLOSS   : num   70 300 150 55 55 190 75 100 55 55 ...
 $ BLDSPASM   : Factor w/ 2 levels "0","1": 1 2 2 1 1 2 1 2 2 1 ...
 $ BMI       : num  13.5 13.7 12 13.3 15 15.8 15.3 15.3 14.8 15.3 ...
 $ CATH      : Factor w/ 2 levels "0","1": 2 2 2 2 2 2 2 2 2 2 ...
 $ CATHPERIOD : num   2 7 5 5 4 7 4 5 5 7 ...
 $ CENTER    : Factor w/ 4 levels "1","2","3","5": 1 1 1 1 1 1 1 1 1 1 ...
 $ CLAVIENCLASS : Factor w/ 4 levels "1","2","3","4": 2 2 2 2 1 2 2 1 2 3 ...
 $ EPIDURAL   : Factor w/ 2 levels "0","1": 1 1 1 1 1 1 2 1 1 1 ...
 $ FUPM      : num   24 24 44 36 120 36 36 37 12 36 ...
 $ GENDER     : Factor w/ 2 levels "0","1": 1 1 2 2 2 1 1 2 2 1 ...
 $ GU30      : Factor w/ 2 levels "0","1": 1 2 1 1 1 1 2 1 2 1 ...
 $ GU90N     : Factor w/ 3 levels "0","1","2": 1 2 2 2 1 2 2 1 2 1 ...
 $ GUER30    : Factor w/ 2 levels "0","1": 1 2 1 1 1 1 2 1 2 1 ...
 $ GUER90    : Factor w/ 2 levels "0","1": 2 2 1 2 1 1 2 1 2 1 ...
 $ HEMAT     : Factor w/ 2 levels "0","1": 1 2 2 2 1 2 2 1 2 1 ...
 $ HISTOINJ   : Factor w/ 2 levels "0","1": 2 1 1 1 1 1 1 1 2 1 ...
 $ HOSPTIME   : num   2 10 5 5 4 5 8 5 5 4 ...
 $ ILEUS     : Factor w/ 2 levels "0","1": 1 1 1 1 1 1 1 1 1 1 ...
 $ INCCOMPBLEED : Factor w/ 2 levels "0","1": 2 1 2 1 1 1 1 1 1 1 ...
 $ INCCOMPINF  : Factor w/ 2 levels "0","1": 1 1 1 1 1 2 1 1 2 1 ...
 $ INCCOMPSCAR : Factor w/ 2 levels "0","1": 1 2 1 1 1 1 1 1 1 1 ...
 $ NARC      : Factor w/ 2 levels "0","1": 1 2 2 1 1 2 1 1 2 1 ...

```

```

$ NONARC      : Factor w/ 2 levels "0","1": 2 2 2 2 2 2 2 2 2 2 ...
$ OPRTIME     : num 76 80 80 83 90 75 90 88 85 80 ...
$ PAIN        : Factor w/ 4 levels "0","1","2","3": 2 4 4 2 2 3 3 2 4 2 ...
$ PREOPEPISODE1: Factor w/ 3 levels "1","2","3": 2 3 3 3 3 3 3 2 3 3 ...
$ PREOPRNPLN  : Factor w/ 2 levels "0","1": 1 1 1 1 1 1 1 1 1 1 ...
$ PREOPRNSCR  : Factor w/ 2 levels "0","1": 1 2 1 2 1 1 1 1 2 1 ...
$ PREOPTYPEU1 : Factor w/ 2 levels "0","1": 2 2 2 2 2 2 2 2 2 2 ...
$ PREOPVURGRADE: Factor w/ 3 levels "3","4","5": 3 3 3 3 2 3 2 2 3 2 ...
$ PROC365     : Factor w/ 2 levels "0","1": 1 1 1 1 1 1 2 1 1 1 ...
$ STENT       : Factor w/ 2 levels "0","1": 1 1 1 1 1 1 1 1 1 1 ...
$ SURGIND     : Factor w/ 11 levels "1","2","3","4",...: 3 3 3 3 4 4 4 3 3 3 ...
$ SURGTECH2   : Factor w/ 3 levels "1","2","3": 3 2 2 2 2 2 2 2 2 2 ...
$ TRT2        : Factor w/ 2 levels "0","1": 2 2 2 2 2 2 2 2 2 2 ...
$ UDR7M       : Factor w/ 2 levels "0","1": 2 2 2 2 1 2 2 2 2 2 ...
$ URCOMPCVUR  : Factor w/ 2 levels "0","1": 1 1 1 1 1 1 1 1 1 1 ...
$ URCOMPINJ   : Factor w/ 2 levels "0","1": 1 1 1 1 1 1 1 1 1 1 ...
$ URCOMPOBST  : Factor w/ 2 levels "0","1": 1 1 1 1 1 1 1 1 2 1 ...
$ URCOMPSTEN  : Factor w/ 2 levels "0","1": 1 1 1 1 1 1 1 1 1 2 ...
$ POSTUTI     : Factor w/ 2 levels "0","1": 2 1 1 1 1 1 2 1 1 2 ...
$ POSTUTIFUP  : num 1 24 44 36 120 36 1 37 12 1 ...
$ POSTUTI_T   : num 1 0 0 0 0 0 1 0 0 1 ...
$ VURPHASE    : Factor w/ 3 levels "1","2","3": 1 1 3 3 1 1 1 1 1 1 ...
$ VURSIDE     : Factor w/ 2 levels "0","1": 1 2 2 2 1 2 2 1 1 2 ...
$ RECGRADE_MAX : Factor w/ 6 levels "0","1","2","3",...: 2 2 1 3 1 3 3 2 3 3 ...
$ SURGIND2    : Factor w/ 5 levels "1","2","3","4",...: 2 2 2 2 1 1 1 2 2 2 ...
$ LUTD        : Factor w/ 2 levels "0","1": 2 1 2 1 1 2 1 2 2 2 ...
$ OBST        : Factor w/ 2 levels "0","1": 1 1 1 1 1 1 1 1 2 2 ...
$ SURG        : Factor w/ 2 levels "0","1": 2 2 2 1 1 2 1 1 2 1 ...
$ GU90        : Factor w/ 2 levels "0","1": 1 2 2 2 1 2 2 1 2 1 ...
$ CATHPERIOD_G : Factor w/ 3 levels "Early removal 3 days",...: 1 2 2 2 2 2 2 2 2 2 ...
$ ANY         : Factor w/ 2 levels "0","1": 2 2 2 1 1 2 1 2 2 2 ...
$ MINOR       : Factor w/ 2 levels "0","1": 2 2 2 1 1 2 1 2 2 2 ...
$ MAJOR       : Factor w/ 2 levels "0","1": 1 1 1 1 1 1 1 1 2 2 ...

```

4 Descriptive analysis: BEFORE weighting and matching

```

# Baseline characteristics
table1<- Out %>%
  mutate(YEARS = factor(YEARS)) %>% # Convert only within this pipeline
  dplyr::select(YEARS, AGS, GENDER, BMI, HISTOINJ, SURGIND2, PREOPVURGRADE, VURSIDE, UDR7M,
    VURPHASE, PREOPTYPEU1, PREOPEPISODE1, PREOPRNPLN, PREOPRNSCR,
    SURGTECH2, CENTER, TRT2
  ) %>%
  tbl_summary(
    by = TRT2, # for stratification
    statistic = list(
      all_continuous() ~ "{median} ({p25}-{p75})",
      all_categorical() ~ "{n} ({p}%)"
    ),
    digits = all_categorical() ~ c(0, 1), # n = 0 decimals, p = 1 decimal
    missing = "ifany", # shows missing only when present
    missing_text = "Missing"
  )

```



```

) %>%
add_overall() %>%
add_p(
  test = list(
    YEARS ~ "chisq.test"
  ),
pvalue_fun = ~style_pvalue(.x, digits = 3)) %>% # 3 decimals
bold_labels()

```

The following warnings were returned during `add\_p()`:

```

! For variable `YEARS` (`TRT2`) and "statistic", "p.value", and "parameter"
  statistics: Chi-squared approximation may be incorrect

```

```
table1
```

```

table1 %>%
  as_flex_table() %>%
  flextable::save_as_docx(path = "output/baseline.docx")

```

```
# Baseline characteristics
```

```

table1b<- Out %>%
  dplyr::select(YEARS, AGS, GENDER, BMI, HISTOINJ, SURGIND2, PREOPVURGRADE, VURSIDE, UDR7M,
    VURPHASE, PREOPTYPEU1, PREOPEPISODE1, PREOPRNPLN, PREOPRNSCR,
    SURGTECH2, CENTER, TRT2
  ) %>%
  tbl_summary(
    by = TRT2, # for stratification
    statistic = list(
      all_continuous() ~ "{median} ({p25}-{p75})",
      all_categorical() ~ "{n} ({p}%)"
    ),
    digits = all_categorical() ~ c(0, 1), # n = 0 decimals, p = 1 decimal
    missing = "ifany", # shows missing only when present
    missing_text = "Missing"
  ) %>%
  add_overall() %>%
  add_p(pvalue_fun = ~style_pvalue(.x, digits = 3)) %>% # 3 decimals
  bold_labels()

```

```
table1b
```

```

table1b %>%
  as_flex_table() %>%
  flextable::save_as_docx(path = "output/baseline2.docx")

```

```
# Procedural factors
```

```

table2<- Out %>%
  dplyr::select(TRT2, OPRTIME, STENT, CATH, CATHPERIOD, CATHPERIOD_G, HOSPTIME) %>%
  tbl_summary(

```

Characteristic	Overall N = 412 ¹	0 N = 197 ¹	1 N = 215 ¹	p-value ²
YEARS				<0.001
2010	5 (1.2%)	0 (0.0%)	5 (2.3%)	
2011	10 (2.4%)	0 (0.0%)	10 (4.7%)	
2012	30 (7.3%)	12 (6.1%)	18 (8.4%)	
2013	45 (10.9%)	22 (11.2%)	23 (10.7%)	
2014	40 (9.7%)	20 (10.2%)	20 (9.3%)	
2015	32 (7.8%)	14 (7.1%)	18 (8.4%)	
2016	45 (10.9%)	15 (7.6%)	30 (14.0%)	
2017	34 (8.3%)	12 (6.1%)	22 (10.2%)	
2018	59 (14.3%)	25 (12.7%)	34 (15.8%)	
2019	36 (8.7%)	22 (11.2%)	14 (6.5%)	
2020	44 (10.7%)	29 (14.7%)	15 (7.0%)	
2021	18 (4.4%)	15 (7.6%)	3 (1.4%)	
2022	14 (3.4%)	11 (5.6%)	3 (1.4%)	
AGS	8.00 (6.00-9.00)	8.00 (4.00-9.00)	8.00 (6.00-9.00)	0.122
GENDER				0.334
0	236 (57.3%)	108 (54.8%)	128 (59.5%)	
1	176 (42.7%)	89 (45.2%)	87 (40.5%)	
BMI	14.1 (13.6-18.1)	15.1 (13.5-20.8)	13.9 (13.6-15.0)	<0.001
HISTOINJ				0.515
0	368 (89.3%)	178 (90.4%)	190 (88.4%)	
1	44 (10.7%)	19 (9.6%)	25 (11.6%)	
SURGIND2				<0.001
1	215 (52.2%)	94 (47.7%)	121 (56.3%)	
2	132 (32.0%)	57 (28.9%)	75 (34.9%)	
3	19 (4.6%)	10 (5.1%)	9 (4.2%)	
4	13 (3.2%)	11 (5.6%)	2 (0.9%)	
5	33 (8.0%)	25 (12.7%)	8 (3.7%)	
PREOPVURGRADE				0.194
3	20 (4.9%)	11 (5.6%)	9 (4.2%)	
4	126 (30.6%)	52 (26.4%)	74 (34.4%)	
5	266 (64.6%)	134 (68.0%)	132 (61.4%)	
VURSIDE				0.608
0	162 (39.3%)	80 (40.6%)	82 (38.1%)	
1	250 (60.7%)	117 (59.4%)	133 (61.9%)	
UDR7M				<0.001
0	97 (23.5%)	32 (16.2%)	65 (30.2%)	
1	315 (76.5%)	165 (83.8%)	150 (69.8%)	
VURPHASE				<0.001
1	226 (54.9%)	112 (56.9%)	114 (53.0%)	
2	69 (16.7%)	1 (0.5%)	68 (31.6%)	
3	117 (28.4%)	84 (42.6%)	33 (15.3%)	
PREOPTYPEU1				0.031
0	8 (1.9%)	7 (3.6%)	1 (0.5%)	
1	404 (98.1%)	190 (96.4%)	214 (99.5%)	
PREOPEPISODE1				<0.001
1	24 (5.8%)	16 (8.1%)	8 (3.7%)	
2	98 (23.8%)	65 (33.0%)	33 (15.3%)	
3	290 (70.4%)	116 (58.9%)	174 (80.9%)	
PREOPRNPLN				<0.001

Characteristic	Overall N = 412 ¹	0 N = 197 ¹	1 N = 215
YEARS	2,016.00 (2,014.00-2,019.00)	2,018.00 (2,014.00-2,020.00)	2,016.00 (2,013.00-2,019.00)
AGS	8.00 (6.00-9.00)	8.00 (4.00-9.00)	8.00 (6.00-9.00)
GENDER			
0	236 (57.3%)	108 (54.8%)	128 (59.5%)
1	176 (42.7%)	89 (45.2%)	87 (40.5%)
BMI	14.1 (13.6-18.1)	15.1 (13.5-20.8)	13.9 (13.6-18.1)
HISTOINJ			
0	368 (89.3%)	178 (90.4%)	190 (88.4%)
1	44 (10.7%)	19 (9.6%)	25 (11.6%)
SURGIND2			
1	215 (52.2%)	94 (47.7%)	121 (56.3%)
2	132 (32.0%)	57 (28.9%)	75 (34.9%)
3	19 (4.6%)	10 (5.1%)	9 (4.2%)
4	13 (3.2%)	11 (5.6%)	2 (0.9%)
5	33 (8.0%)	25 (12.7%)	8 (3.7%)
PREOPVURGRADE			
3	20 (4.9%)	11 (5.6%)	9 (4.2%)
4	126 (30.6%)	52 (26.4%)	74 (34.4%)
5	266 (64.6%)	134 (68.0%)	132 (61.4%)
VURSIDE			
0	162 (39.3%)	80 (40.6%)	82 (38.1%)
1	250 (60.7%)	117 (59.4%)	133 (61.9%)
UDR7M			
0	97 (23.5%)	32 (16.2%)	65 (30.2%)
1	315 (76.5%)	165 (83.8%)	150 (69.8%)
VURPHASE			
1	226 (54.9%)	112 (56.9%)	114 (53.0%)
2	69 (16.7%)	1 (0.5%)	68 (31.6%)
3	117 (28.4%)	84 (42.6%)	33 (15.3%)
PREOPTYPEU1			
0	8 (1.9%)	7 (3.6%)	1 (0.5%)
1	404 (98.1%)	190 (96.4%)	214 (99.5%)
PREOPEPISODE1			
1	24 (5.8%)	16 (8.1%)	8 (3.7%)
2	98 (23.8%)	65 (33.0%)	33 (15.3%)
3	290 (70.4%)	116 (58.9%)	174 (80.9%)
PREOPRNPLN			
0	332 (80.6%)	131 (66.5%)	201 (93.5%)
1	80 (19.4%)	66 (33.5%)	14 (6.5%)
PREOPRNSCR			
0	275 (66.7%)	104 (52.8%)	171 (79.5%)
1	137 (33.3%)	93 (47.2%)	44 (20.5%)
SURGTECH2			
1	85 (20.6%)	15 (7.6%)	70 (32.6%)
2	273 (66.3%)	137 (69.5%)	136 (63.3%)
3	54 (13.1%)	45 (22.8%)	9 (4.2%)
CENTER			
1	290 (70.4%) ¹¹	103 (52.3%)	187 (87.0%)
2	4 (1.0%)	4 (2.0%)	0 (0.0%)
3	37 (9.0%)	31 (15.7%)	6 (2.8%)

Characteristic	Overall N = 412 ¹	0 N = 197 ¹	1 N = 215 ¹	p-value ²
OPRTIME	80 (70-113)	115 (85-150)	70 (67-80)	<0.001
STENT				<0.001
0	270 (65.5%)	92 (46.7%)	178 (82.8%)	
1	142 (34.5%)	105 (53.3%)	37 (17.2%)	
CATH				0.179
0	9 (2.2%)	2 (1.0%)	7 (3.3%)	
1	403 (97.8%)	195 (99.0%)	208 (96.7%)	
CATHPERIOD	4 (1-5)	4 (4-38)	1 (1-4)	<0.001
CATHPERIOD_G				<0.001
Early removal 3 days	164 (39.8%)	12 (6.1%)	152 (70.7%)	
Standard removal 4-7 days	165 (40.0%)	126 (64.0%)	39 (18.1%)	
Delayed removal >7 days	83 (20.1%)	59 (29.9%)	24 (11.2%)	
HOSPTIME	4.00 (2.00-5.50)	5.00 (4.00-6.00)	2.00 (2.00-5.00)	<0.001

¹Median (Q1-Q3); n (%)

²Wilcoxon rank sum test; Pearson's Chi-squared test; Fisher's exact test

```

by = TRT2, # for stratification
statistic = list(
  all_continuous() ~ "{median} ({p25}-{p75})", # or ~ "{mean} ± {sd}",
  all_categorical() ~ "{n} ({p}%)"
),
digits = all_categorical() ~ c(0, 1), # n = 0 decimals, p = 1 decimal
missing = "ifany",
missing_text = "Missing"
) %>%
add_overall() %>%
add_p(pvalue_fun = ~style_pvalue(.x, digits = 3)) %>% # 3 decimals
bold_labels()

table2

table2 %>%
as_flex_table() %>%
flextable::save_as_docx(path = "output/procedural.docx")

# Outcomes
table3<- Out %>%
dplyr::select(TRT2, FUPM, LUTD, OBST, URCOMPINJ, SURG, ILEUS, CLAVIENCLASS, URCOMPVUR,
  GU30, GU90, GU90N, GUER30, GUER90, PROC365,
  BLDLOSS, BLDSPASM, HEMAT,
  POSTUTI) %>%
tbl_summary(
  by = TRT2, # for stratification
  statistic = list(
    all_continuous() ~ "{median} ({p25}-{p75})", # or ~ "{mean} ± {sd}",
    all_categorical() ~ "{n} ({p}%)"
  )
)

```

```

    ),
    digits = all_categorical() ~ c(0, 1), # n = 0 decimals, p = 1 decimal
    missing = "ifany",
    missing_text = "Missing"
  ) %>%
  add_overall() %>%
  add_p(pvalue_fun = ~style_pvalue(.x, digits = 3)) %>% # 3 decimals
  bold_labels()

table3

table3 %>%
  as_flex_table() %>%
  flextable::save_as_docx(path = "output/outcomes.docx")

## Treatment distribution change over time using a chi-square test
Out$Treatment <- factor(
  Out$TRT2,
  levels = c(0,1),
  labels = c("Cohen", "Lich-Gregoir")
)

tab_year <- table(Out$YEARS, Out$Treatment)

chisq.test(tab_year)

      Pearson's Chi-squared test

data:  tab_year
X-squared = 44.138, df = 12, p-value = 1.447e-05

p_value_year <- chisq.test(tab_year)$p.value

p_label <- ifelse(
  p_value_year < 0.001,
  "Chi-square p < 0.001",
  paste0(
    "Chi-square p = ",
    format.pval(p_value_year, digits = 3)
  )
)

p_label

[1] "Chi-square p < 0.001"

# Proportion plot for treatment across

plot_data1 <- Out %>%

```

Characteristic	Overall N = 412 ¹	0 N = 197 ¹	1 N = 215 ¹	p-value ²
FUPM	28 (24-40)	30 (24-48)	26 (24-39)	0.140
LUTD				0.048
0	335 (81.3%)	168 (85.3%)	167 (77.7%)	
1	77 (18.7%)	29 (14.7%)	48 (22.3%)	
OBST				0.002
0	371 (90.0%)	187 (94.9%)	184 (85.6%)	
1	41 (10.0%)	10 (5.1%)	31 (14.4%)	
URCOMPINJ				0.714
0	405 (98.3%)	193 (98.0%)	212 (98.6%)	
1	7 (1.7%)	4 (2.0%)	3 (1.4%)	
SURG				0.175
0	314 (76.2%)	156 (79.2%)	158 (73.5%)	
1	98 (23.8%)	41 (20.8%)	57 (26.5%)	
ILEUS				0.218
0	385 (93.4%)	181 (91.9%)	204 (94.9%)	
1	27 (6.6%)	16 (8.1%)	11 (5.1%)	
CLAVIENCLASS				<0.001
1	201 (62.6%)	102 (76.1%)	99 (52.9%)	
2	83 (25.9%)	19 (14.2%)	64 (34.2%)	
3	36 (11.2%)	13 (9.7%)	23 (12.3%)	
4	1 (0.3%)	0 (0.0%)	1 (0.5%)	
Missing	91	63	28	
URCOMPVUR				0.012
0	367 (97.6%)	159 (95.2%)	208 (99.5%)	
1	9 (2.4%)	8 (4.8%)	1 (0.5%)	
Missing	36	30	6	
GU30				<0.001
0	349 (84.7%)	144 (73.1%)	205 (95.3%)	
1	63 (15.3%)	53 (26.9%)	10 (4.7%)	
GU90				<0.001
0	342 (83.0%)	146 (74.1%)	196 (91.2%)	
1	70 (17.0%)	51 (25.9%)	19 (8.8%)	
GU90N				<0.001
0	261 (63.3%)	87 (44.2%)	174 (80.9%)	
1	70 (17.0%)	51 (25.9%)	19 (8.8%)	
2	81 (19.7%)	59 (29.9%)	22 (10.2%)	
GUER30				<0.001
0	316 (76.7%)	116 (58.9%)	200 (93.0%)	
1	96 (23.3%)	81 (41.1%)	15 (7.0%)	
GUER90				<0.001
0	305 (74.0%)	112 (56.9%)	193 (89.8%)	
1	107 (26.0%)	85 (43.1%)	22 (10.2%)	
PROC365				0.658
0	399 (96.8%)	190 (96.4%)	209 (97.2%)	
1	13 (3.2%)	7 (3.6%)	6 (2.8%)	
BLDLOSS	70 (40-100)	80 (40-100)	55 (40-90)	0.057
Missing	8	6	2	
BLDSPASM				0.008
0	295 (71.6%)	129 (65.5%)	166 (77.2%)	
1	117 (28.4%)	68 (34.5%)	49 (22.8%)	

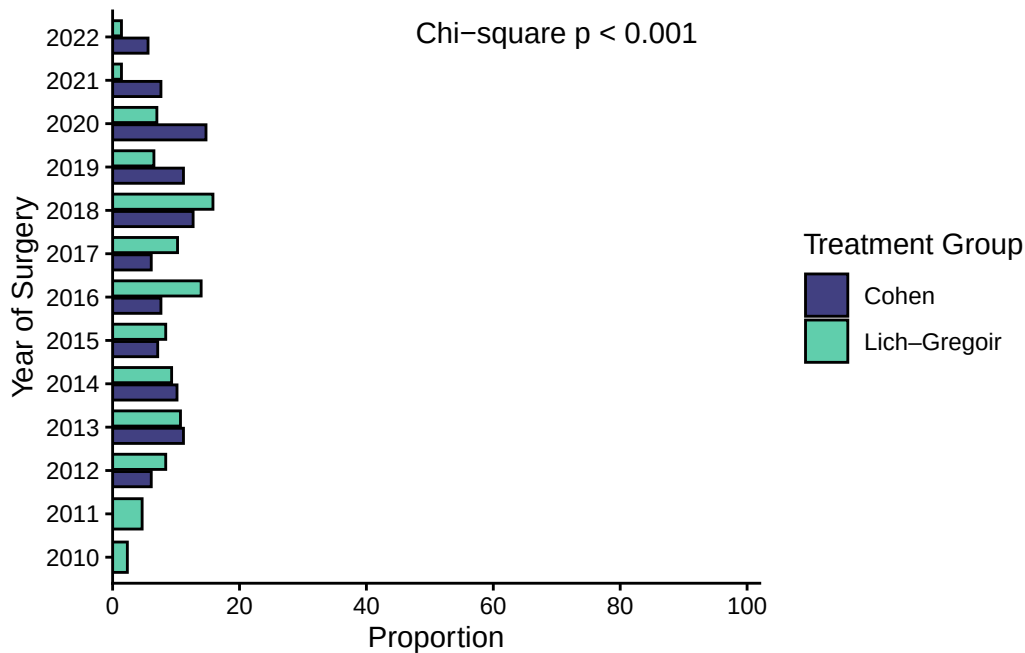
```

count(YEARS, Treatment) %>%
group_by(Treatment) %>%
mutate(
  Percent = 100 * n / sum(n)
) %>%
ungroup()

pt <- ggplot(
  plot_data1,
  aes(
    x = Percent,
    y = factor(YEARS),
    fill = Treatment
  )
) +
geom_col(
  position = position_dodge(width = 0.8),
  width = 0.7,
  colour = "black"
) +
scale_x_continuous(
  limits = c(0, 100),
  breaks = seq(0, 100, 20),
  expand = expansion(mult = c(0, 0.02))
) +
scale_fill_viridis_d(option = "mako", begin = 0.3, end = 0.8) +
labs(
  x = "Proportion",
  y = "Year of Surgery",
  fill = "Treatment Group"
) +
annotate(
  "text",
  x = 70,
  y = Inf,
  label = p_label,
  vjust = 1.5,
  size = 4
) +
theme_classic()

pt

```



```
ggsave(filename = "figures/treatment.png",
        plot = pt,
        width = 8, # width in inches
        height = 10, # height in inches
        dpi = 300) # resolution

##### Postoperative UTI outcome as time-to-event outcome

# Create labeled TRT2 variable only for survival analysis
Out$Treatment <- factor(
  Out$TRT2,
  levels = c(0,1),
  labels = c("Cohen", "Lich-Gregoir")
)

surv_object <- Surv(time = Out$POSTUTIFUP, event = Out$POSTUTI== 1)
fit <- survfit(surv_object ~ Treatment, data = Out)

p_u <- ggsurvplot(
  fit,
  data = Out,
  fun="event", # Show cumulative incidence (1 - survival)
  pval = FALSE,
  pval.size = 4,
  pval.coord = c(0, 0.9), # c(x,y) according to my x and y scale
  conf.int = TRUE,
  risk.table = TRUE,
  risk.table.y.text = FALSE, # to remove text from y-axis in the risk table
  # risk.table.title = "At risk",
  risk.table.fontsize = 4, # controls numbers in table
  xlim = c(0, 135),
```



```

break.time.by = 12,
xlab = "Follow-up time",
ylab = "Cumulative incidence of postoperative symptomatic UTI",
palette = c("#482878", "#2E8B8B"), # instead of "#ADD8E6", "#00008B"
legend.title = "Treatment",
legend.labs = c("Cohen Group", "Lich-Gregoir Group"),
ylim = c(0, 1) # Force y-axis from 0 to 1
)

# calculate P-value from ggsurvplot (pval = TRUE), then turn it off

p_u$plot <- p_u$plot +
  annotate(
    "text",
    x = 20,
    y = 0.90,
    label = "Log rank test p = 0.058",
    size = 4
  )

#-----
# Customize the plot
#-----
p_u$plot <- p_u$plot +
  scale_x_continuous(
    limits = c(0, 135),
    breaks = seq(0, 135, 12)
  ) +
  scale_y_continuous(
    labels = function(x) paste0(round(x * 100, 1), "%"),
    limits = c(0, 1)
  ) +
  theme_classic() + # Many clinical epidemiology journals prefer theme_classic() instead of theme_bw()
  theme(
    axis.title.x = element_text(size = 12),
    axis.title.y = element_text(size = 12),
    axis.text.x = element_text(size = 10),
    axis.text.y = element_text(size = 10),
    legend.title = element_text(size = 11),
    legend.text = element_text(size = 10)
  )

```

Scale for x is already present.

Adding another scale for x, which will replace the existing scale.

Scale for y is already present.

Adding another scale for y, which will replace the existing scale.

```

p_u$plot <- p_u$plot +
  labs(title = "(A) Before overlap weighting")

#-----
# Customize the risk table
#-----

```

```

p_u$table <- p_u$table +
  scale_x_continuous(
    limits = c(0, 135),
    breaks = seq(0, 135, 12)
  ) +
  theme(
    axis.title.x = element_blank(),
    axis.text.x = element_blank(),
    axis.ticks.x = element_blank(),
    axis.line.x = element_blank(),
    axis.line.y = element_blank(),
    plot.title = element_text(size = 12),
    text = element_text(size = 10),
    axis.title.y = element_text(size = 12)
  )

```

Scale for x is already present.

Adding another scale for x, which will replace the existing scale.

```

#-----
# Combine plot and risk table using patchwork
#-----
final_plot <- p_u$plot / p_u$table +
  plot_layout(heights = c(4, 1))

#-----
# Save the figure
#-----
ggsave(
  filename = "figures/UTI.png",
  plot = final_plot,
  width = 8,
  height = 6,
  dpi = 300
)

```

Ignoring unknown labels:

```
* colour : "Treatment"
```

```
# The estimated 1-year cumulative probability of UTI
```

```
s <- summary(fit, times = 12)
```

```

uti_1yr <- data.frame(
  Treatment = s$strata,
  UTI_1yr = 1 - s$surv,
  Lower_95CI = 1 - s$upper,
  Upper_95CI = 1 - s$lower
)

```

```
uti_1yr$UTI_1yr <- 100 * uti_1yr$UTI_1yr
uti_1yr$Lower_95CI <- 100 * uti_1yr$Lower_95CI
uti_1yr$Upper_95CI <- 100 * uti_1yr$Upper_95CI

uti_1yr
```

	Treatment	UTI_1yr	Lower_95CI	Upper_95CI
1	Treatment=Cohen	22.86901	16.77081	28.52039
2	Treatment=Lich-Gregoir	30.69767	24.24980	36.59671

MAIN ANALYSIS

1 Do weighting: using ATO

```
set.seed(123)

w.out <- weightit(
  TRT2 ~ AGS + YEARS + GENDER + BMI + PREOPRNPLN + HISTOINJ +
    SURGIND2 + PREOPVURGRADE + VURSIDE + UDR7M + VURPHASE +
    PREOPEPISODE1 + PREOPTYPEU1 + PREOPRNSCR + SURGTECH2,
  data = Out,
  method = "ps",
  estimand = "ATO",
  stabilize = FALSE
)
Out$weights <- w.out$weights # Add weights to the dataset
write.csv(Out, "data/weighted_data.csv", row.names = FALSE)

summary(w.out$weights)

      Min.    1st Qu.    Median      Mean   3rd Qu.      Max.
0.0007522 0.0688498 0.2352262 0.2909478 0.4649644 0.9527184

# Table for balance
bal <- cobalt::bal.tab(w.out, un = TRUE, ESS = TRUE, m.threshold = 0.1)
bal
```

Balance Measures

	Type	Diff.Un	Diff.Adj	M.Threshold
prop.score	Distance	1.8173	-0.0102	Balanced, <0.1
AGS	Contin.	0.2801	-0.0000	Balanced, <0.1
YEARS	Contin.	-0.4813	0.0000	Balanced, <0.1
GENDER	Binary	-0.0471	-0.0000	Balanced, <0.1
BMI	Contin.	-0.6526	-0.0000	Balanced, <0.1
PREOPRNPLN	Binary	-0.2699	0.0000	Balanced, <0.1
HISTOINJ	Binary	0.0198	-0.0000	Balanced, <0.1
SURGIND2_1	Binary	0.0856	0.0000	Balanced, <0.1
SURGIND2_2	Binary	0.0595	-0.0000	Balanced, <0.1
SURGIND2_3	Binary	-0.0089	-0.0000	Balanced, <0.1

SURGIND2_4	Binary	-0.0465	-0.0000	Balanced, <0.1
SURGIND2_5	Binary	-0.0897	-0.0000	Balanced, <0.1
PREOPVURGRADE_3	Binary	-0.0140	-0.0000	Balanced, <0.1
PREOPVURGRADE_4	Binary	0.0802	-0.0000	Balanced, <0.1
PREOPVURGRADE_5	Binary	-0.0662	0.0000	Balanced, <0.1
VURSIDE	Binary	0.0247	-0.0000	Balanced, <0.1
UDR7M	Binary	-0.1399	0.0000	Balanced, <0.1
VURPHASE_1	Binary	-0.0383	-0.0000	Balanced, <0.1
VURPHASE_2	Binary	0.3112	0.0000	Balanced, <0.1
VURPHASE_3	Binary	-0.2729	-0.0000	Balanced, <0.1
PREOPEPISODE1_1	Binary	-0.0440	-0.0000	Balanced, <0.1
PREOPEPISODE1_2	Binary	-0.1765	-0.0000	Balanced, <0.1
PREOPEPISODE1_3	Binary	0.2205	0.0000	Balanced, <0.1
PREOPTYPEU1	Binary	0.0309	0.0000	Balanced, <0.1
PREOPRNSCR	Binary	-0.2674	-0.0000	Balanced, <0.1
SURGTECH2_1	Binary	0.2494	0.0000	Balanced, <0.1
SURGTECH2_2	Binary	-0.0629	-0.0000	Balanced, <0.1
SURGTECH2_3	Binary	-0.1866	-0.0000	Balanced, <0.1

Balance tally for mean differences

	count
Balanced, <0.1	28
Not Balanced, >0.1	0

Variable with the greatest mean difference

Variable	Diff.Adj	M.Threshold
BMI	-0	Balanced, <0.1

Effective sample sizes

	Control	Treated
Unadjusted	197.	215.
Adjusted	126.08	113.73

```
bal_df <- as.data.frame(bal$Balance)
smd_table <- bal_df %>%
  mutate(
    Variable = rownames(bal_df),
    SMD_Before = round(abs(Diff.Un), 3),
    SMD_After = round(abs(Diff.Adj), 3)
  ) %>%
  dplyr::select(Variable, SMD_Before, SMD_After)

smd_table %>%
  flextable() %>%
  save_as_docx(path = "output/SMD_w.docx")
```

Figure for balance - Love plot

```
pl <- love.plot(
  w.out,
  threshold = 0.1,
  abs = TRUE,
  stats = "mean.diffs",
  var.order = "unadjusted",
```

```

drop.distance = TRUE,
sample.names = c("Before overlap weighting", "After overlap weighting"),
size = 4,
var.names = c(
  AGS = "Age at surgery",
  YEARS = "Year of surgery",
  BMI = "Body mass index",
  GENDER = "Female gender",
  PREOPRNPLN = "Preop pyelonephritis",
  HISTOINJ = "History of injection",
  VURSIDE = "Right-sided VUR",
  UDR7M = "Ureteral dilation 7mm",
  PREOPTYPEU1 = "Preop febrile UTIs",
  PREOPRNSCR = "Preop renal scarring",
  VURPHASE_1 = "Reflux: voiding phase",
  VURPHASE_2 = "Reflux: filling phase",
  VURPHASE_3 = "Reflux: filling + voiding",
  SURGTECH2_1 = "Surgeon experience 100 operations",
  SURGTECH2_2 = "Surgeon experience 50-100 operations",
  SURGTECH2_3 = "Surgeon experience <50 operations",
  PREOPEPISODE1_1 = "Preop UTI episodes: 1",
  PREOPEPISODE1_2 = "Preop UTI episodes: 2",
  PREOPEPISODE1_3 = "Preop UTI episodes: 3",
  SURGIND2_1 = "Indication: breakthrough UTI",
  SURGIND2_2 = "Indication: high-grade reflux (III-V)",
  SURGIND2_3 = "Indication: poor compliance to medical treatment",
  SURGIND2_4 = "Indication: ineffective/poorly tolerated CAP",
  SURGIND2_5 = "Indication: other",
  PREOPVURGRADE_3 = "Preop VUR grade: III",
  PREOPVURGRADE_4 = "Preop VUR grade: IV",
  PREOPVURGRADE_5 = "Preop VUR grade: V"
)
)

```

Warning: Standardized mean differences and raw mean differences are present in the same plot. Use the `stars` argument to distinguish between them and appropriately label the x-axis. See `love.plot()` for details.

```

pl <- pl +
  scale_color_manual(
    values = c(
      "Before overlap weighting" = "#0072B2",
      "After overlap weighting" = "#D55E00"
    )
  ) +
  labs(
    x = "Absolute Standardized Differences",
    subtitle = "Overlap population of all children"
  ) +
  theme(
    plot.margin = margin(10, 10, 10, 10),
    axis.text = element_text(size = 16),
    axis.title = element_text(size = 16),

```

```

plot.title = element_text(size = 16),
legend.text = element_text(size = 16),
legend.title = element_text(size = 16),
axis.ticks = element_line(size = 0.5, color = "black"),
axis.line = element_line(size = 0.5, color = "black"),
panel.background = element_blank(),
panel.grid.major = element_blank(),
panel.grid.minor = element_blank(),
panel.border = element_blank(),
axis.ticks.length = unit(0.3, "cm"),
panel.spacing = unit(2, "lines")
)

```

Scale for colour is already present.

Adding another scale for colour, which will replace the existing scale.

Warning: The `size` argument of `element\_line()` is deprecated as of ggplot2 3.4.0.
 i Please use the `linewidth` argument instead.

```
ggsave("figures/love_plot_w.png", plot = pl, width = 12, height = 10, dpi = 300)
```

```

# Compute mean SMD
smd_df <- bal$Balance
mean_smd_before <- mean(abs(smd_df$Diff.Un), na.rm = TRUE) # calculate mean SMD before matching
mean_smd_after <- mean(abs(smd_df$Diff.Adj), na.rm = TRUE) # calculate mean SMD before matching
mean_smd <- data.frame(
  Mean_SMD_Before = mean_smd_before,
  Mean_SMD_After = mean_smd_after
)

```

```
mean_smd
```

```

  Mean_SMD_Before Mean_SMD_After
1      0.2158582    0.0003644521

```

```

# how much effective sample size remains (overall ESS: calculated manually)?
summary(w.out$weights)

```

```

      Min.    1st Qu.    Median      Mean    3rd Qu.      Max.
0.0007522 0.0688498 0.2352262 0.2909478 0.4649644 0.9527184

```

```
bal.tab(w.out)$Effective.Sample.Size
```

```
NULL
```

```
sum(w.out$weights)^2 / sum(w.out$weights^2)
```

```
[1] 239.181
```

```

# propensity score distribution
summary(w.out$ps)

      Min.   1st Qu.   Median     Mean   3rd Qu.     Max.
0.006799 0.218309 0.504244 0.521845 0.803450 0.999248

pl2 <-plot(summary(w.out, type = "hist"))
ggsave ("figures/propensity_w.png", plot = pl2, width = 12, height = 10, dpi = 300)

# Distribution of overlap weights: histogram and density

## Histogram
summary(w.out$weights)

      Min.   1st Qu.   Median     Mean   3rd Qu.     Max.
0.0007522 0.0688498 0.2352262 0.2909478 0.4649644 0.9527184

png("figures/weights_histo_w.png",
    width = 6,
    height = 4,
    units = "in",
    res = 300)

hist(
  w.out$weights,
  breaks = 30,
  col = "lightblue",
  border = "black",
  main = "Distribution of Overlap Weights",
  xlab = "Overlap Weight"
)
dev.off()

pdf
2

# Before-After weighting
p_ps <- bal.plot(
  w.out,
  var.name = "prop.score",
  type = "density",
  which = "both"
)

p_ps <- p_ps +
  scale_color_viridis_d(
    option = "mako",
    begin = 0.3,
    end = 0.8,
    labels = c("Cohen group", "Lich-Gregoir group")
  )

```

```

) +
scale_fill_viridis_d(
  option = "mako",
  begin = 0.3,
  end = 0.8,
  labels = c("Cohen group", "Lich-Gregoir group")
)

```

Scale for fill is already present.

Adding another scale for fill, which will replace the existing scale.

```

ggsave(
  "figures/PS_density_before_after_ATO.png",
  p_ps,
  width = 10,
  height = 7,
  dpi = 300
)

```

2 Descriptive analysis: AFTER weighting

Create survey design object using overlap weights

```

design_ato <- svydesign(
  ids = ~1,
  data = Out,
  weights = ~weights
)

```

Weighted baseline characteristics

```

table1w <- design_ato %>%
  tbl_svysummary(
    by = TRT2,
    include = c(
      YEARS, AGS, GENDER, BMI, HISTOINJ, SURGIND2, PREOPVURGRADE, VURSIDE, UDR7M,
      VURPHASE, PREOPTYPEU1, PREOPEPISODE1, PREOPRNPLN, PREOPRNSCR,
      SURGTECH2, CENTER
    ),
    statistic = list(
      all_continuous() ~ "{median} ({p25}-{p75})",
      all_categorical() ~ "{n} ({p}%"
    ),
    digits = all_categorical() ~ c(0, 1),
    missing = "ifany",
    missing_text = "Missing"
  ) %>%
  add_overall() %>%
  bold_labels()

```

```
table1w
```


Characteristic	Overall N = 120 ^I	0 N = 60 ^I	1 N = 60
YEARS	2,017.00 (2,014.00-2,019.00)	2,017.00 (2,014.00-2,019.00)	2,017.00 (2,014.00-2,019.00)
AGS	8.00 (6.00-9.00)	8.00 (7.00-9.00)	7.00 (6.00-9.00)
GENDER			
0	65 (54.1%)	32 (54.1%)	32 (54.1%)
1	55 (45.9%)	28 (45.9%)	28 (45.9%)
BMI	14.1 (13.5-17.4)	14.0 (13.2-17.7)	14.1 (13.6-16.9)
HISTOINJ			
0	104 (87.1%)	52 (87.1%)	52 (87.1%)
1	16 (12.9%)	8 (12.9%)	8 (12.9%)
SURGIND2			
1	55 (46.2%)	28 (46.2%)	28 (46.2%)
2	48 (40.1%)	24 (40.1%)	24 (40.1%)
3	6 (5.2%)	3 (5.2%)	3 (5.2%)
4	2 (1.9%)	1 (1.9%)	1 (1.9%)
5	8 (6.6%)	4 (6.6%)	4 (6.6%)
PREOPVURGRADE			
3	7 (5.9%)	4 (5.9%)	4 (5.9%)
4	35 (29.0%)	17 (29.0%)	17 (29.0%)
5	78 (65.1%)	39 (65.1%)	39 (65.1%)
VURSIDE			
0	45 (37.4%)	22 (37.4%)	22 (37.4%)
1	75 (62.6%)	38 (62.6%)	38 (62.6%)
UDR7M			
0	28 (23.5%)	14 (23.5%)	14 (23.5%)
1	92 (76.5%)	46 (76.5%)	46 (76.5%)
VURPHASE			
1	87 (72.9%)	44 (72.9%)	44 (72.9%)
2	2 (1.5%)	1 (1.5%)	1 (1.5%)
3	31 (25.5%)	15 (25.5%)	15 (25.5%)
PREOPTYPEU1			
0	1 (1.2%)	1 (1.2%)	1 (1.2%)
1	118 (98.8%)	59 (98.8%)	59 (98.8%)
PREOPEPISODE1			
1	7 (5.7%)	3 (5.7%)	3 (5.7%)
2	35 (29.5%)	18 (29.5%)	18 (29.5%)
3	78 (64.9%)	39 (64.9%)	39 (64.9%)
PREOPRNPLN			
0	101 (84.2%)	50 (84.2%)	50 (84.2%)
1	19 (15.8%)	9 (15.8%)	9 (15.8%)
PREOPRNSCR			
0	84 (69.8%)	42 (69.8%)	42 (69.8%)
1	36 (30.2%)	18 (30.2%)	18 (30.2%)
SURGTECH2			
1	7 (6.2%)	4 (6.2%)	4 (6.2%)
2	99 (82.3%)	49 (82.3%)	49 (82.3%)
3	14 (11.5%)	7 (11.5%)	7 (11.5%)
CENTER			
1	87 (72.9%) ²⁵	42 (69.4%)	46 (76.4%)
2	1 (0.9%)	1 (1.8%)	0 (0.0%)
3	9 (7.1%)	5 (8.5%)	3 (5.8%)

```

table1w %>%
  as_flex_table() %>%
  flextable::save_as_docx(
    path = "output/baseline_w.docx"
  )

# Weighted procedural factors
table2w<- design_ato %>%
  tbl_svsummary(
    by = TRT2, # for stratification
    include = c (
      OPRTIME, STENT, CATH, CATHPERIOD, CATHPERIOD_G, HOSPTIME
    ),
    statistic = list(
      all_continuous() ~ "{median} ({p25}-{p75})", # or ~ "{mean} ± {sd}",
      all_categorical() ~ "{n} ({p}%)"
    ),
    digits = all_categorical() ~ c(0, 1), # n = 0 decimals, p = 1 decimal
    missing = "ifany",
    missing_text = "Missing"
  ) %>%
  add_overall() %>%
  add_p(pvalue_fun = ~style_pvalue(.x, digits = 3)) %>% # 3 decimals
  bold_labels()

```

The following warnings were returned during `add\_p()`:

```

! For variable `CATH` (`TRT2`) and "statistic" and "p.value" statistics:
  Chi-squared approximation may be incorrect

```

```
table2w
```

```

table2w %>%
  as_flex_table() %>%
  flextable::save_as_docx(path = "output/procedural_w.docx")

# Outcomes
table3w<- design_ato %>%
  tbl_svsummary(
    by = TRT2,
    include = c (
      FUPM, LUTD, OBST, URCOMPINJ, SURG, ILEUS, CLAVIENCLASS, URCOMPCVUR,
      GU30, GU90, GU90N, GUER30, GUER90, PROC365,
      BLDLOSS, BLDSPASM, HEMAT, POSTUTI
    ),
    statistic = list(
      all_continuous() ~ "{median} ({p25}-{p75})", # or ~ "{mean} ± {sd}",
      all_categorical() ~ "{n} ({p}%)"
    ),
    digits = all_categorical() ~ c(0, 1), # n = 0 decimals, p = 1 decimal
    missing = "ifany",

```

Characteristic	Overall N = 120 ¹	0 N = 60 ¹	1 N = 60 ¹	p-value ²
OPRTIME	85 (70-130)	130 (95-160)	70 (68-80)	<0.001
STENT				0.075
0	78 (64.7%)	35 (59.1%)	42 (70.4%)	
1	42 (35.3%)	25 (40.9%)	18 (29.6%)	
CATH				0.555
0	2 (2.1%)	1 (1.5%)	2 (2.6%)	
1	117 (97.9%)	59 (98.5%)	58 (97.4%)	
CATHPERIOD	4 (3-5)	4 (4-5)	3 (1-5)	<0.001
CATHPERIOD_G				<0.001
Early removal 3 days	38 (31.8%)	5 (8.1%)	33 (55.6%)	
Standard removal 4-7 days	58 (48.5%)	43 (71.6%)	15 (25.4%)	
Delayed removal >7 days	24 (19.7%)	12 (20.3%)	11 (19.1%)	
HOSPTIME	5.00 (3.00-6.00)	5.00 (5.00-6.00)	3.00 (2.00-5.00)	<0.001

¹Median (Q1-Q3); n (%)

²Design-based KruskalWallis test; Pearson's X²: Rao & Scott adjustment

```

    missing_text = "Missing"
  ) %>%
  add_overall() %>%
  add_p(pvalue_fun = ~style_pvalue(.x, digits = 3)) %>% # 3 decimals
  bold_labels()

```

The following warnings were returned during `add\_p()`:

```

! For variable `CLAVIENCLASS` (`TRT2`) and "statistic" and "p.value"
  statistics: Chi-squared approximation may be incorrect
! For variable `URCOMPINJ` (`TRT2`) and "statistic" and "p.value" statistics:
  Chi-squared approximation may be incorrect

```

table3w

```

table3w %>%
  as_flex_table() %>%
  flextable::save_as_docx(path = "output/outcomes_w.docx")

```

Postoperative UTI outcome as time-to-event outcome: Weighted survival curve

```

fit_w <- survfit(
  Surv(POSTUTIFUP, POSTUTI == 1) ~ Treatment,
  data = Out,
  weights = w.out$weights
)

```

```

p_u_w <- ggsurvplot(
  fit_w,

```

Characteristic	Overall N = 120 ¹	0 N = 60 ¹	1 N = 60 ¹	p-value ²
FUPM	26 (24-38)	30 (24-39)	24 (13-36)	0.007
LUTD				0.696
0	99 (82.7%)	50 (83.6%)	49 (81.8%)	
1	21 (17.3%)	10 (16.4%)	11 (18.2%)	
OBST				0.028
0	108 (90.2%)	57 (94.3%)	52 (86.0%)	
1	12 (9.8%)	3 (5.7%)	8 (14.0%)	
URCOMPINJ				0.165
0	117 (97.8%)	58 (96.5%)	59 (99.1%)	
1	3 (2.2%)	2 (3.5%)	1 (0.9%)	
SURG				0.469
0	86 (71.5%)	42 (69.3%)	44 (73.7%)	
1	34 (28.5%)	18 (30.7%)	16 (26.3%)	
ILEUS				0.004
0	111 (92.7%)	53 (88.1%)	58 (97.3%)	
1	9 (7.3%)	7 (11.9%)	2 (2.7%)	
CLAVIENCLASS				0.002
1	57 (60.7%)	35 (72.6%)	22 (48.2%)	
2	25 (26.7%)	7 (15.5%)	18 (38.5%)	
3	11 (12.1%)	6 (11.9%)	6 (12.2%)	
4	1 (0.5%)	0 (0.0%)	1 (1.1%)	
Missing	26	12	14	
URCOMPVCUR				<0.001
0	108 (97.3%)	52 (94.7%)	56 (99.8%)	
1	3 (2.7%)	3 (5.3%)	0 (0.2%)	
Missing	9	5	3	
GU30				<0.001
0	97 (81.1%)	40 (66.6%)	57 (95.6%)	
1	23 (18.9%)	20 (33.4%)	3 (4.4%)	
GU90				<0.001
0	92 (77.0%)	40 (66.3%)	53 (87.7%)	
1	28 (23.0%)	20 (33.7%)	7 (12.3%)	
GU90N				<0.001
0	69 (58.0%)	28 (46.1%)	42 (69.9%)	
1	28 (23.0%)	20 (33.7%)	7 (12.3%)	
2	23 (19.1%)	12 (20.3%)	11 (17.8%)	
GUER30				<0.001
0	84 (70.3%)	29 (48.0%)	56 (92.6%)	
1	36 (29.7%)	31 (52.0%)	4 (7.4%)	
GUER90				<0.001
0	79 (66.3%)	27 (45.6%)	52 (86.9%)	
1	40 (33.7%)	33 (54.4%)	8 (13.1%)	
PROC365				0.630
0	114 (95.3%)	57 (94.6%)	58 (96.0%)	
1	6 (4.7%)	3 (5.4%)	2 (4.0%)	
BLDLOSS	70 (40-100)	90 (50-100)	50 (40-80)	<0.001
Missing	1	0	1	
BLDSPASM				0.213
0	82 (68.1%)	39 (64.3%)	43 (71.8%)	
1	38 (31.9%)	21 (35.7%)	17 (28.2%)	

```

data = Out,
fun = "event",
pval = FALSE,
pval.size = 4,
pval.coord = c(0, 0.9), # c(x,y) according to my x and y scale
conf.int = TRUE,
risk.table = TRUE,
risk.table.y.text = FALSE, # to remove text from y-axis in the risk table
risk.table.fontsize = 4, # controls numbers in table
xlim = c(0,135),
break.time.by = 12,
xlab = "Follow-up time",
ylab = "Cumulative incidence of postoperative symptomatic UTI",
palette = c("#482878", "#2E8B8B"), # instead of "#ADD8E6", "#00008B"
legend.title = "Treatment",
legend.labs = c("Cohen Group", "Lich-Gregoir Group"),
ylim = c(0,1)
)

cox_w <- svycoxph(
  Surv(POSTUTIFUP, POSTUTI == 1) ~ Treatment,
  design = design_ato
)

summary(cox_w)

```

Independent Sampling design (with replacement)
 svydesign(ids = ~1, data = Out, weights = ~weights)

Call:
 svycoxph(formula = Surv(POSTUTIFUP, POSTUTI == 1) ~ Treatment,
 design = design_ato)

n= 412, number of events= 112

	coef	exp(coef)	se(coef)	robust se	z	Pr(> z)
TreatmentLich-Gregoir	-0.2859	0.7513	0.1903	0.2483	-1.151	0.25

	exp(coef)	exp(-coef)	lower .95	upper .95
TreatmentLich-Gregoir	0.7513	1.331	0.4618	1.222

Concordance= 0.54 (se = 0.035)
 Likelihood ratio test= NA on 1 df, p=NA
 Wald test = 1.33 on 1 df, p=0.2
 Score (logrank) test = NA on 1 df, p=NA

(Note: the likelihood ratio and score tests assume independence of observations within a cluster, the Wald and robust score tests do not).

```

p_u_w$plot <- p_u_w$plot +
  annotate(
    "text",
    x = 20,

```

```

    y = 0.90,
    label = "Weighted Cox p = 0.250",
    size = 4
  )

#-----
# Customize the plot
#-----
p_u_w$plot <- p_u_w$plot +
  scale_x_continuous(
    limits = c(0, 135),
    breaks = seq(0, 135, 12)
  ) +
  scale_y_continuous(
    labels = function(x) paste0(round(x * 100, 1), "%"),
    limits = c(0, 1)
  ) +
  theme_classic() + # Many clinical epidemiology journals prefer theme_classic() instead of theme_bw()
  theme(
    axis.title.x = element_text(size = 12),
    axis.title.y = element_text(size = 12),
    axis.text.x = element_text(size = 10),
    axis.text.y = element_text(size = 10),
    legend.title = element_text(size = 11),
    legend.text = element_text(size = 10)
  )

```

Scale for x is already present.

Adding another scale for x, which will replace the existing scale.

Scale for y is already present.

Adding another scale for y, which will replace the existing scale.

```

p_u_w$plot <- p_u_w$plot +
  labs(title = "(B) After overlap weighting")

#-----
# Customize the risk table
#-----
p_u_w$table <- p_u_w$table +
  scale_x_continuous(
    limits = c(0, 135),
    breaks = seq(0, 135, 12)
  ) +
  theme(
    axis.title.x = element_blank(),
    axis.text.x = element_blank(),
    axis.ticks.x = element_blank(),
    axis.line.x = element_blank(),
    axis.line.y = element_blank(),
    plot.title = element_text(size = 12),
    text = element_text(size = 10),
    axis.title.y = element_text(size = 12)
  )

```

```

)

Scale for x is already present.
Adding another scale for x, which will replace the existing scale.

#-----
# Combine plot and risk table
#-----
final_plot_w <- p_u_w$plot / p_u_w$table +
  plot_layout(heights = c(4, 1))

#-----
# Save the figure
#-----
ggsave(
  filename = "figures/UTI_w.png",
  plot = final_plot_w,
  width = 8,
  height = 6,
  dpi = 300
)

Ignoring unknown labels:
* colour : "Treatment"

# Combine UTI before and after weighting plots

UTI_combined <- final_plot | final_plot_w

ggsave(
  filename = "figures/UTI_ba_w.png",
  plot = UTI_combined,
  width = 16,
  height = 6,
  dpi = 300
)

Ignoring unknown labels:
* colour : "Treatment"
Ignoring unknown labels:
* colour : "Treatment"

# Estimated 1-year survival
s_w <- summary(fit_w, times = 12, extend = TRUE) # # The extend = TRUE is useful if there isn't an even

# Estimated 1-year cumulative probability of UTI
uti_1yr_w <- data.frame(
  Treatment = s_w$strata,
  UTI_1yr = 100 * (1 - s_w$surv),
  Lower_95CI = 100 * (1 - s_w$upper),

```

```
Upper_95CI = 100 * (1 - s_w$lower)
)
```

```
uti_1yr_w
```

	Treatment	UTI_1yr	Lower_95CI	Upper_95CI
1	Treatment=Cohen	30.61186	21.73584	38.48124
2	Treatment=Lich-Gregoir	23.81598	15.91515	30.97443

3 Regression models

Block 1

```
# GEE model after overlap weighting - Accounting for center-clustering
```

```
# Block 1: technique effect on postoperative outcomes = Univariable analyses
```

```
outcomes <- c(
  "LUTD",
  "OBST",
  "URCOMPINJ",
  "SURG",
  "ILEUS",
  "GU30",
  "GU90",
  "PROC365"
)
```

```
# convert outcomes to numeric
Out[outcomes] <- lapply(
  Out[outcomes],
  function(x) as.numeric(as.character(x))
)
```

```
# Apply GEE model after overlap weighting for all binary outcomes
```

```
gee_results <- lapply(outcomes, function(y) {
  formula <- as.formula(
    paste(y, "~ TRT2")
  )
  model <- geeglm(
    formula,
    data = Out,
    id = CENTER,
    weights = weights,
    family = binomial(link = "logit"),
    corstr = "exchangeable"
  )
  tidy(model, exponentiate = TRUE, conf.int = TRUE) # estimate = OR because exponentiate = TRUE
})
names(gee_results) <- outcomes
```



```

# Combine results into one table
gee_table <- do.call(
  rbind,
  lapply(names(gee_results), function(x){
    tmp <- gee_results[[x]]
    tmp <- tmp %>%
      filter(grepl("^TRT2", term)) %>%
      mutate(outcome = x)
    tmp
  })
)

# Create final manuscript table
gee_table_final <- gee_table %>%
  mutate(
    Effect_measure = "OR",
    Estimate = sprintf("%.2f", estimate),
    CI = paste0(
      sprintf("%.2f", conf.low),
      "-",
      sprintf("%.2f", conf.high)
    ),
    P_value = p.value
  ) %>%
  dplyr::select(
    Outcome = outcome,
    Effect_measure,
    Estimate,
    CI,
    P_value
  )

# weighted Cox model for POSTUTI

cox_uti <- svycoxph(
  Surv(POSTUTIFUP, POSTUTI == 1) ~ TRT2 + strata(CENTER),
  design = design_ato
)

cox_summary <- summary(cox_uti)

Independent Sampling design (with replacement)
svydesign(ids = ~1, data = Out, weights = ~weights)

cox_row <- grep(
  "^TRT2",
  rownames(cox_summary$coefficients)
)

POSTUTI_table <- data.frame(
  Outcome = "POSTUTI",
  Effect_measure = "HR",
  Estimate = sprintf(

```

```

      "%.2f",
      cox_summary$coefficients[
        cox_row,
        "exp(coef)"
      ]
    ),
    CI = paste0(
      sprintf("%.2f",
        cox_summary$conf.int[cox_row, "lower .95"]),
      "-",
      sprintf("%.2f",
        cox_summary$conf.int[cox_row, "upper .95"])
    ),
    P_value = cox_summary$coefficients[
      cox_row,
      "Pr(>|z|)"
    ]
  )

# Combine final results
final_results <- rbind(
  gee_table_final,
  POSTUTI_table
)

# Multiple correction

## Bonferroni correction
final_results$p_bonferroni <- p.adjust(
  final_results$P_value,
  method = "bonferroni"
)

## Holm-Bonferroni
final_results$p_holm <- p.adjust(
  final_results$P_value,
  method = "holm"
)

## Benjamini-Hochberg FDR
final_results$p_FDR <- p.adjust(
  final_results$P_value,
  method = "BH"
)

final_results <- final_results %>%
  mutate(
    across(
      c(P_value, p_bonferroni, p_holm, p_FDR),
      ~case_when(
        .x < 0.001 ~ "<0.001",

```

```

      TRUE ~ sprintf("%.3f", .x)
    )
  )
)

```

```
final_results
```

```
# A tibble: 9 x 8
```

	Outcome <chr>	Effect_measure <chr>	Estimate <chr>	CI <chr>	P_value <chr>	p_bonferroni <chr>	p_holm <chr>	p_FDR <chr>
1	LUTD	OR	1.16	0.80-1.67	0.446	1.000	0.446	0.446
2	OBST	OR	3.67	2.21-6.10	<0.001	<0.001	<0.001	<0.001
3	URCOMPINJ	OR	0.43	0.27-0.69	<0.001	0.004	0.003	0.001
4	SURG	OR	0.72	0.58-0.90	0.004	0.038	0.021	0.008
5	ILEUS	OR	0.10	0.04-0.25	<0.001	<0.001	<0.001	<0.001
6	GU30	OR	0.02	0.00-1.87	0.090	0.814	0.362	0.129
7	GU90	OR	0.20	0.16-0.26	<0.001	<0.001	<0.001	<0.001
8	PROC365	OR	0.79	0.60-1.05	0.109	0.982	0.362	0.129
9	POSTUTI	HR	0.67	0.41-1.10	0.114	1.000	0.362	0.129

```

final_results %>%
  flextable() %>%
  theme_booktabs() %>%
  autofit() %>%
  align(
    j = c("Estimate", "CI", "P_value"),
    align = "center",
    part = "all"
  ) %>%
  save_as_docx(
    path = "output/treatment_unadj_w_norec.docx"
  )

```

```
# Evaluate whether CENTER distribution is balanced after weighting, despite that CENTER was not included
```

```

bal.tab(
  w.out,
  data = Out,
  add1 = ~ CENTER
)

```

```
Balance Measures
```

	Type	Diff.Adj
prop.score	Distance	-0.0102
AGS	Contin.	-0.0000
YEARS	Contin.	0.0000
GENDER	Binary	-0.0000
BMI	Contin.	-0.0000
PREOPRNPLN	Binary	0.0000
HISTOINJ	Binary	-0.0000
SURGIND2_1	Binary	0.0000
SURGIND2_2	Binary	-0.0000

SURGIND2_3	Binary	-0.0000
SURGIND2_4	Binary	-0.0000
SURGIND2_5	Binary	-0.0000
PREOPVURGRADE_3	Binary	-0.0000
PREOPVURGRADE_4	Binary	-0.0000
PREOPVURGRADE_5	Binary	0.0000
VURSIDE	Binary	-0.0000
UDR7M	Binary	0.0000
VURPHASE_1	Binary	-0.0000
VURPHASE_2	Binary	0.0000
VURPHASE_3	Binary	-0.0000
PREOPEPISODE1_1	Binary	-0.0000
PREOPEPISODE1_2	Binary	-0.0000
PREOPEPISODE1_3	Binary	0.0000
PREOPTYPEU1	Binary	0.0000
PREOPRNSCR	Binary	-0.0000
SURGTECH2_1	Binary	0.0000
SURGTECH2_2	Binary	-0.0000
SURGTECH2_3	Binary	-0.0000
CENTER_1	Binary	0.0703
CENTER_2	Binary	-0.0185
CENTER_3	Binary	-0.0273
CENTER_5	Binary	-0.0245

Effective sample sizes

	Control	Treated
Unadjusted	197.	215.
Adjusted	126.08	113.73

Results showed that all CENTER SMD < 0.05 → excellent balance

Therefore, after ATO weighting, CENTER is well balanced between treatment groups.

Block 2a - without VUR recurrence

Block 2: exposure-effect analysis (stent) on postoperative outcomes = Multivariable analyses

```

outcomes <- c(
  "LUTD",
  "OBST",
  "URCOMPINJ",
  "SURG",
  "ILEUS",
  "GU30",
  "GU90",
  "PROC365"
)

# convert outcomes to numeric
Out[outcomes] <- lapply(
  Out[outcomes],
  function(x) as.numeric(as.character(x))
)

```

```

# Predictors to include in the outcome models
predictors <- c("TRT2", "STENT")

# Apply GEE model after overlap weighting for all binary outcomes
gee_results <- lapply(outcomes, function(y) {
  formula <- as.formula(
    paste(y, "~",
          paste(predictors, collapse = " + ")
    )
  )
  model <- geeglm(
    formula,
    data = Out,
    id = CENTER,
    weights = weights,
    family = binomial(link = "logit"),
    corstr = "exchangeable"
  )
  tidy(model, exponentiate = TRUE, conf.int = TRUE) # estimate = OR because exponentiate = TRUE
})
names(gee_results) <- outcomes

# Combine results into one table
gee_table <- do.call(
  rbind,
  lapply(names(gee_results), function(x){
    tmp <- gee_results[[x]]
    tmp %>%
      mutate(outcome = x)
  })
)

# Create final manuscript table
gee_table_final <- gee_table %>%
  mutate(
    Effect_measure = "OR",
    Estimate = sprintf("%.2f", estimate),
    CI = paste0(
      sprintf("%.2f", conf.low),
      "-",
      sprintf("%.2f", conf.high)
    ),
    P_value = p.value
  ) %>%
  dplyr::select(
    Outcome = outcome,
    Predictor = term,
    Effect_measure,
    Estimate,
    CI,
    P_value
  )

```

```

# weighted Cox model for POSTUTI

cox_uti <- svycoxph(
  Surv(POSTUTIFUP, POSTUTI == 1) ~ TRT2 + STENT + strata(CENTER),
  design = design_ato
)

cox_summary <- summary(cox_uti)

Independent Sampling design (with replacement)
svydesign(ids = ~1, data = Out, weights = ~weights)

cox_row <- rownames(cox_summary$coefficients)

POSTUTI_table <- data.frame(
  Outcome = "POSTUTI",
  Predictor = rownames(cox_summary$coefficients),
  Effect_measure = "HR",
  Estimate = sprintf(
    "%.2f",
    cox_summary$coefficients[
      cox_row,
      "exp(coef)"
    ]
  ),
  CI = paste0(
    sprintf("%.2f",
      cox_summary$conf.int[cox_row, "lower .95"]),
    "-",
    sprintf("%.2f",
      cox_summary$conf.int[cox_row, "upper .95"])
  ),
  P_value = cox_summary$coefficients[
    cox_row,
    "Pr(>|z|)"
  ]
)

# Combine final results
final_results <- rbind(
  gee_table_final,
  POSTUTI_table
)

final_results

# A tibble: 26 x 6
  Outcome Predictor Effect_measure Estimate CI P_value
* <chr> <chr> <chr> <chr> <chr> <dbl>
1 LUTD (Intercept) OR 0.32 0.24-0.42 2.78e-15

```

2	LUTD	TRT21	OR	1.01	0.69-1.50	9.43e- 1
3	LUTD	STENT1	OR	0.35	0.14-0.84	1.85e- 2
4	OBST	(Intercept)	OR	0.02	0.01-0.06	2.05e-13
5	OBST	TRT21	OR	3.71	2.50-5.52	8.75e-11
6	OBST	STENT1	OR	1.09	0.50-2.40	8.24e- 1
7	URCOMPINJ	(Intercept)	OR	0.03	0.02-0.05	0
8	URCOMPINJ	TRT21	OR	0.36	0.27-0.49	6.58e-11
9	URCOMPINJ	STENT1	OR	2.03	0.52-7.91	3.06e- 1
10	SURG	(Intercept)	OR	1.04	0.75-1.45	8.01e- 1

i 16 more rows

```
final_results <- final_results %>%
  mutate(
    across(
      c(P_value),
      ~case_when(
        .x < 0.001 ~ "<0.001",
        TRUE ~ sprintf("%.3f", .x)
      )
    )
  )
```

```
final_results %>%
  flextable() %>%
  theme_booktabs() %>%
  autofit() %>%
  align(
    j = c("Estimate", "CI", "P_value"),
    align = "center",
    part = "all"
  ) %>%
  save_as_docx(
    path = "output/treatment_adj_all_wa_norec.docx"
  )
```

```
# Multiple correction
```

```
# Extract TRT2 rows only
```

```
gee_trt2 <- gee_table_final %>%
  filter(grepl("^TRT2", Predictor))
```

```
POSTUTI_trt2 <- POSTUTI_table %>%
  filter(grepl("^TRT2", Predictor))
```

```
# Combine only treatment effects
```

```
treatment_results <- rbind(
  gee_trt2,
  POSTUTI_trt2
)
```

```
# Apply Correction
```

```

treatment_results$p_bonferroni <- p.adjust(
  as.numeric(treatment_results$P_value),
  method = "bonferroni"
) # Bonferroni correction

treatment_results$p_holm <- p.adjust(
  as.numeric(treatment_results$P_value),
  method = "holm"
) # Holm-Bonferroni

treatment_results$p_FDR <- p.adjust(
  as.numeric(treatment_results$P_value),
  method = "BH"
) # Benjamini-Hochberg FDR

treatment_results <- treatment_results %>%
  mutate(
    across(
      c(P_value, p_bonferroni, p_holm, p_FDR),
      ~case_when(
        .x < 0.001 ~ "<0.001",
        TRUE ~ sprintf("%.3f", .x)
      )
    )
  )

treatment_results

# A tibble: 9 x 9
  Outcome Predictor Effect_measure Estimate CI P_value p_bonferroni p_holm
  <chr> <chr> <chr> <chr> <chr> <chr> <chr>
1 LUTD TRT21 OR 1.01 0.69-- 0.943 1.000 0.943
2 OBST TRT21 OR 3.71 2.50-- <0.001 <0.001 <0.001
3 URCOMPINJ TRT21 OR 0.36 0.27-- <0.001 <0.001 <0.001
4 SURG TRT21 OR 0.63 0.50-- <0.001 0.002 0.001
5 ILEUS TRT21 OR 0.20 0.17-- <0.001 <0.001 <0.001
6 GU30 TRT21 OR 0.03 0.01-- <0.001 <0.001 <0.001
7 GU90 TRT21 OR 0.18 0.15-- <0.001 <0.001 <0.001
8 PROC365 TRT21 OR 0.71 0.56-- 0.003 0.031 0.010
9 POSTUTI TRT21 HR 0.67 0.41-- 0.114 1.000 0.228
# i 1 more variable: p_FDR <chr>

treatment_results %>%
  flextable() %>%
  theme_booktabs() %>%
  autofit() %>%
  align(
    j = c("Estimate", "CI", "P_value"),
    align = "center",
    part = "all"
  ) %>%
  save_as_docx(

```



```

    path = "output/treatment_adj_wa_norec.docx"
  )

# ADDITIONAL MULTIPLE COMPARISONS

# Extract STENT rows only

gee_st <- gee_table_final %>%
  filter(grepl("^STENT", Predictor))

POSTUTI_st <- POSTUTI_table %>%
  filter(grepl("^STENT", Predictor))

# Combine only stent effects
st_results <- rbind(
  gee_st,
  POSTUTI_st
)

# Apply Correction

st_results$p_bonferroni <- p.adjust(
  as.numeric(st_results$P_value),
  method = "bonferroni"
) # Bonferroni correction

st_results$p_holm <- p.adjust(
  as.numeric(st_results$P_value),
  method = "holm"
) # Holm-Bonferroni

st_results$p_FDR <- p.adjust(
  as.numeric(st_results$P_value),
  method = "BH"
) # Benjamini-Hochberg FDR

st_results <- st_results %>%
  mutate(
    across(
      c(P_value, p_bonferroni, p_holm, p_FDR),
      ~case_when(
        .x < 0.001 ~ "<0.001",
        TRUE ~ sprintf("%.3f", .x)
      )
    )
  )

st_results

# A tibble: 9 x 9
  Outcome Predictor Effect_measure Estimate CI P_value p_bonferroni p_holm

```

	<chr>	<chr>	<chr>	<chr>	<chr>	<chr>	<chr>
1	LUTD	STENT1	OR	0.35	0.14--	0.019	0.167
2	OBST	STENT1	OR	1.09	0.50--	0.824	1.000
3	URCOMPINJ	STENT1	OR	2.03	0.52--	0.306	1.000
4	SURG	STENT1	OR	0.11	0.03--	<0.001	0.007
5	ILEUS	STENT1	OR	0.06	0.01--	<0.001	0.002
6	GU30	STENT1	OR	0.25	0.09--	0.007	0.065
7	GU90	STENT1	OR	0.22	0.07--	0.013	0.120
8	PROC365	STENT1	OR	0.50	0.11--	0.387	1.000
9	POSTUTI	STENT1	HR	0.93	0.42--	0.854	1.000

i 1 more variable: p_FDR <chr>

```

st_results %>%
  flextable() %>%
  theme_booktabs() %>%
  autofit() %>%
  align(
    j = c("Estimate", "CI", "P_value"),
    align = "center",
    part = "all"
  ) %>%
  save_as_docx(
    path = "output/stent_adj_wa_norec.docx"
  )

```

Block 2ai - without VUR recurrence + with interaction (secondary effect-modification analysis)

Block 2: exposure-effect analysis (stent) on postoperative outcomes with interaction = Multivariable

```

outcomes <- c(
  "LUTD",
  "OBST",
  "URCOMPINJ",
  "SURG",
  "ILEUS",
  "GU30",
  "GU90",
  "PROC365"
)

# convert outcomes to numeric
Out[outcomes] <- lapply(
  Out[outcomes],
  function(x) as.numeric(as.character(x))
)

# Predictors to include in the outcome models
predictors <- "TRT2 * STENT"

# Apply GEE model after overlap weighting for all binary outcomes
gee_results <- lapply(outcomes, function(y) {

```

```

formula <- as.formula(
  paste(y, "~",
        paste(predictors, collapse = " + "))
)
model <- geeglm(
  formula,
  data = Out,
  id = CENTER,
  weights = weights,
  family = binomial(link = "logit"),
  corstr = "exchangeable"
)
tidy(model, exponentiate = TRUE, conf.int = TRUE) # estimate = OR because exponentiate = TRUE
})

```

```

Warning in eval(family$initialize): non-integer #successes in a binomial glm!
Warning in eval(family$initialize): non-integer #successes in a binomial glm!
Warning in eval(family$initialize): non-integer #successes in a binomial glm!
Warning in eval(family$initialize): non-integer #successes in a binomial glm!
Warning in eval(family$initialize): non-integer #successes in a binomial glm!
Warning in eval(family$initialize): non-integer #successes in a binomial glm!
Warning in eval(family$initialize): non-integer #successes in a binomial glm!
Warning in eval(family$initialize): non-integer #successes in a binomial glm!

```

```

names(gee_results) <- outcomes

# Combine results into one table
gee_table <- do.call(
  rbind,
  lapply(names(gee_results), function(x){
    tmp <- gee_results[[x]]
    tmp %>%
      mutate(outcome = x)
  })
)

# Create final manuscript table
gee_table_final <- gee_table %>%
  mutate(
    Effect_measure = "OR",
    Estimate = sprintf("%.2f", estimate),
    CI = paste0(
      sprintf("%.2f", conf.low),
      "-",
      sprintf("%.2f", conf.high)
    ),
    P_value = p.value
  ) %>%
  dplyr::select(
    Outcome = outcome,
    Predictor = term,
    Effect_measure,

```

```

      Estimate,
      CI,
      P_value
    )

# weighted Cox model for POSTUTI

cox_uti <- svycoxph(
  Surv(POSTUTIFUP, POSTUTI == 1) ~ TRT2*STENT + strata(CENTER),
  design = design_ato
)

cox_summary <- summary(cox_uti)

Independent Sampling design (with replacement)
svydesign(ids = ~1, data = Out, weights = ~weights)

cox_row <- rownames(cox_summary$coefficients)

POSTUTI_table <- data.frame(
  Outcome = "POSTUTI",
  Predictor = rownames(cox_summary$coefficients),
  Effect_measure = "HR",
  Estimate = sprintf(
    "%.2f",
    cox_summary$coefficients[
      cox_row,
      "exp(coef)"
    ]
  ),
  CI = paste0(
    sprintf("%.2f",
      cox_summary$conf.int[cox_row, "lower .95"]),
    "-",
    sprintf("%.2f",
      cox_summary$conf.int[cox_row, "upper .95"])
  ),
  P_value = cox_summary$coefficients[
    cox_row,
    "Pr(>|z|)"
  ]
)

# Combine final results
final_results <- rbind(
  gee_table_final,
  POSTUTI_table
)

final_results

```

```

# A tibble: 35 x 6
  Outcome Predictor Effect_measure Estimate CI P_value
* <chr> <chr> <chr> <chr> <chr> <dbl>
1 LUTD (Intercept) OR 0.29 0.23-0.37 0
2 LUTD TRT21 OR 1.21 0.93-1.56 0.149
3 LUTD STENT1 OR 0.49 0.22-1.10 0.0833
4 LUTD TRT21:STENT1 OR 0.42 0.17-1.07 0.0690
5 OBST (Intercept) OR 0.01 0.00-0.36 0.0107
6 OBST TRT21 OR 6.75 0.37-124.14 0.199
7 OBST STENT1 OR 2.42 0.37-15.91 0.358
8 OBST TRT21:STENT1 OR 0.33 0.02-5.31 0.434
9 URCOMPINJ (Intercept) OR 0.02 0.02-0.02 0
10 URCOMPINJ TRT21 OR 0.00 0.00-0.00 0
# i 25 more rows

final_results <- final_results %>%
  mutate(
    across(
      c(P_value),
      ~case_when(
        .x < 0.001 ~ "<0.001",
        TRUE ~ sprintf("%.3f", .x)
      )
    )
  )

final_results %>%
  flextable() %>%
  theme_booktabs() %>%
  autofit() %>%
  align(
    j = c("Estimate", "CI", "P_value"),
    align = "center",
    part = "all"
  ) %>%
  save_as_docx(
    path = "output/treatment_adj_all_wa_norec_i.docx"
  )

# Multiple correction

# Extract TRT2 rows only

gee_trt2 <- gee_table_final %>%
  filter(grepl("^TRT2", Predictor))

POSTUTI_trt2 <- POSTUTI_table %>%
  filter(grepl("^TRT2", Predictor))

# Combine only treatment effects
treatment_results <- rbind(
  gee_trt2,

```

```

    POSTUTI_trt2
  )

# Apply Correction

treatment_results$p_bonferroni <- p.adjust(
  as.numeric(treatment_results$P_value),
  method = "bonferroni"
) # Bonferroni correction

treatment_results$p_holm <- p.adjust(
  as.numeric(treatment_results$P_value),
  method = "holm"
) # Holm-Bonferroni

treatment_results$p_FDR <- p.adjust(
  as.numeric(treatment_results$P_value),
  method = "BH"
) # Benjamini-Hochberg FDR

treatment_results <- treatment_results %>%
  mutate(
    across(
      c(P_value, p_bonferroni, p_holm, p_FDR),
      ~case_when(
        .x < 0.001 ~ "<0.001",
        TRUE ~ sprintf("%.3f", .x)
      )
    )
  )

treatment_results

# A tibble: 18 x 9
  Outcome Predictor Effect_measure Estimate CI P_value p_bonferroni p_holm
  <chr> <chr> <chr> <chr> <chr> <chr> <chr> <chr>
1 LUTD TRT21 OR 1.21 0.93~ 0.149 1.000 0.745
2 LUTD TRT21:ST~ OR 0.42 0.17~ 0.069 1.000 0.414
3 OBST TRT21 OR 6.75 0.37~ 0.199 1.000 0.763
4 OBST TRT21:ST~ OR 0.33 0.02~ 0.434 1.000 0.868
5 URCOMPINJ TRT21 OR 0.00 0.00~ <0.001 <0.001 <0.001
6 URCOMPINJ TRT21:ST~ OR Inf Inf~ <0.001 <0.001 <0.001
7 SURG TRT21 OR 0.81 0.69~ 0.013 0.226 0.113
8 SURG TRT21:ST~ OR 0.06 0.02~ <0.001 <0.001 <0.001
9 ILEUS TRT21 OR 0.20 0.17~ <0.001 <0.001 <0.001
10 ILEUS TRT21:ST~ OR 0.00 0.00~ <0.001 <0.001 <0.001
11 GU30 TRT21 OR 0.04 0.01~ <0.001 <0.001 <0.001
12 GU30 TRT21:ST~ OR 0.00 0.00~ <0.001 <0.001 <0.001
13 GU90 TRT21 OR 0.17 0.14~ <0.001 <0.001 <0.001
14 GU90 TRT21:ST~ OR 1.55 1.02~ 0.040 0.725 0.322
15 PROC365 TRT21 OR 0.47 0.36~ <0.001 <0.001 <0.001
16 PROC365 TRT21:ST~ OR 4.70 1.03~ 0.046 0.826 0.322

```

17	POSTUTI	TRT21	HR	0.70	0.42~	0.191	1.000	0.763
18	POSTUTI	TRT21:ST~	HR	0.65	0.10~	0.656	1.000	0.868

i 1 more variable: p_FDR <chr>

```

treatment_results %>%
  flextable() %>%
  theme_booktabs() %>%
  autofit() %>%
  align(
    j = c("Estimate", "CI", "P_value"),
    align = "center",
    part = "all"
  ) %>%
  save_as_docx(
    path = "output/treatment_adj_wa_norec_i.docx"
  )

# ADDITIONAL MULTIPLE COMPARISONS

# Extract STENT rows only

gee_st <- gee_table_final %>%
  filter(grepl("^STENT", Predictor))

POSTUTI_st <- POSTUTI_table %>%
  filter(grepl("^STENT", Predictor))

# Combine only stent effects
st_results <- rbind(
  gee_st,
  POSTUTI_st
)

# Apply Correction

st_results$p_bonferroni <- p.adjust(
  as.numeric(st_results$P_value),
  method = "bonferroni"
) # Bonferroni correction

st_results$p_holm <- p.adjust(
  as.numeric(st_results$P_value),
  method = "holm"
) # Holm-Bonferroni

st_results$p_FDR <- p.adjust(
  as.numeric(st_results$P_value),
  method = "BH"
) # Benjamini-Hochberg FDR

st_results <- st_results %>%
  mutate(

```

```

    across(
      c(P_value, p_bonferroni, p_holm, p_FDR),
      ~case_when(
        .x < 0.001 ~ "<0.001",
        TRUE ~ sprintf("%.3f", .x)
      )
    )
  )
)

st_results

# A tibble: 9 x 9
  Outcome Predictor Effect_measure Estimate CI      P_value p_bonferroni p_holm
  <chr>      <chr>      <chr>      <chr> <chr> <chr>      <chr>      <chr>
1 LUTD      STENT1      OR          0.49  0.22-- 0.083    0.749      0.333
2 OBST      STENT1      OR          2.42  0.37-- 0.358    1.000      0.717
3 URCOMPINJ STENT1      OR          3.09  0.76-- 0.115    1.000      0.344
4 SURG      STENT1      OR          0.29  0.13-- 0.003    0.027      0.024
5 ILEUS     STENT1      OR          0.07  0.02-- <0.001   0.002      0.002
6 GU30      STENT1      OR          0.25  0.09-- 0.010    0.086      0.061
7 GU90      STENT1      OR          0.20  0.06-- 0.009    0.078      0.061
8 PROC365   STENT1      OR          0.26  0.09-- 0.018    0.165      0.091
9 POSTUTI   STENT1      HR          1.04  0.44-- 0.936    1.000      0.936
# i 1 more variable: p_FDR <chr>

st_results %>%
  flextable() %>%
  theme_booktabs() %>%
  autofit() %>%
  align(
    j = c("Estimate", "CI", "P_value"),
    align = "center",
    part = "all"
  ) %>%
  save_as_docx(
    path = "output/stent_adj_wa_norec_i.docx"
  )

```

Block 2ai - without VUR recurrence + with interaction (to calculate OR for two levels)
 (secondary effect-modification analysis)

```
## Manual calculation
```

```

outcomes <- c(
  "LUTD",
  "OBST",
  "URCOMPINJ",
  "SURG",
  "ILEUS",
  "GU30",
  "GU90",
  "PROC365"
)

```



```

)

# Convert outcomes to numeric
Out[outcomes] <- lapply(
  Out[outcomes],
  function(x) as.numeric(as.character(x))
)

# Function to extract treatment effects within each stent stratum
get_gee_interaction_results <- function(y) {

  formula <- as.formula(
    paste(y, "~ TRT2 * STENT")
  )

  model <- geeglm(
    formula,
    data = Out,
    id = CENTER,
    weights = weights,
    family = binomial(link = "logit"),
    corstr = "exchangeable"
  )

  # Coefficients
  beta <- coef(model)

  # Robust covariance matrix
  V <- vcov(model)

  # -----
  # LG vs Cohen among NO-STENT patients
  # -----

  b_no_stent <- beta["TRT21"]

  se_no_stent <- sqrt(
    V["TRT21", "TRT21"]
  )

  OR_no_stent <- exp(b_no_stent)

  CI_no_stent <- exp(
    b_no_stent + c(-1, 1) * 1.96 * se_no_stent
  )

  # -----
  # LG vs Cohen among STENTED patients
  # = TRT21 + TRT21:STENT1
  # -----

  b_stent <- beta["TRT21"] +

```

```

    beta["TRT21:STENT1"]

# Variance of the linear combination:
# Var(beta1 + beta3)

var_stent <-
  V["TRT21", "TRT21"] +
  V["TRT21:STENT1", "TRT21:STENT1"] +
  2 * V["TRT21", "TRT21:STENT1"]

se_stent <- sqrt(var_stent)

OR_stent <- exp(b_stent)

CI_stent <- exp(
  b_stent + c(-1, 1) * 1.96 * se_stent
)

# -----
# Interaction test
# -----

b_interaction <- beta["TRT21:STENT1"]

se_interaction <- sqrt(
  V["TRT21:STENT1", "TRT21:STENT1"]
)

z_interaction <- b_interaction / se_interaction

p_interaction <- 2 * pnorm(
  abs(z_interaction),
  lower.tail = FALSE
)

# -----
# Return manuscript-ready results
# -----

data.frame(
  Outcome = y,

  `LG vs Cohen, No Stent` =
    sprintf(
      "%.2f (%.2f-%.2f)",
      OR_no_stent,
      CI_no_stent[1],
      CI_no_stent[2]
    ),

  `LG vs Cohen, Stent` =
    sprintf(
      "%.2f (%.2f-%.2f)",

```

```

      OR_stent,
      CI_stent[1],
      CI_stent[2]
    ),

    P_interaction = p_interaction,

    stringsAsFactors = FALSE
  )
}

# Run for all 8 outcomes
interaction_results <- do.call(
  rbind,
  lapply(outcomes, get_gee_interaction_results)
)

Warning in eval(family$initialize): non-integer #successes in a binomial glm!
Warning in eval(family$initialize): non-integer #successes in a binomial glm!
Warning in eval(family$initialize): non-integer #successes in a binomial glm!

Warning in sqrt(var_stent): NaNs produced

Warning in eval(family$initialize): non-integer #successes in a binomial glm!
Warning in eval(family$initialize): non-integer #successes in a binomial glm!
Warning in eval(family$initialize): non-integer #successes in a binomial glm!
Warning in eval(family$initialize): non-integer #successes in a binomial glm!
Warning in eval(family$initialize): non-integer #successes in a binomial glm!

interaction_results

      Outcome LG.vs.Cohen..No.Stent LG.vs.Cohen..Stent P_interaction
TRT21:STENT1      LUTD      1.21 (0.93-1.56) 0.51 (0.19-1.34) 6.900074e-02
TRT21:STENT11     OBST      6.75 (0.37-124.15) 2.23 (1.11-4.46) 4.341585e-01
TRT21:STENT12 URCOMPINJ      0.00 (0.00-0.00) 0.51 (NaN-NaN) 3.528042e-97
TRT21:STENT13     SURG      0.81 (0.69-0.96) 0.05 (0.02-0.14) 9.737343e-07
TRT21:STENT14     ILEUS      0.20 (0.17-0.22) 0.00 (0.00-0.00) 0.000000e+00
TRT21:STENT15      GU30      0.04 (0.01-0.09) 0.00 (0.00-0.00) 1.199019e-94
TRT21:STENT16      GU90      0.17 (0.14-0.21) 0.26 (0.18-0.38) 4.028779e-02
TRT21:STENT17  PROC365      0.47 (0.36-0.61) 2.21 (0.56-8.69) 4.587269e-02

# POSTUTI

cox_uti <- svycoxph(
  Surv(POSTUTIFUP, POSTUTI == 1) ~ TRT2 * STENT +
    strata(CENTER),
  design = design_ato
)

# Coefficients
beta_cox <- coef(cox_uti)

```

```

# Covariance matrix
V_cox <- vcov(cox_uti)

# -----
# LG vs Cohen among NO-STENT patients
# -----

b_no_stent_cox <- beta_cox["TRT21"]

se_no_stent_cox <- sqrt(
  V_cox["TRT21", "TRT21"]
)

HR_no_stent <- exp(b_no_stent_cox)

CI_no_stent_cox <- exp(
  b_no_stent_cox +
  c(-1, 1) * 1.96 * se_no_stent_cox
)

# -----
# LG vs Cohen among STENTED patients
# = TRT21 + TRT21:STENT1
# -----

b_stent_cox <- beta_cox["TRT21"] +
  beta_cox["TRT21:STENT1"]

var_stent_cox <-
  V_cox["TRT21", "TRT21"] +
  V_cox["TRT21:STENT1", "TRT21:STENT1"] +
  2 * V_cox["TRT21", "TRT21:STENT1"]

se_stent_cox <- sqrt(var_stent_cox)

HR_stent <- exp(b_stent_cox)

CI_stent_cox <- exp(
  b_stent_cox +
  c(-1, 1) * 1.96 * se_stent_cox
)

# -----
# Interaction p-value
# -----

b_interaction_cox <- beta_cox["TRT21:STENT1"]

se_interaction_cox <- sqrt(
  V_cox["TRT21:STENT1", "TRT21:STENT1"]
)

```

```

z_interaction_cox <-
  b_interaction_cox / se_interaction_cox

p_interaction_cox <- 2 * pnorm(
  abs(z_interaction_cox),
  lower.tail = FALSE
)

# Create POSTUTI result
POSTUTI_interaction <- data.frame(
  Outcome = "POSTUTI",

  `LG vs Cohen, No Stent` =
    sprintf(
      "%.2f (%.2f-%.2f)",
      HR_no_stent,
      CI_no_stent_cox[1],
      CI_no_stent_cox[2]
    ),

  `LG vs Cohen, Stent` =
    sprintf(
      "%.2f (%.2f-%.2f)",
      HR_stent,
      CI_stent_cox[1],
      CI_stent_cox[2]
    ),

  P_interaction = p_interaction_cox
)

POSTUTI_interaction

      Outcome LG.vs.Cohen..No.Stent LG.vs.Cohen..Stent P_interaction
TRT21:STENT1 POSTUTI      0.70 (0.42-1.19)    0.46 (0.08-2.76)      0.656211

```

4 PAIN - Non-epidural Cohort

4.1 Filter & weight the new population

```

# Filter dataset to only those who did not receive EPIDURAL
Outp <- Out %>%
  filter(EPIDURAL == "0") %>%
  mutate(
    PAIN = factor(
      PAIN,
      levels = c("0", "1", "2", "3"),
      labels = c(
        "No pain",
        "Mild pain (score 1-3)",
        "Moderate pain (score 4-6)",

```

```

        "Severe pain (score 7-10)"
    ),
    ordered = TRUE
)
)

# Do weighting: using ATO (Recommended)

set.seed(123)

w.outp <- weightit(
  TRT2 ~ AGS + YEARS + GENDER + BMI + PREOPRNPLN + HISTOINJ +
    SURGIND2 + PREOPVURGRADE + VURSIDE + UDR7M + VURPHASE +
    PREOPEPISODE1 + PREOPTYPEU1 + PREOPRNSCR + SURGTECH2,
  data = Outp,
  method = "ps",
  estimand = "ATO",
  stabilize = FALSE # used in ATE
)
Outp$weights <- w.outp$weights # Add weights to the dataset
write.csv(Outp, "data/weighted_nonepidural_data.csv", row.names = FALSE)

summary(w.outp$weights)

      Min.      1st Qu.      Median      Mean      3rd Qu.      Max.
0.0007272 0.0555583 0.2395282 0.2875446 0.4552000 0.9418707

# Table for balance

balp <- cobalt::bal.tab(w.outp, un = TRUE, ESS = TRUE, m.threshold = 0.1)
balp

```

Balance Measures

	Type	Diff.Un	Diff.Adj	M.Threshold
prop.score	Distance	1.8310	-0.0259	Balanced, <0.1
AGS	Contin.	0.3389	0.0000	Balanced, <0.1
YEARS	Contin.	-0.5189	-0.0000	Balanced, <0.1
GENDER	Binary	-0.0397	-0.0000	Balanced, <0.1
BMI	Contin.	-0.6798	-0.0000	Balanced, <0.1
PREOPRNPLN	Binary	-0.2680	-0.0000	Balanced, <0.1
HISTOINJ	Binary	0.0111	-0.0000	Balanced, <0.1
SURGIND2_1	Binary	0.0673	0.0000	Balanced, <0.1
SURGIND2_2	Binary	0.0864	-0.0000	Balanced, <0.1
SURGIND2_3	Binary	-0.0123	-0.0000	Balanced, <0.1
SURGIND2_4	Binary	-0.0520	-0.0000	Balanced, <0.1
SURGIND2_5	Binary	-0.0894	-0.0000	Balanced, <0.1
PREOPVURGRADE_3	Binary	-0.0179	-0.0000	Balanced, <0.1
PREOPVURGRADE_4	Binary	0.0999	-0.0000	Balanced, <0.1
PREOPVURGRADE_5	Binary	-0.0820	0.0000	Balanced, <0.1
VURSIDE	Binary	0.0233	-0.0000	Balanced, <0.1
UDR7M	Binary	-0.1485	0.0000	Balanced, <0.1
VURPHASE_1	Binary	-0.0702	-0.0000	Balanced, <0.1

VURPHASE_2	Binary	0.3261	0.0000	Balanced, <0.1
VURPHASE_3	Binary	-0.2559	-0.0000	Balanced, <0.1
PREOPEPISODE1_1	Binary	-0.0509	-0.0000	Balanced, <0.1
PREOPEPISODE1_2	Binary	-0.1592	-0.0000	Balanced, <0.1
PREOPEPISODE1_3	Binary	0.2101	0.0000	Balanced, <0.1
PREOPTYPEU1	Binary	0.0120	0.0000	Balanced, <0.1
PREOPRNSCR	Binary	-0.2746	-0.0000	Balanced, <0.1
SURGTECH2_1	Binary	0.2538	0.0000	Balanced, <0.1
SURGTECH2_2	Binary	-0.0625	-0.0000	Balanced, <0.1
SURGTECH2_3	Binary	-0.1913	-0.0000	Balanced, <0.1

Balance tally for mean differences

	count
Balanced, <0.1	28
Not Balanced, >0.1	0

Variable with the greatest mean difference

Variable	Diff.Adj	M.Threshold
YEARS	-0	Balanced, <0.1

Effective sample sizes

	Control	Treated
Unadjusted	178.	205.
Adjusted	113.59	105.65

```
balp_df <- as.data.frame(balp$Balance)
smd_table <- balp_df %>%
  mutate(
    Variable = rownames(balp_df),
    SMD_Before = round(abs(Diff.Un), 3),
    SMD_After = round(abs(Diff.Adj), 3)
  ) %>%
  dplyr::select(Variable, SMD_Before, SMD_After)
```

```
smd_table %>%
  flextable() %>%
  save_as_docx(path = "output/SMD_pw.docx")
```

Figure for balance - Love plot

```
plp <- love.plot(
  w.outp,
  threshold = 0.1,
  abs = TRUE,
  stats = "mean.diffs",
  var.order = "unadjusted",
  drop.distance = TRUE,
  sample.names = c("Before Overlap Weighting", "After Overlap Weighting"),
  var.names = c(
    AGS = "Age at surgery",
    YEARS = "Year of surgery",
    BMI = "Body mass index",
    GENDER = "Female gender",
    PREOPRNPLN = "Preop pyelonephritis",
```

```

HISTOINJ = "History of injection",
VURSIDE = "Right-sided VUR",
UDR7M = "Ureteral dilation 7mm",
PREOPTYPEU1 = "Preop febrile UTIs",
PREOPRNSCR = "Preop renal scarring",
VURPHASE_1 = "Reflux: voiding phase",
VURPHASE_2 = "Reflux: filling phase",
VURPHASE_3 = "Reflux: filling + voiding",
SURGTECH2_1 = "Surgeon experience 100 operations",
SURGTECH2_2 = "Surgeon experience 50-100 operations",
SURGTECH2_3 = "Surgeon experience <50 operations",
PREOPEPISODE1_1 = "Preop UTI episodes: 1",
PREOPEPISODE1_2 = "Preop UTI episodes: 2",
PREOPEPISODE1_3 = "Preop UTI episodes: 3",
SURGIND2_1 = "Indication: breakthrough UTI",
SURGIND2_2 = "Indication: high-grade reflux (III-V)",
SURGIND2_3 = "Indication: poor compliance to medical treatment",
SURGIND2_4 = "Indication: ineffective/poorly tolerated CAP",
SURGIND2_5 = "Indication: other",
PREOPVURGRADE_3 = "Preop VUR grade: III",
PREOPVURGRADE_4 = "Preop VUR grade: IV",
PREOPVURGRADE_5 = "Preop VUR grade: V"
)
)

```

Warning: Standardized mean differences and raw mean differences are present in the same plot. Use the `stars` argument to distinguish between them and appropriately label the x-axis. See `love.plot()` for details.

```

plp <- plp +
  labs(
    x = "Absolute Standardized Differences",
    subtitle = "Overlap population of children who did not receive epidural analgesia"
  )
  theme(
    plot.margin = margin(10, 10, 10, 10),
    axis.text = element_text(size = 16),
    axis.title = element_text(size = 16),
    plot.title = element_text(size = 16),
    legend.text = element_text(size = 16),
    legend.title = element_text(size = 16),
    axis.ticks = element_line(size = 0.5, color = "black"),
    axis.line = element_line(size = 0.5, color = "black"),
    panel.background = element_blank(),
    panel.grid.major = element_blank(),
    panel.grid.minor = element_blank(),
    panel.border = element_blank(),
    axis.ticks.length = unit(0.3, "cm"),
    panel.spacing = unit(2, "lines")
  )

```

```

<theme> List of 14
 $ axis.title      : <ggplot2::element_text>

```



```

..@ family      : NULL
..@ face        : NULL
..@ italic      : chr NA
..@ fontweight  : num NA
..@ fontwidth   : num NA
..@ colour      : NULL
..@ size        : num 16
..@ hjust       : NULL
..@ vjust       : NULL
..@ angle       : NULL
..@ lineheight  : NULL
..@ margin      : NULL
..@ debug       : NULL
..@ inherit.blank: logi FALSE
$ axis.text     : <ggplot2::element_text>
..@ family      : NULL
..@ face        : NULL
..@ italic      : chr NA
..@ fontweight  : num NA
..@ fontwidth   : num NA
..@ colour      : NULL
..@ size        : num 16
..@ hjust       : NULL
..@ vjust       : NULL
..@ angle       : NULL
..@ lineheight  : NULL
..@ margin      : NULL
..@ debug       : NULL
..@ inherit.blank: logi FALSE
$ axis.ticks    : <ggplot2::element_line>
..@ colour      : chr "black"
..@ linewidth   : num 0.5
..@ linetype    : NULL
..@ lineend     : NULL
..@ linejoin    : NULL
..@ arrow       : logi FALSE
..@ arrow.fill  : chr "black"
..@ inherit.blank: logi FALSE
$ axis.ticks.length: 'simpleUnit' num 0.3cm
..- attr(*, "unit")= int 1
$ axis.line     : <ggplot2::element_line>
..@ colour      : chr "black"
..@ linewidth   : num 0.5
..@ linetype    : NULL
..@ lineend     : NULL
..@ linejoin    : NULL
..@ arrow       : logi FALSE
..@ arrow.fill  : chr "black"
..@ inherit.blank: logi FALSE
$ legend.text   : <ggplot2::element_text>
..@ family      : NULL
..@ face        : NULL
..@ italic      : chr NA
..@ fontweight  : num NA

```

```

..@ fontwidth      : num NA
..@ colour         : NULL
..@ size           : num 16
..@ hjust          : NULL
..@ vjust          : NULL
..@ angle           : NULL
..@ lineheight     : NULL
..@ margin         : NULL
..@ debug          : NULL
..@ inherit.blank: logi FALSE
$ legend.title     : <ggplot2::element_text>
..@ family         : NULL
..@ face           : NULL
..@ italic         : chr NA
..@ fontweight     : num NA
..@ fontwidth      : num NA
..@ colour         : NULL
..@ size           : num 16
..@ hjust          : NULL
..@ vjust          : NULL
..@ angle           : NULL
..@ lineheight     : NULL
..@ margin         : NULL
..@ debug          : NULL
..@ inherit.blank: logi FALSE
$ panel.background : <ggplot2::element_blank>
$ panel.border      : <ggplot2::element_blank>
$ panel.spacing     : 'simpleUnit' num 2lines
..- attr(*, "unit")= int 3
$ panel.grid.major  : <ggplot2::element_blank>
$ panel.grid.minor  : <ggplot2::element_blank>
$ plot.title        : <ggplot2::element_text>
..@ family         : NULL
..@ face           : NULL
..@ italic         : chr NA
..@ fontweight     : num NA
..@ fontwidth      : num NA
..@ colour         : NULL
..@ size           : num 16
..@ hjust          : NULL
..@ vjust          : NULL
..@ angle           : NULL
..@ lineheight     : NULL
..@ margin         : NULL
..@ debug          : NULL
..@ inherit.blank: logi FALSE
$ plot.margin       : <ggplot2::margin> num [1:4] 10 10 10 10
@ complete: logi FALSE
@ validate: logi TRUE

ggsave ("figures/love_plot_pw.png", plot = plp, width = 12, height = 10, dpi = 300)

# Compute mean SMD
smdp_df <- balp$Balance

```

```

mean_smdp_before <- mean(abs(smdp_df$Diff.Un), na.rm = TRUE) # calculate mean SMD before matching
mean_smdp_after <- mean(abs(smdp_df$Diff.Adj), na.rm = TRUE) # calculate mean SMD before matching
mean_smdp <- data.frame(
  Mean_SMDp_Before = mean_smdp_before,
  Mean_SMDp_After = mean_smdp_after
)

mean_smdp

  Mean_SMDp_Before Mean_SMDp_After
1          0.2226086      0.0009262931

# how much effective sample size remains (overall ESS: calculated manually)?
summary(w.outp$weights)

      Min.    1st Qu.    Median      Mean   3rd Qu.      Max.
0.0007272 0.0555583 0.2395282 0.2875446 0.4552000 0.9418707

bal.tab(w.outp)$Effective.Sample.Size

NULL

sum(w.outp$weights)^2 / sum(w.outp$weights^2)

[1] 218.9503

# propensity score distribution
summary(w.outp$ps)

      Min.    1st Qu.    Median      Mean   3rd Qu.      Max.
0.006964 0.240110 0.543399 0.535248 0.819747 0.999273

pl2p <-plot(summary(w.outp, type = "hist"))
ggsave("figures/propensity_pw.png", plot = pl2p, width = 12, height = 10, dpi = 300)

# Distribution of overlap weights: histogram and density

## Histogram
summary(w.outp$weights)

      Min.    1st Qu.    Median      Mean   3rd Qu.      Max.
0.0007272 0.0555583 0.2395282 0.2875446 0.4552000 0.9418707

png("figures/weights_histo_pw.png",
    width = 6,
    height = 4,
    units = "in",
    res = 300)

```

```

hist(
  w.outp$weights,
  breaks = 30,
  col = "lightblue",
  border = "black",
  main = "Distribution of Overlap Weights",
  xlab = "Overlap Weight"
)
dev.off()

pdf
2

# Distributional balance for propensity score (purpose: to evaluate overlap/positivity):

# Before-After weighting
p_ps_pain <- bal.plot(
  w.outp,
  var.name = "prop.score",
  type = "density",
  which = "both"
)

p_ps_pain <- p_ps_pain +
  scale_color_viridis_d(
    option = "mako",
    begin = 0.3,
    end = 0.8,
    labels = c("Cohen group", "Lich-Gregoir group")
  ) +
  scale_fill_viridis_d(
    option = "mako",
    begin = 0.3,
    end = 0.8,
    labels = c("Cohen group", "Lich-Gregoir group")
  )

Scale for fill is already present.
Adding another scale for fill, which will replace the existing scale.

ggsave(
  "figures/PS_density_before_after_pATO.png",
  p_ps_pain,
  width = 10,
  height = 7,
  dpi = 300
)

```

4.2 Descriptive analysis

```

# Baseline characteristics in the population for pain
table1p <- Outp %>%
  dplyr::select(YEARS, AGS, GENDER, BMI, HISTOINJ, SURGIND2, PREOPVURGRADE, VURSIDE, UDR7M,
    VURPHASE, PREOPTYPEU1, PREOPEPISODE1, PREOPRNPLN, PREOPRNSCR,
    SURGTECH2, CENTER, TRT2
  ) %>%
  tbl_summary(
    by = TRT2, # for stratification
    statistic = list(
      all_continuous() ~ "{median} ({p25}-{p75})",
      all_categorical() ~ "{n} ({p}%)"
    ),
    digits = all_categorical() ~ c(0, 1), # n = 0 decimals, p = 1 decimal
    missing = "ifany", # shows missing only when present
    missing_text = "Missing"
  ) %>%
  add_overall() %>%
  add_p(pvalue_fun = ~style_pvalue(.x, digits = 3)) %>% # 3 decimals
  bold_labels()

table1p

table1p %>%
  as_flex_table() %>%
  flextable::save_as_docx(path = "output/baseline_p.docx")

# Create survey design object using overlap weights
design_pato <- svydesign(
  ids = ~1,
  data = Outp,
  weights = ~weights
)

# Weighted baseline characteristics
table1pw <- design_pato %>%
  tbl_svysummary(
    by = TRT2, # for stratification
    include = c(
      YEARS, AGS, GENDER, BMI, HISTOINJ, SURGIND2, PREOPVURGRADE, VURSIDE, UDR7M,
      VURPHASE, PREOPTYPEU1, PREOPEPISODE1, PREOPRNPLN, PREOPRNSCR,
      SURGTECH2, CENTER # generally we should not include TRT2 in include list
    ),
    statistic = list(
      all_continuous() ~ "{median} ({p25}-{p75})",
      all_categorical() ~ "{n} ({p}%)"
    ),
    digits = all_categorical() ~ c(0, 1),
    missing = "ifany", # shows missing only when present
    missing_text = "Missing"
  ) %>%
  add_overall() %>%
  bold_labels()

```

Characteristic	Overall N = 383 ¹	0 N = 178 ¹	1 N = 205
YEARS	2,017.00 (2,014.00-2,019.00)	2,018.00 (2,014.00-2,020.00)	2,016.00 (2,013.00-2,019.00)
AGS	7.00 (6.00-9.00)	8.00 (4.00-9.00)	7.00 (6.00-9.00)
GENDER			
0	219 (57.2%)	98 (55.1%)	121 (59.0%)
1	164 (42.8%)	80 (44.9%)	84 (41.0%)
BMI	14.1 (13.5-17.7)	14.9 (13.4-21.0)	13.9 (13.6-14.4)
HISTOINJ			
0	342 (89.3%)	160 (89.9%)	182 (88.8%)
1	41 (10.7%)	18 (10.1%)	23 (11.2%)
SURGIND2			
1	201 (52.5%)	87 (48.9%)	114 (55.6%)
2	121 (31.6%)	48 (27.0%)	73 (35.6%)
3	19 (5.0%)	10 (5.6%)	9 (4.4%)
4	13 (3.4%)	11 (6.2%)	2 (1.0%)
5	29 (7.6%)	22 (12.4%)	7 (3.4%)
PREOPVURGRADE			
3	20 (5.2%)	11 (6.2%)	9 (4.4%)
4	113 (29.5%)	43 (24.2%)	70 (34.1%)
5	250 (65.3%)	124 (69.7%)	126 (61.5%)
VURSIDE			
0	148 (38.6%)	71 (39.9%)	77 (37.6%)
1	235 (61.4%)	107 (60.1%)	128 (62.4%)
UDR7M			
0	95 (24.8%)	30 (16.9%)	65 (31.7%)
1	288 (75.2%)	148 (83.1%)	140 (68.3%)
VURPHASE			
1	218 (56.9%)	108 (60.7%)	110 (53.7%)
2	69 (18.0%)	1 (0.6%)	68 (33.2%)
3	96 (25.1%)	69 (38.8%)	27 (13.2%)
PREOPTYPEU1			
0	4 (1.0%)	3 (1.7%)	1 (0.5%)
1	379 (99.0%)	175 (98.3%)	204 (99.5%)
PREOPEPISODE1			
1	24 (6.3%)	16 (9.0%)	8 (3.9%)
2	90 (23.5%)	57 (32.0%)	33 (16.1%)
3	269 (70.2%)	105 (59.0%)	164 (80.0%)
PREOPRNPLN			
0	311 (81.2%)	119 (66.9%)	192 (93.7%)
1	72 (18.8%)	59 (33.1%)	13 (6.3%)
PREOPRNSCR			
0	265 (69.2%)	97 (54.5%)	168 (82.0%)
1	118 (30.8%)	81 (45.5%)	37 (18.0%)
SURGTECH2			
1	80 (20.9%)	13 (7.3%)	67 (32.7%)
2	254 (66.3%)	124 (69.7%)	130 (63.4%)
3	49 (12.8%)	41 (23.0%)	8 (3.9%)
CENTER			
1	282 (73.6%) ⁶²	99 (55.6%)	183 (89.3%)
2	0 (0.0%)	0 (0.0%)	0 (0.0%)
3	37 (9.7%)	31 (17.4%)	6 (2.9%)

```
table1pw
```

```
table1pw %>%
  as_flex_table() %>%
  flextable::save_as_docx(
    path = "output/baseline_pw.docx"
  )

# Weighted procedural factors
table2p<- design_pato %>%
  tbl_svsummary(
    by = TRT2, # for stratification
    include = c (
      OPRTIME, STENT, CATH, CATHPERIOD, CATHPERIOD_G, HOSPTIME
    ),
    statistic = list(
      all_continuous() ~ "{median} ({p25}-{p75})", # or ~ "{mean} ± {sd}",
      all_categorical() ~ "{n} ({p}%)"
    ),
    digits = all_categorical() ~ c(0, 1), # n = 0 decimals, p = 1 decimal
    missing = "ifany",
    missing_text = "Missing"
  ) %>%
  add_overall() %>%
  add_p(pvalue_fun = ~style_pvalue(.x, digits = 3)) %>% # 3 decimals
  bold_labels()
```

The following warnings were returned during `add\_p()`:

! For variable `CATH` (`TRT2`) and "statistic" and "p.value" statistics:
Chi-squared approximation may be incorrect

```
table2p
```

```
table2p %>%
  as_flex_table() %>%
  flextable::save_as_docx(path = "output/procedural_pw.docx")
```

```
# Weighted pain
```

```
# Before weighting
```

```
table4 <- Outp %>%
  dplyr::select(PAIN, NARC, NONARC, Treatment) %>%
  tbl_summary(
    by = Treatment,
    statistic = list(
      all_categorical() ~ "{n} ({p}%)"
    ),
    label = list(
```

Characteristic	Overall N = 110 ^I	0 N = 55 ^I	1 N = 55
YEARS	2,017.00 (2,014.00-2,019.00)	2,017.00 (2,014.00-2,019.00)	2,017.00 (2,015.00-2,019.00)
AGS	8.00 (6.00-9.00)	8.00 (6.00-9.00)	7.00 (6.00-9.00)
GENDER			
0	60 (54.2%)	30 (54.2%)	30 (54.2%)
1	50 (45.8%)	25 (45.8%)	25 (45.8%)
BMI	14.0 (13.4-16.3)	13.9 (13.2-17.5)	14.1 (13.6-14.6)
HISTOINJ			
0	96 (86.8%)	48 (86.8%)	48 (86.8%)
1	15 (13.2%)	7 (13.2%)	7 (13.2%)
SURGIND2			
1	51 (46.5%)	26 (46.5%)	26 (46.5%)
2	44 (39.8%)	22 (39.8%)	22 (39.8%)
3	6 (5.6%)	3 (5.6%)	3 (5.6%)
4	2 (1.9%)	1 (1.9%)	1 (1.9%)
5	7 (6.2%)	3 (6.2%)	3 (6.2%)
PREOPVURGRADE			
3	7 (6.5%)	4 (6.5%)	4 (6.5%)
4	30 (27.6%)	15 (27.6%)	15 (27.6%)
5	72 (65.8%)	36 (65.8%)	36 (65.8%)
VURSIDE			
0	39 (35.7%)	20 (35.7%)	20 (35.7%)
1	71 (64.3%)	35 (64.3%)	35 (64.3%)
UDR7M			
0	27 (24.8%)	14 (24.8%)	14 (24.8%)
1	83 (75.2%)	41 (75.2%)	41 (75.2%)
VURPHASE			
1	84 (76.4%)	42 (76.4%)	42 (76.4%)
2	2 (1.6%)	1 (1.6%)	1 (1.6%)
3	24 (22.0%)	12 (22.0%)	12 (22.0%)
PREOPTYPEU1			
0	1 (0.8%)	0 (0.8%)	0 (0.8%)
1	109 (99.2%)	55 (99.2%)	55 (99.2%)
PREOPEPISODE1			
1	7 (6.1%)	3 (6.1%)	3 (6.1%)
2	33 (30.3%)	17 (30.3%)	17 (30.3%)
3	70 (63.6%)	35 (63.6%)	35 (63.6%)
PREOPRNPLN			
0	93 (84.6%)	47 (84.6%)	47 (84.6%)
1	17 (15.4%)	8 (15.4%)	8 (15.4%)
PREOPRNSCR			
0	80 (73.0%)	40 (73.0%)	40 (73.0%)
1	30 (27.0%)	15 (27.0%)	15 (27.0%)
SURGTECH2			
1	6 (5.4%)	3 (5.4%)	3 (5.4%)
2	92 (83.3%)	46 (83.3%)	46 (83.3%)
3	12 (11.3%)	6 (11.3%)	6 (11.3%)
CENTER			
1	84 (76.3%) ⁶⁴	40 (72.9%)	44 (79.6%)
2	0 (0.0%)	0 (0.0%)	0 (0.0%)
3	9 (8.0%)	5 (9.5%)	4 (6.4%)

Characteristic	Overall N = 110 ¹	0 N = 55 ¹	1 N = 55 ¹	p-value ²
OPRTIME	85 (70-135)	135 (90-160)	70 (67-80)	<0.001
STENT				0.062
0	74 (67.1%)	34 (61.0%)	40 (73.1%)	
1	36 (32.9%)	21 (39.0%)	15 (26.9%)	
CATH				0.516
0	2 (2.3%)	1 (1.6%)	2 (2.9%)	
1	108 (97.7%)	54 (98.4%)	53 (97.1%)	
CATHPERIOD	4 (3-5)	4 (4-5)	3 (1-5)	<0.001
CATHPERIOD_G				<0.001
Early removal 3 days	36 (33.1%)	4 (6.6%)	33 (59.5%)	
Standard removal 4-7 days	56 (50.5%)	42 (75.8%)	14 (25.1%)	
Delayed removal >7 days	18 (16.5%)	10 (17.6%)	8 (15.4%)	
HOSPTIME	5.00 (3.00-6.00)	5.00 (5.00-6.00)	3.00 (2.00-5.00)	<0.001

¹Median (Q1-Q3); n (%)

²Design-based KruskalWallis test; Pearson's X²: Rao & Scott adjustment

```

    PAIN ~ "Pain score",
    NARC ~ "Narcotic use",
    NONARC ~ "Non-narcotic use"
  ),
  digits = all_categorical() ~ c(0, 1),
  missing = "ifany",
  missing_text = "Missing"
) %>%
add_overall() %>%
add_p(pvalue_fun = ~style_pvalue(.x, digits = 3)) %>%
bold_labels()

# After ATO weighting

design_atop <- svydesign(
  ids = ~1,
  data = Outp,
  weights = ~weights
)

# Weighted pain characteristics
table4w <- design_atop %>%
tbl_svysummary(
  by = Treatment,      # for stratification
  include = c(
    PAIN, NARC, NONARC
  ),
  statistic = list(
    all_categorical() ~ "{n} ({p}%)",
  ),

```

```

label = list(
  PAIN ~ "Pain score",
  NARC ~ "Narcotic use",
  NONARC ~ "Non-narcotic use"
),
digits = all_categorical() ~ c(0, 1),
missing = "ifany", # shows missing only when present
missing_text = "Missing"
) %>%
add_overall() %>%
add_p(pvalue_fun = ~style_pvalue(.x, digits = 3)) %>%
bold_labels()

```

The following warnings were returned during `add\_p()`:

! For variable `NONARC` (`Treatment`) and "statistic" and "p.value" statistics:

Chi-squared approximation may be incorrect

! For variable `PAIN` (`Treatment`) and "statistic" and "p.value" statistics:

Chi-squared approximation may be incorrect

```
# MERGE TABLES
```

```

table_combined <- tbl_merge(
  tbls = list(table4, table4w),
  tab_spanner = c("**Before overlap weighting**", "**After overlap weighting**")
) %>%
  modify_caption(
    "Table 4. Comparison of pain intensity and analgesic use between treatment groups before and after"
  ) %>%
  modify_source_note(
    "Analysis restricted to patients not receiving epidural analgesia."
  )

```

```
table_combined
```

```

table_combined %>%
  as_flex_table() %>%
  flextable::save_as_docx(path = "output/pain_w.docx")

```

4.3 Plot: BEFORE-AFTER weighting

```

# label datasets
before <- Outp %>%
  mutate(
    plot_weight = 1,
    Status = "Before overlap weighting (N=383), p<0.001"
  )

after <- Outp %>%
  mutate(
    plot_weight = weights,
    Status = "After overlap weighting (ESS=219), p<0.001"
  )

```

Characteristic	Before overlap weighting			p-value
	Overall N = 383 ¹	Cohen N = 178 ¹	Lich–Gregoir N = 205 ¹	
Pain score				<0.001
No pain	3 (0.8%)	0 (0.0%)	3 (1.5%)	
Mild pain (score 1-3)	127 (33.2%)	45 (25.3%)	82 (40.0%)	
Moderate pain (score 4-6)	123 (32.1%)	53 (29.8%)	70 (34.1%)	
Severe pain (score 7-10)	130 (33.9%)	80 (44.9%)	50 (24.4%)	
Narcotic use				0.2
0	215 (56.1%)	105 (59.0%)	110 (53.7%)	
1	168 (43.9%)	73 (41.0%)	95 (46.3%)	
Non-narcotic use				0.2
0	2 (0.5%)	2 (1.1%)	0 (0.0%)	
1	381 (99.5%)	176 (98.9%)	205 (100.0%)	

¹n (%)

²Fisher's exact test; Pearson's Chi-squared test

³Pearson's X²: Rao & Scott adjustment

Analysis restricted to patients not receiving epidural analgesia.

```
# Combine
plot_data <- bind_rows(before, after)

# pain labels
plot_data <- plot_data %>%
  mutate(
    PAIN_plot = factor(
      PAIN,
      levels = levels(PAIN),
      labels = c("No pain", "Mild pain", "Moderate pain", "Severe pain")
    )
  )

# To define order of data
plot_data <- plot_data %>%
  mutate(
    Status = factor(
      Status,
      levels = c(
        "Before overlap weighting (N=383), p<0.001",
        "After overlap weighting (ESS=219), p<0.001"
      )
    )
  )

# 100% stacked grouped bar plot

# summary
plot_summary1 <- plot_data %>%
```

```

group_by(Status, Treatment, PAIN) %>%
  summarise(
    weighted_n = sum(plot_weight, na.rm = TRUE),
    .groups = "drop"
  ) %>%
group_by(Status, Treatment) %>%
  mutate(
    Percent = 100 * weighted_n / sum(weighted_n)
  )

s1 <- ggplot(plot_summary1,
             aes(x = Treatment, y = Percent, fill = PAIN)) +
  geom_bar(stat = "identity", position = "fill", width = 0.7) +
  facet_wrap(~Status) +
  labs(
    x = "
    Treatment groups",
    y = "Percentage of patients not receiving epidural analgesia (%)",
    fill = "1-Day postoperative pain score"
  ) +
  scale_fill_manual(values = c(
    "No pain" = "#f5f5f5", # Intead of pure white "#ffffff"
    "Mild pain (score 1-3)" = "#fee8c8",
    "Moderate pain (score 4-6)" = "#fdbb84",
    "Severe pain (score 7-10)" = "#e34a33"
  )) +
  scale_y_continuous(labels = scales::percent_format(scale = 100)) +
  theme_minimal() +
  theme(
    panel.grid = element_blank(),
    axis.title = element_text(size = 12),
    axis.text = element_text(size = 11),
    panel.border = element_rect(color = "black", fill = NA),
    strip.text = element_text(face = "bold"),
    panel.spacing = unit(0, "lines")
  )

ggsave(
  filename = "figures/pain_stacked_w.png",
  plot = s1,
  width = 9,
  height = 6,
  dpi = 300
)

```

4.4 Regression

```

# -----
### Moderate-severe pain vs. Other levels
# -----

Outp <- Outp %>%
  mutate(

```

```

PAIN_ms = factor(
  ifelse(
    PAIN %in% c(
      "Moderate pain (score 4-6)",
      "Severe pain (score 7-10)"
    ),
    1,
    0
  ),
  levels = c(0, 1),
  labels = c(
    "No/Mild pain (score 0-3)",
    "Moderate-Severe pain (score 4-10)"
  )
)

Outp$PAIN_ms <- ifelse(
  Outp$PAIN_ms == "Moderate-Severe pain (score 4-10)",
  1,
  0
)

# -----
### Exploratory analysis: Severe pain vs. Other levels
# -----

Outp <- Outp %>%
  mutate(
    PAIN_s = factor(
      ifelse(
        PAIN == "Severe pain (score 7-10)",
        1,
        0
      ),
      levels = c(0, 1),
      labels = c(
        "No severe pain (score 0-6)",
        "Severe pain (score 7-10)"
      )
    )
  )

# a numeric binary outcome:
Outp$PAIN_s <- ifelse(
  Outp$PAIN_s == "Severe pain (score 7-10)",
  1,
  0
)

#-----
# Block 1

```

```

#-----

outcomes <- c(
  "PAIN_ms",
  "PAIN_s"
)

# convert outcomes to numeric
Outp[outcomes] <- lapply(
  Outp[outcomes],
  function(x) as.numeric(as.character(x))
)

# Apply GEE model after overlap weighting for all binary outcomes
gee_results <- lapply(outcomes, function(y) {
  formula <- as.formula(
    paste(y, "~ TRT2")
  )
  model <- geeglm(
    formula,
    data = Outp,
    id = CENTER,
    weights = weights,
    family = binomial(link = "logit"),
    corstr = "exchangeable"
  )
  tidy(model, exponentiate = TRUE, conf.int = TRUE) # estimate = OR because exponentiate = TRUE
})

Warning in eval(family$initialize): non-integer #successes in a binomial glm!
Warning in eval(family$initialize): non-integer #successes in a binomial glm!

names(gee_results) <- outcomes

# Combine results into one table
gee_table <- do.call(
  rbind,
  lapply(names(gee_results), function(x){
    tmp <- gee_results[[x]]
    tmp <- tmp %>%
      filter(grepl("^TRT2", term)) %>%
      mutate(outcome = x)
    tmp
  })
)

# Create final manuscript table
gee_table_final <- gee_table %>%
  mutate(
    Effect_measure = "OR",
    Estimate = sprintf("%.2f", estimate),
    CI = paste0(

```

```

        sprintf("%.2f", conf.low),
        "-",
        sprintf("%.2f", conf.high)
    ),
    P_value = ifelse(p.value < 0.001, "<0.001", sprintf("%.3f", p.value))
) %>%
dplyr::select(
    Outcome = outcome,
    Effect_measure,
    Estimate,
    CI,
    P_value
)

gee_table_final

# A tibble: 2 x 5
  Outcome Effect_measure Estimate CI      P_value
<chr>    <chr>          <chr> <chr>    <chr>
1 PAIN_ms OR            0.32  0.14-0.73 0.007
2 PAIN_s  OR            0.20  0.15-0.26 <0.001

gee_table_final %>%
  flextable() %>%
  theme_booktabs() %>%
  autofit() %>%
  align(
    j = c("Estimate", "CI", "P_value"),
    align = "center",
    part = "all"
  ) %>%
  save_as_docx(
    path = "output/pain_unadj_w.docx"
  )

#-----
# Block 2
#-----

outcomes <- c(
  "PAIN_ms",
  "PAIN_s"
)

# convert outcomes to numeric
Outp[outcomes] <- lapply(
  Outp[outcomes],
  function(x) as.numeric(as.character(x))
)

# Predictors to include in the outcome models

```

```

predictors <- c("TRT2", "STENT")

# Apply GEE model after overlap weighting for all binary outcomes
gee_results <- lapply(outcomes, function(y) {
  formula <- as.formula(
    paste(y, "~",
          paste(predictors, collapse = " + ")
    )
  )
  model <- geeglm(
    formula,
    data = Outp,
    id = CENTER,
    weights = weights,
    family = binomial(link = "logit"),
    corstr = "exchangeable"
  )
  tidy(model, exponentiate = TRUE, conf.int = TRUE) # estimate = OR because exponentiate = TRUE
})

Warning in eval(family$initialize): non-integer #successes in a binomial glm!
Warning in eval(family$initialize): non-integer #successes in a binomial glm!

names(gee_results) <- outcomes

# Combine results into one table
gee_table <- do.call(
  rbind,
  lapply(names(gee_results), function(x){
    tmp <- gee_results[[x]]
    tmp %>%
      mutate(outcome = x)
  })
)

# Create final manuscript table
gee_table_final <- gee_table %>%
  mutate(
    Effect_measure = "OR",
    Estimate = sprintf("%.2f", estimate),
    CI = paste0(
      sprintf("%.2f", conf.low),
      "-",
      sprintf("%.2f", conf.high)
    ),
    P_value = ifelse(p.value < 0.001, "<0.001", sprintf("%.3f", p.value))
  ) %>%
  dplyr::select(
    Outcome = outcome,
    Predictor = term,
    Effect_measure,
    Estimate,

```



```

      CI,
      P_value
    )

gee_table_final

# A tibble: 6 x 6
  Outcome Predictor Effect_measure Estimate CI P_value
  <chr>   <chr>      <chr>      <chr>   <chr>   <chr>
1 PAIN_ms (Intercept) OR          3.78   1.37-10.37 0.010
2 PAIN_ms TRT21      OR          0.32   0.15-0.69 0.004
3 PAIN_ms STENT1     OR          0.87   0.45-1.68 0.680
4 PAIN_s  (Intercept) OR          3.74   2.46-5.70 <0.001
5 PAIN_s  TRT21      OR          0.14   0.10-0.19 <0.001
6 PAIN_s  STENT1     OR          0.08   0.02-0.27 <0.001

gee_table_final %>%
  flextable() %>%
  theme_booktabs() %>%
  autofit() %>%
  align(
    j = c("Estimate", "CI", "P_value"),
    align = "center",
    part = "all"
  ) %>%
  save_as_docx(
    path = "output/pain_adj_w.docx"
  )

#-----
# Block 2ai- Interaction
#-----

outcomes <- c(
  "PAIN_ms",
  "PAIN_s"
)

# convert outcomes to numeric
Outp[outcomes] <- lapply(
  Outp[outcomes],
  function(x) as.numeric(as.character(x))
)

# Function to extract treatment effects within each stent stratum
get_gee_interaction_results <- function(y) {

  formula <- as.formula(
    paste(y, "~ TRT2 * STENT")
  )

```

```

model <- geeglm(
  formula,
  data = Outp,
  id = CENTER,
  weights = weights,
  family = binomial(link = "logit"),
  corstr = "exchangeable"
)

# Coefficients
beta <- coef(model)

# Robust covariance matrix
V <- vcov(model)

# -----
# LG vs Cohen among NO-STENT patients
# -----

b_no_stent <- beta["TRT21"]

se_no_stent <- sqrt(
  V["TRT21", "TRT21"]
)

OR_no_stent <- exp(b_no_stent)

CI_no_stent <- exp(
  b_no_stent + c(-1, 1) * 1.96 * se_no_stent
)

# -----
# LG vs Cohen among STENTED patients
# = TRT21 + TRT21:STENT1
# -----

b_stent <- beta["TRT21"] +
  beta["TRT21:STENT1"]

# Variance of the linear combination:
# Var(beta1 + beta3)

var_stent <-
  V["TRT21", "TRT21"] +
  V["TRT21:STENT1", "TRT21:STENT1"] +
  2 * V["TRT21", "TRT21:STENT1"]

se_stent <- sqrt(var_stent)

OR_stent <- exp(b_stent)

CI_stent <- exp(
  b_stent + c(-1, 1) * 1.96 * se_stent
)

```

```

)

# -----
# Interaction test
# -----

b_interaction <- beta["TRT21:STENT1"]

se_interaction <- sqrt(
  V["TRT21:STENT1", "TRT21:STENT1"]
)

z_interaction <- b_interaction / se_interaction

p_interaction <- 2 * pnorm(
  abs(z_interaction),
  lower.tail = FALSE
)

# -----
# Return manuscript-ready results
# -----

data.frame(
  Outcome = y,

  `LG vs Cohen, No Stent` =
    sprintf(
      "%.2f (%.2f-%.2f)",
      OR_no_stent,
      CI_no_stent[1],
      CI_no_stent[2]
    ),

  `LG vs Cohen, Stent` =
    sprintf(
      "%.2f (%.2f-%.2f)",
      OR_stent,
      CI_stent[1],
      CI_stent[2]
    ),

  P_interaction = p_interaction,

  stringsAsFactors = FALSE
)
}

# Run for all 8 outcomes
interaction_results <- do.call(
  rbind,
  lapply(outcomes, get_gee_interaction_results)
)

```

```
)
```

```
Warning in eval(family$initialize): non-integer #successes in a binomial glm!  
Warning in eval(family$initialize): non-integer #successes in a binomial glm!
```

```
interaction_results
```

	Outcome	LG.vs.Cohen..No.Stent	LG.vs.Cohen..Stent	P_interaction
TRT21:STENT1	PAIN_ms	0.17 (0.14-0.20)	0.85 (0.29-2.48)	0.002771835
TRT21:STENT11	PAIN_s	0.14 (0.10-0.18)	0.14 (0.05-0.40)	0.965328289

SENSITIVITY ANALYSIS 1

1 E-value for outcomes

```
# E-value explains away a specific treatment-outcome association.
```

```
evaluate_data <- data.frame(
```

```
  Outcome = c(  
    "OBST",  
    "URCOMPINJ",  
    "SURG",  
    "ILEUS",  
    "GU90",  
    "OBSTadj",  
    "URCOMPINJadj",  
    "SURGadj",  
    "ILEUSadj",  
    "GU90adj",  
    "GU30adj",  
    "PROC365adj",  
    "PAIN_ms",  
    "PAIN_s",  
    "PAIN_msadj",  
    "PAIN_sadj"  
  ),
```

```
  OR = c(  
    3.67,  
    0.43,  
    0.72,  
    0.10,  
    0.20,  
    3.71,  
    0.36,  
    0.63,  
    0.20,  
    0.18,  
    0.03,  
    0.71,  
    0.32,
```

```

    0.20,
    0.32,
    0.14
  ),
  CI_low = c(
    2.21,
    0.27,
    0.58,
    0.04,
    0.16,
    2.50,
    0.27,
    0.50,
    0.17,
    0.15,
    0.01,
    0.56,
    0.14,
    0.15,
    0.15,
    0.10
  ),
  CI_high = c(
    6.10,
    0.69,
    0.90,
    0.25,
    0.26,
    5.52,
    0.49,
    0.81,
    0.24,
    0.21,
    0.08,
    0.89,
    0.73,
    0.26,
    0.69,
    0.19
  )
)

# Specify rare outcome assumption
evaluate_data <- evaluate_data %>%
  mutate(
    rare = ifelse(
      Outcome %in% c("SURG", "GU90", "SURGadj", "GU90adj", "GU30adj", "Pain_ms", "Pain_s", "Pain_msadj")
      FALSE,
      TRUE
    )
  )
)

```

```

# Calculate E-values
evaluate_results <- evaluate_data %>%
  rowwise() %>%
  mutate(
    OR_input = ifelse(OR < 1, 1/OR, OR),
    CI_low_input = ifelse(OR < 1, 1/CI_high, CI_low),
    CI_high_input = ifelse(OR < 1, 1/CI_low, CI_high),

    E_value = list(
      evalues.OR(
        est = OR_input,
        lo = CI_low_input,
        hi = CI_high_input,
        rare = rare
      )
    )
  ) %>%
  ungroup()

evaluate_table <- evaluate_results %>%
  mutate(
    E_value_point = sapply(
      E_value,
      function(x) x["E-values", "point"]
    ),
    E_value_CI = sapply(
      E_value,
      function(x) x["E-values", "lower"]
    )
  ) %>%
  dplyr::select(
    Outcome,
    OR,
    CI_low,
    CI_high,
    E_value_point,
    E_value_CI
  )

evaluate_table

# A tibble: 16 x 6
  Outcome      OR CI_low CI_high E_value_point E_value_CI
  <chr>      <dbl> <dbl> <dbl>         <dbl>      <dbl>
1 OBST         3.67  2.21  6.1           6.80       3.85
2 URCOMPINJ    0.43  0.27  0.69          4.08       2.26
3 SURG         0.72  0.58  0.9           1.64       1.29
4 ILEUS         0.1   0.04  0.25          19.5       7.46
5 GU90         0.2   0.16  0.26           3.90       3.33
6 OBSTadj      3.71  2.5   5.52           6.88       4.44
7 URCOMPINJadj 0.36  0.27  0.49           5          3.50
8 SURGadj      0.63  0.5   0.81           1.83       1.46
9 ILEUSadj     0.2   0.17  0.24           9.47       7.80

```

10	GU90adj	0.18	0.15	0.21	4.15	3.79
11	GU30adj	0.03	0.01	0.08	11.0	6.53
12	PROC365adj	0.71	0.56	0.89	2.17	1.50
13	PAIN_ms	0.32	0.14	0.73	5.70	2.08
14	PAIN_s	0.2	0.15	0.26	9.47	7.15
15	PAIN_msadj	0.32	0.15	0.69	5.70	2.26
16	PAIN_sadj	0.14	0.1	0.19	13.8	10

```

evaluate_table_formatted <- evaluate_table %>%
  mutate(
    OR = sprintf("%.2f", OR),
    CI = paste0(
      sprintf("%.2f", CI_low),
      "-",
      sprintf("%.2f", CI_high)
    ),
    E_value_point = sprintf("%.2f", E_value_point),
    E_value_CI = sprintf("%.2f", E_value_CI)
  ) %>%
  dplyr::select(
    Outcome,
    OR,
    CI,
    E_value_point,
    E_value_CI
  )

```

```

evaluate_table_formatted %>%
  flextable() %>%
  theme_booktabs() %>%
  autofit() %>%
  align(
    j = c("OR", "CI", "E_value_point", "E_value_CI"),
    align = "center",
    part = "all"
  ) %>%
  save_as_docx(
    path = "output/E_value_stent.docx"
  )

```

1.1. Plot E-value

```
# E-value sensitivity contour plot for Obstruction risk in overlap weighting before adjustment to procedure
```

```

# OBST values
RR_est <- 3.67
RR_CI <- 2.21

E_value_point <- RR_est + sqrt(RR_est*(RR_est-1))
E_value_CI <- RR_CI + sqrt(RR_CI*(RR_CI-1))

# Function for contour

```

```

evaluate_curve <- function(RR){

  x <- seq(RR + 0.001, 20, length.out = 5000)

  y <- RR * (x - 1)/(x - RR)

  data.frame(
    RR_EU = x,
    RR_UY = y
  )
}

curve_est <- evaluate_curve(RR_est)
curve_CI <- evaluate_curve(RR_CI)

# Create plot
evaluate_plot <- ggplot() +

  # Point estimate curve
  geom_line(
    data = curve_est,
    aes(
      x = RR_EU,
      y = RR_UY
    ),
    color = "red",
    linewidth = 0.8
  ) +

  # CI curve
  geom_line(
    data = curve_CI,
    aes(
      x = RR_EU,
      y = RR_UY
    ),
    color = "blue",
    linewidth = 0.8
  ) +

  # E-value point
  geom_point(
    aes(
      x = E_value_point,
      y = E_value_point
    ),
    color = "red",
    size = 2.5
  ) +

  # CI E-value point
  geom_point(

```



```

aes(
  x = E_value_CI,
  y = E_value_CI
),
color = "blue",
size = 2.5
) +

# Labels
annotate(
  "text",
  x = E_value_point,
  y = E_value_point + 0.35,
  label = paste0(
    "E value: (",
    round(E_value_point,2),
    ", ",
    round(E_value_point,2),
    ")"
  ),
  color = "black",
  size = 4,
  hjust = 0
) +

annotate(
  "text",
  x = E_value_CI,
  y = E_value_CI - 0.25,
  label = paste0(
    "E value (CI): (",
    round(E_value_CI,2),
    ", ",
    round(E_value_CI,2),
    ")"
  ),
  color = "black",
  size = 4,
  hjust = 0
) +

scale_x_continuous(
  limits = c(1,20),
  breaks = seq(5,20,5)
) +

scale_y_continuous(
  limits = c(1,20),
  breaks = seq(0,20,5)
) +

labs(
  x = "Risk ratio for exposure-confounder relationship",

```

```

  y = "Risk ratio for confounder-outcome relationship"
) +

theme_gray(base_size = 14)

# OBST values
RR_est_a <- 3.71
RR_CI_a  <- 2.50

E_value_point_a <- RR_est_a + sqrt(RR_est_a*(RR_est_a-1))
E_value_CI_a <- RR_CI_a + sqrt(RR_CI_a*(RR_CI_a-1))

# Function for contour
evaluate_curve <- function(RR){

  x <- seq(RR + 0.001, 20, length.out = 5000)

  y <- RR * (x - 1)/(x - RR)

  data.frame(
    RR_EU = x,
    RR_UY = y
  )
}

curve_est_a <- evaluate_curve(RR_est_a)
curve_CI_a  <- evaluate_curve(RR_CI_a)

# Create plot
evaluate_plot_a <- ggplot() +

  # Point estimate curve
  geom_line(
    data = curve_est_a,
    aes(
      x = RR_EU,
      y = RR_UY
    ),
    color = "red",
    linewidth = 0.8
  ) +

  # CI curve
  geom_line(
    data = curve_CI_a,
    aes(
      x = RR_EU,
      y = RR_UY
    ),

```

```

    color = "blue",
    linewidth = 0.8
) +

# E-value point
geom_point(
  aes(
    x = E_value_point_a,
    y = E_value_point_a
  ),
  color = "red",
  size = 2.5
) +

# CI E-value point
geom_point(
  aes(
    x = E_value_CI_a,
    y = E_value_CI_a
  ),
  color = "blue",
  size = 2.5
) +

# Labels
annotate(
  "text",
  x = E_value_point_a,
  y = E_value_point_a + 0.35,
  label = paste0(
    "E value: (",
    round(E_value_point_a,2),
    ", ",
    round(E_value_point_a,2),
    ")"
  ),
  color = "black",
  size = 4,
  hjust = 0
) +

annotate(
  "text",
  x = E_value_CI_a,
  y = E_value_CI_a - 0.25,
  label = paste0(
    "E value (CI): (",
    round(E_value_CI_a,2),
    ", ",
    round(E_value_CI_a,2),
    ")"
  ),
  color = "black",

```

```

    size = 4,
    hjust = 0
  ) +

  scale_x_continuous(
    limits = c(1,20),
    breaks = seq(5,20,5)
  ) +

  scale_y_continuous(
    limits = c(1,20),
    breaks = seq(0,20,5)
  ) +

  labs(
    x = "Risk ratio for exposure-confounder relationship",
    y = "Risk ratio for confounder-outcome relationship"
  ) +

  theme_gray(base_size = 14)

#-----
# Combine plot and risk table using patchwork
#-----
combined_plot <- (evaluate_plot | evaluate_plot_a) +
  plot_annotation(
    tag_levels = "A"
  )

#-----
# Save the figure
#-----
ggsave(
  filename = "figures/Evalue_OBST.png",
  plot = combined_plot,
  width = 20,
  height = 10,
  dpi = 300
)

```

Warning: Removed 184 rows containing missing values or values outside the scale range (``geom_line()``).

Warning: Removed 42 rows containing missing values or values outside the scale range (``geom_line()``).

Warning: Removed 190 rows containing missing values or values outside the scale range (``geom_line()``).

Warning: Removed 61 rows containing missing values or values outside the scale range (``geom_line()``).

SENSITIVITY ANALYSIS 2

1 Do matching 1:1 without replacement

```
# Fit matching model using propensity score (calculated inside MatchIt): 1:1 matching

set.seed(123)

m.out <- matchit(
  TRT2 ~ AGS + YEARS + GENDER + BMI + PREOPRNPLN + HISTOINJ +
    SURGIND2 + PREOPVURGRADE + VURSIDE + UDR7M + VURPHASE +
    PREOPEPISODE1 + PREOPTYPEU1 + PREOPRNSCR + SURGTECH2,
  data = Out,
  method = "nearest",
  ratio = 1, # 1:1 matching (1 treated : 1 control)
  distance = "logit",
  caliper = 0.2
)

mdata <- match.data(
  m.out,
  weights = "weights_psm"
)

readr::write_csv(mdata, "data/matched_data.csv")

# Table for balance
balm <- cobalt::bal.tab(m.out, un = TRUE, m.threshold = 0.1)
balm
```

Balance Measures

	Type	Diff.Un	Diff.Adj	M.Threshold
distance	Distance	1.5815	0.1163	
AGS	Contin.	0.3507	0.0642	Balanced, <0.1
YEARS	Contin.	-0.4776	0.0109	Balanced, <0.1
GENDER	Binary	-0.0471	0.0206	Balanced, <0.1
BMI	Contin.	-0.8345	0.0397	Balanced, <0.1
PREOPRNPLN	Binary	-0.2699	-0.0103	Balanced, <0.1
HISTOINJ	Binary	0.0198	-0.0206	Balanced, <0.1
SURGIND2_1	Binary	0.0856	0.0206	Balanced, <0.1
SURGIND2_2	Binary	0.0595	0.0103	Balanced, <0.1
SURGIND2_3	Binary	-0.0089	-0.0103	Balanced, <0.1
SURGIND2_4	Binary	-0.0465	0.0000	Balanced, <0.1
SURGIND2_5	Binary	-0.0897	-0.0206	Balanced, <0.1
PREOPVURGRADE_3	Binary	-0.0140	-0.0206	Balanced, <0.1
PREOPVURGRADE_4	Binary	0.0802	0.0619	Balanced, <0.1
PREOPVURGRADE_5	Binary	-0.0662	-0.0412	Balanced, <0.1
VURSIDE	Binary	0.0247	-0.0412	Balanced, <0.1
UDR7M	Binary	-0.1399	-0.0309	Balanced, <0.1
VURPHASE_1	Binary	-0.0383	0.0000	Balanced, <0.1

VURPHASE_2	Binary	0.3112	-0.0103	Balanced, <0.1
VURPHASE_3	Binary	-0.2729	0.0103	Balanced, <0.1
PREOPEPISODE1_1	Binary	-0.0440	-0.0206	Balanced, <0.1
PREOPEPISODE1_2	Binary	-0.1765	-0.0309	Balanced, <0.1
PREOPEPISODE1_3	Binary	0.2205	0.0515	Balanced, <0.1
PREOPTYPEU1	Binary	0.0309	-0.0103	Balanced, <0.1
PREOPRNSCR	Binary	-0.2674	-0.0309	Balanced, <0.1
SURGTECH2_1	Binary	0.2494	0.0412	Balanced, <0.1
SURGTECH2_2	Binary	-0.0629	0.0000	Balanced, <0.1
SURGTECH2_3	Binary	-0.1866	-0.0412	Balanced, <0.1

Balance tally for mean differences

	count
Balanced, <0.1	27
Not Balanced, >0.1	0

Variable with the greatest mean difference

Variable	Diff.Adj	M.Threshold
AGS	0.0642	Balanced, <0.1

Sample sizes

	Control	Treated
All	197	215
Matched	97	97
Unmatched	100	118

```
bal_dfm <- as.data.frame(balm$Balance)
smd_tablem <- bal_dfm %>%
  mutate(
    Variable = rownames(bal_dfm),
    SMD_Before = round(abs(Diff.Un), 3),
    SMD_After = round(abs(Diff.Adj), 3)
  ) %>%
  select(Variable, SMD_Before, SMD_After)

smd_tablem %>%
  flextable() %>%
  save_as_docx(path = "output/SMD_psm.docx")
```

```
# how much effective sample size remains (overall ESS: calculated manually)?
summary(m.out$weights)
```

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.0000	0.0000	0.4709	1.0000	1.0000

```
bal.tab(m.out)$Effective.Sample.Size
```

NULL

```
sum(m.out$weights)^2 / sum(m.out$weights^2)
```

```
[1] 194
```

```

# Figure for balance - Love plot
plm <- love.plot(
  m.out,
  threshold = 0.1,
  abs = TRUE,
  stats = "mean.diffs",
  var.order = "unadjusted",
  drop.distance = TRUE,
  sample.names = c("Before matching", "After matching"),
  size = 4,
  var.names = c(
    AGS = "Age at surgery",
    YEARS = "Year of surgery",
    BMI = "Body mass index",
    GENDER = "Female gender",
    PREOPRNPLN = "Preop pyelonephritis",
    HISTOINJ = "History of injection",
    VURSIDE = "Right-sided VUR",
    UDR7M = "Ureteral dilation 7mm",
    PREOPTYPEU1 = "Preop febrile UTIs",
    PREOPRNSCR = "Preop renal scarring",
    VURPHASE_1 = "Reflux: voiding phase",
    VURPHASE_2 = "Reflux: filling phase",
    VURPHASE_3 = "Reflux: filling + voiding",
    SURGTECH2_1 = "Surgeon experience 100 operations",
    SURGTECH2_2 = "Surgeon experience 50-100 operations",
    SURGTECH2_3 = "Surgeon experience <50 operations",
    PREOPEPISODE1_1 = "Preop UTI episodes: 1",
    PREOPEPISODE1_2 = "Preop UTI episodes: 2",
    PREOPEPISODE1_3 = "Preop UTI episodes: 3",
    SURGIND2_1 = "Indication: breakthrough UTI",
    SURGIND2_2 = "Indication: high-grade reflux (III-V)",
    SURGIND2_3 = "Indication: poor compliance to medical treatment",
    SURGIND2_4 = "Indication: ineffective/poorly tolerated CAP",
    SURGIND2_5 = "Indication: other",
    PREOPVURGRADE_3 = "Preop VUR grade: III",
    PREOPVURGRADE_4 = "Preop VUR grade: IV",
    PREOPVURGRADE_5 = "Preop VUR grade: V"
  )
)

plm <- plm +
  scale_color_manual(
    values = c(
      "Before matching" = "#0072B2",
      "After matching" = "#D55E00"
    )
  ) +
  labs(
    x = "Absolute Standardized Differences",
    subtitle = "Matched population"
  ) +
  theme(

```

```

plot.margin = margin(10, 10, 10, 10),
axis.text = element_text(size = 16),
axis.title = element_text(size = 16),
plot.title = element_text(size = 16),
legend.text = element_text(size = 16),
legend.title = element_text(size = 16),
axis.ticks = element_line(size = 0.5, color = "black"),
axis.line = element_line(size = 0.5, color = "black"),
panel.background = element_blank(),
panel.grid.major = element_blank(),
panel.grid.minor = element_blank(),
panel.border = element_blank(),
axis.ticks.length = unit(0.3, "cm"),
panel.spacing = unit(2, "lines")
)

```

Scale for colour is already present.

Adding another scale for colour, which will replace the existing scale.

```
ggsave ("figures/love_plot_m.png", plot = plm, width = 12, height = 10, dpi = 300)
```

```

# Compute mean SMD
smd_df_m <- balm$Balance
mean_smd_before_m <- mean(abs(smd_df_m$Diff.Un), na.rm = TRUE) # calculate mean SMD before matching
mean_smd_after_m <- mean(abs(smd_df_m$Diff.Adj), na.rm = TRUE) # calculate mean SMD before matching
mean_smd_m <- data.frame(
  Mean_SMD_Before = mean_smd_before_m,
  Mean_SMD_After = mean_smd_after_m
)

```

```
mean_smd_m
```

```

  Mean_SMD_Before Mean_SMD_After
1      0.2163242      0.02813468

```

```

# propensity score distribution: MatchIt does not store the propensity scores in m.out$ps. Instead, the
summary(m.out$distance)

```

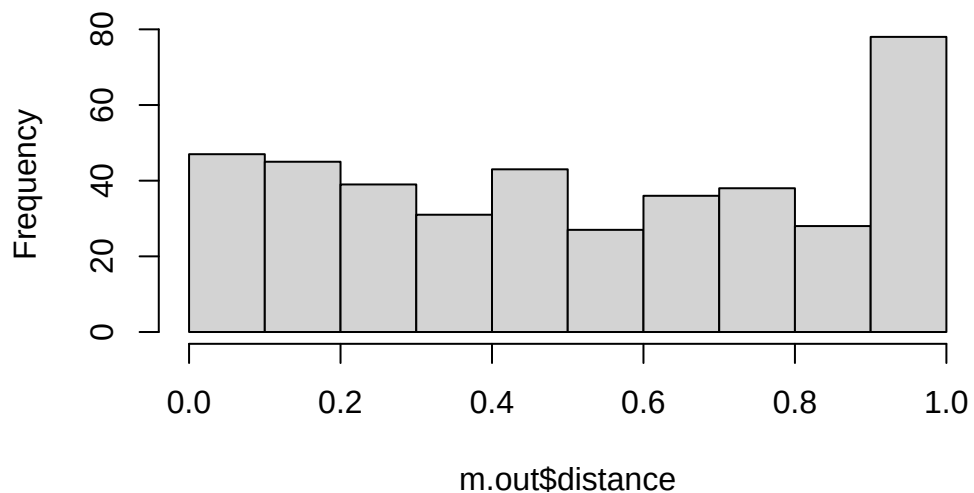
```

      Min.   1st Qu.   Median     Mean   3rd Qu.     Max.
0.006799 0.218309 0.504244 0.521845 0.803450 0.999248

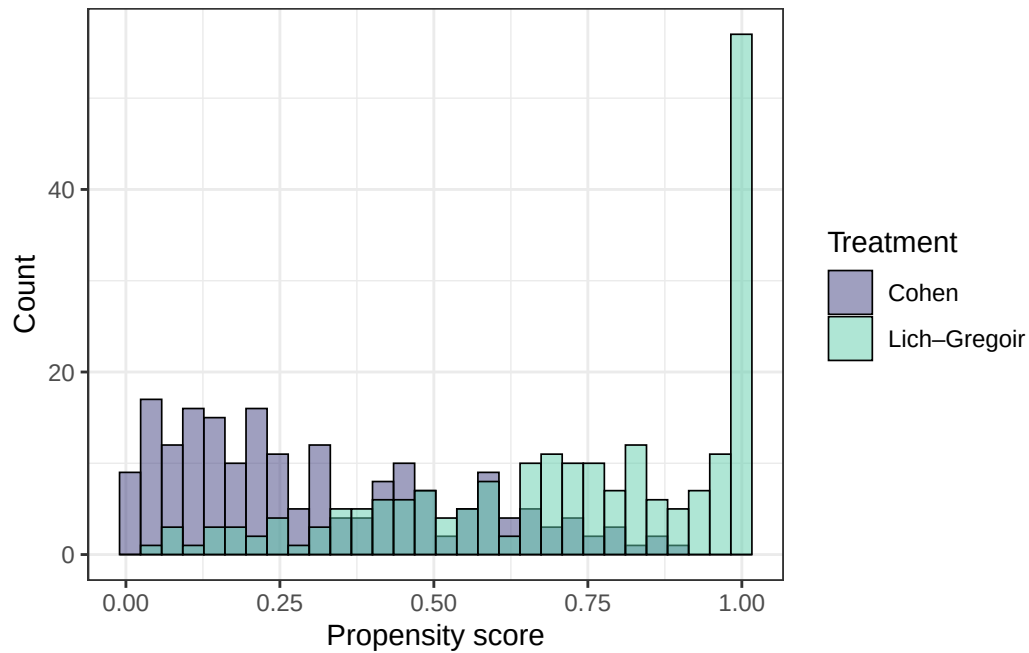
```

```
hist(m.out$distance)
```


Histogram of m.out\$distance



```
ggplot(
  data.frame(
    ps = m.out$distance,
    Treatment = factor(Out$Treatment)
  ),
  aes(ps, fill = Treatment)
) +
  geom_histogram(alpha = 0.5, bins = 30, position = "identity", color = "black", linewidth = 0.3) +
  scale_fill_viridis_d(
    option = "mako",
    begin = 0.3,
    end = 0.8,
    labels = c("Cohen", "Lich-Gregoir")
  ) +
  labs(
    x = "Propensity score",
    y = "Count",
    fill = "Treatment"
  ) +
  theme_bw()
```



```
ggsave("figures/propensity_psm.png",
        width = 12,
        height = 10,
        dpi = 300)

# Distributional balance for propensity score

# Before-After matching
p_psm <- bal.plot(
  m.out,
  var.name = "distance",
  type = "density",
  which = "both"
)

p_psm <- p_psm +
  scale_color_viridis_d(
    option = "mako",
    begin = 0.3,
    end = 0.8,
    labels = c("Cohen group", "Lich-Gregoir group")
  ) +
  scale_fill_viridis_d(
    option = "mako",
    begin = 0.3,
    end = 0.8,
    labels = c("Cohen group", "Lich-Gregoir group")
  )
```

Scale for fill is already present.
Adding another scale for fill, which will replace the existing scale.

```

ggsave(
  "figures/PS_density_before_after_m.png",
  p_psm,
  width = 10,
  height = 6,
  dpi = 300
)

```

2 Descriptive analysis: AFTER matching

```

# mdata is the dataset with added weights_psm

# Convert variables to appropriate data types for analysis
mdata <- mdata %>%
  mutate(
    across(c(PID, QCODE), as.character),

    across(c(AGS, YEARS,BLDLOSS, BMI, CATHPERIOD, FUPM, HOSPTIME,OPRTIME,
      POSTUTIFUP, POSTUTI_T), as.numeric),

    across(c(ANASTOMO,BLDCompleak,BLDCompRetn,
      BLDCompVoid,BLDSPASM,CATH,CENTER,CLAVIENCLASS,EPIDURAL,GENDER,
      GU30, GU90N, GUER30, GUER90, HEMAT,
      HISTOINJ,ILEUS, INCCOMPBLEED, INCCOMPINF, INCCOMPSCAR, NARC,
      NONARC, PAIN, PREOPEPISODE1, PREOPRNPLN, PREOPRNSCR, PREOPTYPEU1,
      PREOPVURGRADE, PROC365, STENT, SURGIND, SURGTECH2, TRT2, UDR7M,
      URCOMPVUR, URCOMPINJ, URCOMPOBST, URCOMPSTEN, POSTUTI, VURPHASE,
      VURSIDE, RECGRADE_MAX, SURGIND2, LUTD, OBST, SURG, GU90,
      CATHPERIOD_G, ANY, MINOR, MAJOR), as.factor)
  )

str(mdata) # 194 obs. of 71 variables

Classes 'matchdata' and 'data.frame': 194 obs. of 69 variables:
 $ PID          : chr  "P1" "P3" "P6" "P7" ...
 $ QCODE        : chr  "102" "3" "8" "9" ...
 $ RECO11       : int   0 0 1 1 0 1 0 0 0 0 ...
 $ AGS          : num   5 7 9 10 8 11 5 8 8 7 ...
 $ YEARS        : num   2011 2017 2018 2015 2017 ...
 $ ANASTOMO     : Factor w/ 2 levels "0","1": 1 1 1 1 1 1 1 1 1 1 ...
 $ BLDCompleak  : Factor w/ 2 levels "0","1": 1 1 1 1 1 1 1 1 1 1 ...
 $ BLDCompRetn  : Factor w/ 2 levels "0","1": 1 2 2 1 2 2 1 2 1 1 ...
 $ BLDCompVoid  : Factor w/ 2 levels "0","1": 2 1 1 1 1 1 1 1 1 1 ...
 $ BLDLOSS      : num   70 150 190 75 100 55 50 80 80 50 ...
 $ BLDSPASM     : Factor w/ 2 levels "0","1": 1 2 2 1 2 1 2 1 2 1 ...
 $ BMI          : num   13.5 12 15.8 15.3 15.3 15.3 13.7 15 15 14.4 ...
 $ CATH         : Factor w/ 2 levels "0","1": 2 2 2 2 2 2 2 2 2 2 ...
 $ CATHPERIOD   : num    2 5 7 4 5 7 1 5 5 4 ...
 $ CENTER       : Factor w/ 4 levels "1","2","3","5": 1 1 1 1 1 1 1 1 1 1 ...
 $ CLAVIENCLASS : Factor w/ 4 levels "1","2","3","4": 2 2 2 2 1 3 1 2 1 2 ...
 $ EPIDURAL     : Factor w/ 2 levels "0","1": 1 1 1 2 1 1 1 1 1 1 ...

```

```

$ FUPM      : num  24 44 36 36 37 36 24 36 24 24 ...
$ GENDER    : Factor w/ 2 levels "0","1": 1 2 1 1 2 1 2 2 2 1 ...
$ GU30      : Factor w/ 2 levels "0","1": 1 1 1 2 1 1 1 1 1 1 ...
$ GU90N     : Factor w/ 3 levels "0","1","2": 1 2 2 2 1 1 1 1 1 1 ...
$ GUER30    : Factor w/ 2 levels "0","1": 1 1 1 2 1 1 1 1 1 1 ...
$ GUER90    : Factor w/ 2 levels "0","1": 2 1 1 2 1 1 1 1 1 1 ...
$ HEMAT     : Factor w/ 2 levels "0","1": 1 2 2 2 1 1 1 1 1 1 ...
$ HISTOINJ  : Factor w/ 2 levels "0","1": 2 1 1 1 2 1 1 1 1 1 ...
$ HOSPTIME  : num  2 5 5 8 5 4 2 5 5 5 ...
$ ILEUS     : Factor w/ 2 levels "0","1": 1 1 1 1 1 1 1 1 1 1 ...
$ INCCOMPBLEED : Factor w/ 2 levels "0","1": 2 2 1 1 1 1 1 2 1 1 ...
$ INCCOMPINF : Factor w/ 2 levels "0","1": 1 1 2 1 1 1 1 1 1 1 ...
$ INCCOMPSCAR : Factor w/ 2 levels "0","1": 1 1 1 1 1 1 1 1 1 1 ...
$ NARC      : Factor w/ 2 levels "0","1": 1 2 2 1 1 1 1 1 1 1 ...
$ NONARC    : Factor w/ 2 levels "0","1": 2 2 2 2 2 2 2 2 2 2 ...
$ OPRTIME   : num  76 80 75 90 88 80 68 75 78 76 ...
$ PAIN      : Factor w/ 4 levels "0","1","2","3": 2 4 3 3 2 2 2 2 2 2 ...
$ PREOPEPISODE1 : Factor w/ 3 levels "1","2","3": 2 3 3 3 2 3 3 3 3 3 ...
$ PREOPRNPLN : Factor w/ 2 levels "0","1": 1 1 1 1 1 1 1 1 1 1 ...
$ PREOPRNSCR : Factor w/ 2 levels "0","1": 1 1 1 1 1 1 1 1 1 1 ...
$ PREOPTYPEU1 : Factor w/ 2 levels "0","1": 2 2 2 2 2 2 2 2 2 2 ...
$ PREOPVURGRADE : Factor w/ 3 levels "3","4","5": 3 3 3 2 2 2 3 3 2 3 ...
$ PROC365   : Factor w/ 2 levels "0","1": 1 1 1 2 1 1 1 1 1 1 ...
$ STENT     : Factor w/ 2 levels "0","1": 1 1 1 1 1 1 2 1 1 1 ...
$ SURGIND   : Factor w/ 11 levels "1","2","3","4",...: 3 3 4 4 3 3 4 3 3 4 ...
$ SURGTECH2 : Factor w/ 3 levels "1","2","3": 3 2 2 2 2 2 2 2 2 2 ...
$ TRT2      : Factor w/ 2 levels "0","1": 2 2 2 2 2 2 2 2 2 2 ...
$ UDR7M     : Factor w/ 2 levels "0","1": 2 2 2 2 2 2 2 2 2 2 ...
$ URCOMPCVUR : Factor w/ 2 levels "0","1": 1 1 1 1 1 1 1 1 1 1 ...
$ URCOMPINJ  : Factor w/ 2 levels "0","1": 1 1 1 1 1 1 1 1 1 1 ...
$ URCOMPOBST : Factor w/ 2 levels "0","1": 1 1 1 1 1 1 1 1 1 1 ...
$ URCOMPSTEN : Factor w/ 2 levels "0","1": 1 1 1 1 1 2 1 1 1 2 ...
$ POSTUTI    : Factor w/ 2 levels "0","1": 2 1 1 2 1 2 1 1 1 1 ...
$ POSTUTIFUP : num  1 44 36 1 37 1 24 36 24 24 ...
$ POSTUTI_T  : num  1 0 0 1 0 1 0 0 0 0 ...
$ VURPHASE  : Factor w/ 3 levels "1","2","3": 1 3 1 1 1 1 1 1 1 1 ...
$ VURSIDE   : Factor w/ 2 levels "0","1": 1 2 2 2 1 2 1 2 1 1 ...
$ RECGRADE_MAX : Factor w/ 6 levels "0","1","2","3",...: 2 1 3 3 2 3 2 2 1 2 ...
$ SURGIND2   : Factor w/ 5 levels "1","2","3","4",...: 2 2 1 1 2 2 1 2 2 1 ...
$ LUTD      : Factor w/ 2 levels "0","1": 2 2 2 1 2 2 1 2 1 1 ...
$ OBST      : Factor w/ 2 levels "0","1": 1 1 1 1 1 2 1 1 1 2 ...
$ SURG      : Factor w/ 2 levels "0","1": 2 2 2 1 1 1 1 2 1 1 ...
$ GU90      : Factor w/ 2 levels "0","1": 1 2 2 2 1 1 1 1 1 1 ...
$ CATHPERIOD_G : Factor w/ 3 levels "Early removal 3 days",...: 1 2 2 2 2 2 1 2 2 2 ...
$ ANY       : Factor w/ 2 levels "0","1": 2 2 2 1 2 2 1 2 1 2 ...
$ MINOR     : Factor w/ 2 levels "0","1": 2 2 2 1 2 2 1 2 1 1 ...
$ MAJOR     : Factor w/ 2 levels "0","1": 1 1 1 1 1 2 1 1 1 2 ...
$ Treatment : Factor w/ 2 levels "Cohen","Lich-Gregoir": 2 2 2 2 2 2 2 2 2 2 ...
$ weights   : num  0.578 0.393 0.577 0.26 0.282 ...
$ distance  : num  0.422 0.607 0.423 0.74 0.718 ...
$ weights_psm : num  1 1 1 1 1 1 1 1 1 1 ...
$ subclass  : Factor w/ 97 levels "1","2","3","4",...: 1 2 3 4 5 6 7 8 9 10 ...

```

```

# Matched baseline characteristics
table1m<- mdata %>%
  select(YEARS, AGS, GENDER, BMI, HISTOINJ, SURGIND2, PREOPVURGRADE, VURSIDE, UDR7M,
         VURPHASE, PREOPTYPEU1, PREOPEPISODE1, PREOPRNPLN, PREOPRNSCR,
         SURGTECH2, CENTER, TRT2
  ) %>%
  tbl_summary(
    by = TRT2, # for stratification
    statistic = list(
      all_continuous() ~ "{median} ({p25}-{p75})",
      all_categorical() ~ "{n} ({p}%)"
    ),
    digits = all_categorical() ~ c(0, 1), # n = 0 decimals, p = 1 decimal
    missing = "ifany", # shows missing only when present
    missing_text = "Missing"
  ) %>%
  add_overall() %>%
  bold_labels()

table1m <- table1m %>%
  modify_caption(
    "Baseline characteristics after 1:1 propensity score matching"
  )

table1m %>%
  as_flex_table() %>%
  flextable::save_as_docx(path = "output/baseline_m.docx")

# Matched procedural factors
table2m<- mdata %>%
  select(TRT2, OPRTIME, STENT, CATH, CATHPERIOD, CATHPERIOD_G, HOSPTIME) %>%
  tbl_summary(
    by = TRT2, # for stratification
    statistic = list(
      all_continuous() ~ "{median} ({p25}-{p75})", # or ~ "{mean} ± {sd}",
      all_categorical() ~ "{n} ({p}%)"
    ),
    digits = all_categorical() ~ c(0, 1), # n = 0 decimals, p = 1 decimal
    missing = "ifany",
    missing_text = "Missing"
  ) %>%
  add_overall() %>%
  add_p(pvalue_fun = ~style_pvalue(.x, digits = 3)) %>% # 3 decimals
  bold_labels()

table2m

table2m %>%
  as_flex_table() %>%
  flextable::save_as_docx(path = "output/procedural_m.docx")

# Outcomes

```

Characteristic	Overall N = 194 ¹	0 N = 97 ¹	1 N = 97 ¹	p-value ²
OPRTIME	88 (70-135)	130 (100-159)	70 (68-80)	<0.001
STENT				0.127
0	130 (67.0%)	60 (61.9%)	70 (72.2%)	
1	64 (33.0%)	37 (38.1%)	27 (27.8%)	
CATH				>0.999
0	4 (2.1%)	2 (2.1%)	2 (2.1%)	
1	190 (97.9%)	95 (97.9%)	95 (97.9%)	
CATHPERIOD	4 (2-5)	4 (4-5)	3 (1-5)	<0.001
CATHPERIOD_G				<0.001
Early removal 3 days	63 (32.5%)	8 (8.2%)	55 (56.7%)	
Standard removal 4-7 days	94 (48.5%)	72 (74.2%)	22 (22.7%)	
Delayed removal >7 days	37 (19.1%)	17 (17.5%)	20 (20.6%)	
HOSPTIME	5.00 (3.00-6.00)	5.00 (5.00-6.00)	3.00 (2.00-5.00)	<0.001

¹Median (Q1-Q3); n (%)

²Wilcoxon rank sum test; Pearson's Chi-squared test; Fisher's exact test

```
table3m<- mdata %>%
  select(TRT2, FUPM, LUTD, OBST, URCOMPINJ, SURG, ILEUS, CLAVIENCLASS, URCOMPCVUR,
         GU30, GU90, GU90N, GUER30, GUER90, PROC365,
         BLDLOSS, BLDSPASM, HEMAT,
         POSTUTI) %>%
  tbl_summary(
    by = TRT2, # for stratification
    statistic = list(
      all_continuous() ~ "{median} ({p25}-{p75})", # or ~ "{mean} ± {sd}",
      all_categorical() ~ "{n} ({p}%)"
    ),
    digits = all_categorical() ~ c(0, 1), # n = 0 decimals, p = 1 decimal
    missing = "ifany",
    missing_text = "Missing"
  ) %>%
  add_overall() %>%
  add_p(pvalue_fun = ~style_pvalue(.x, digits = 3)) %>% # 3 decimals
  bold_labels()
```

table3m

```
table3m %>%
  as_flex_table() %>%
  flextable::save_as_docx(path = "output/outcomes_m.docx")
```

```
##### Postoperative UTI outcome as time-to-event outcome: Weighted survival curve
fit_psm <- survfit(
  Surv(POSTUTIFUP, POSTUTI == 1) ~ Treatment,
  data = mdata
)
```

Characteristic	Overall N = 194 ¹	0 N = 97 ¹	1 N = 97 ¹	p-value ²
FUPM	27 (24-37)	30 (24-38)	25 (24-36)	0.048
LUTD				0.460
0	158 (81.4%)	81 (83.5%)	77 (79.4%)	
1	36 (18.6%)	16 (16.5%)	20 (20.6%)	
OBST				0.059
0	174 (89.7%)	91 (93.8%)	83 (85.6%)	
1	20 (10.3%)	6 (6.2%)	14 (14.4%)	
URCOMPINJ				0.621
0	190 (97.9%)	94 (96.9%)	96 (99.0%)	
1	4 (2.1%)	3 (3.1%)	1 (1.0%)	
SURG				0.749
0	140 (72.2%)	69 (71.1%)	71 (73.2%)	
1	54 (27.8%)	28 (28.9%)	26 (26.8%)	
ILEUS				<0.001
0	180 (92.8%)	84 (86.6%)	96 (99.0%)	
1	14 (7.2%)	13 (13.4%)	1 (1.0%)	
CLAVIENCLASS				0.013
1	93 (60.0%)	56 (70.9%)	37 (48.7%)	
2	42 (27.1%)	14 (17.7%)	28 (36.8%)	
3	19 (12.3%)	9 (11.4%)	10 (13.2%)	
4	1 (0.6%)	0 (0.0%)	1 (1.3%)	
Missing	39	18	21	
URCOMPVCUR				0.026
0	178 (97.3%)	84 (94.4%)	94 (100.0%)	
1	5 (2.7%)	5 (5.6%)	0 (0.0%)	
Missing	11	8	3	
GU30				<0.001
0	156 (80.4%)	63 (64.9%)	93 (95.9%)	
1	38 (19.6%)	34 (35.1%)	4 (4.1%)	
GU90				<0.001
0	149 (76.8%)	64 (66.0%)	85 (87.6%)	
1	45 (23.2%)	33 (34.0%)	12 (12.4%)	
GU90N				0.001
0	114 (58.8%)	47 (48.5%)	67 (69.1%)	
1	45 (23.2%)	33 (34.0%)	12 (12.4%)	
2	35 (18.0%)	17 (17.5%)	18 (18.6%)	
GUER30				<0.001
0	137 (70.6%)	46 (47.4%)	91 (93.8%)	
1	57 (29.4%)	51 (52.6%)	6 (6.2%)	
GUER90				<0.001
0	130 (67.0%)	44 (45.4%)	86 (88.7%)	
1	64 (33.0%)	53 (54.6%)	11 (11.3%)	
PROC365				>0.999
0	184 (94.8%)	92 (94.8%)	92 (94.8%)	
1	10 (5.2%)	5 (5.2%)	5 (5.2%)	
BLDLOSS	70 (40-100)	100 (65-110)	50 (40-80)	<0.001
Missing	3	1	2	
BLDSPASM				0.877
0	133 (68.6%)	66 (68.0%)	67 (69.1%)	
1	61 (31.4%)	31 (32.0%)	30 (30.9%)	

```

logrank_test <- survdiff(
  Surv(POSTUTIFUP, POSTUTI == 1) ~ TRT2,
  data = mdata
)

p_value <- 1 - pchisq(
  logrank_test$chisq,
  df = length(logrank_test$n) - 1
)

p_label <- paste0("Log-rank test p = ", format.pval(p_value, digits = 3, eps = 0.001))

p_u_psm <- ggsvplot(
  fit_psm,
  data = mdata,
  fun = "event",
  pval = p_label,
  pval.size = 4,
  pval.coord = c(0, 0.9), # c(x,y) according to my x and y scale
  conf.int = TRUE,
  risk.table = TRUE,
  risk.table.y.text = FALSE, # to remove text from y-axis in the risk table
  risk.table.fontsize = 4, # controls numbers in table
  xlim = c(0,135),
  break.time.by = 12,
  xlab = "Follow-up time",
  ylab = "Cumulative incidence of postoperative symptomatic UTI",
  palette = c("#ADD8E6", "#00008B"),
  legend.title = "Treatment",
  legend.labs = c("Cohen Group", "Lich-Gregoir Group"),
  ylim = c(0,1)
)

#-----
# Customize the plot
#-----
p_u_psm$plot <- p_u_psm$plot +
  scale_x_continuous(
    limits = c(0, 135),
    breaks = seq(0, 135, 12)
  ) +
  scale_y_continuous(
    labels = function(x) paste0(round(x * 100, 1), "%"),
    limits = c(0, 1)
  ) +
  theme_classic() + # Many clinical epidemiology journals prefer theme_classic() instead of theme_bw()
  theme(
    axis.title.x = element_text(size = 12),
    axis.title.y = element_text(size = 12),
    axis.text.x = element_text(size = 10),
    axis.text.y = element_text(size = 10),
    legend.title = element_text(size = 11),
    legend.text = element_text(size = 10)
  )

```


)

Scale for x is already present.

Adding another scale for x, which will replace the existing scale.

Scale for y is already present.

Adding another scale for y, which will replace the existing scale.

```
#-----
# Customize the risk table
#-----
p_u_psm$table <- p_u_psm$table +
  scale_x_continuous(
    limits = c(0, 135),
    breaks = seq(0, 135, 12)
  ) +
  theme(
    axis.title.x = element_blank(),
    axis.text.x  = element_blank(),
    axis.ticks.x = element_blank(),
    axis.line.x  = element_blank(),
    axis.line.y  = element_blank(),
    plot.title   = element_text(size = 12),
    text         = element_text(size = 10),
    axis.title.y = element_text(size = 12)
  )
```

Scale for x is already present.

Adding another scale for x, which will replace the existing scale.

```
#-----
# Combine plot and risk table
#-----
final_plot_psm <- p_u_psm$plot / p_u_psm$table +
  plot_layout(heights = c(4, 1))

#-----
# Save the figure
#-----
ggsave(
  filename = "figures/UTI_m.png",
  plot = final_plot_psm,
  width = 8,
  height = 6,
  dpi = 300
)
```

Ignoring unknown labels:

* colour : "Treatment"

```
# Estimated 1-year survival
s_psm <- summary(fit_psm, times = 12, extend = TRUE)
```

```
# Estimated 1-year cumulative probability of UTI
uti_1yr_psm <- data.frame(
  Treatment = s_psm$strata,
  UTI_1yr = 100 * (1 - s_psm$surv),
  Lower_95CI = 100 * (1 - s_psm$upper),
  Upper_95CI = 100 * (1 - s_psm$lower)
)
```

```
uti_1yr_psm
```

	Treatment	UTI_1yr	Lower_95CI	Upper_95CI
1	Treatment=Cohen	30.94322	21.09943	39.55889
2	Treatment=Lich-Gregoir	27.83505	18.34135	36.22501

3 Regression

Block 1 - without VUR recurrence

```
# GEE model after matching - Accounting for center-clustering
```

```
# Block 1: technique effect on postoperative outcomes = Univariable analyses
```

```
outcomes <- c(
  "LUTD",
  "OBST",
  "URCOMPINJ",
  "SURG",
  "ILEUS",
  "GU30",
  "GU90",
  "PROC365"
)
```

```
# convert outcomes to numeric
mdata[outcomes] <- lapply(
  mdata[outcomes],
  function(x) as.numeric(as.character(x))
)
```

```
# Apply GEE model after matching for all binary outcomes
gee_results_psm <- lapply(outcomes, function(y) {
  formula <- as.formula(
    paste(y, "~ TRT2")
  )
  model <- geeglm(
    formula,
    data = mdata,
    id = CENTER,
    weights = weights_psm,      # weights have no effect because weights are all equal
  )
})
```

```

    family = binomial(link = "logit"),
    corstr = "exchangeable"
  )
  tidy(model, exponentiate = TRUE, conf.int = TRUE)
})
names(gee_results_psm) <- outcomes

# Combine results into one table
gee_table_psm <- do.call(
  rbind,
  lapply(names(gee_results_psm), function(x){
    tmp <- gee_results_psm[[x]]
    tmp <- tmp %>%
      filter(grepl("^TRT2", term)) %>%
      mutate(outcome = x)
    tmp
  })
)

# Create final manuscript table
gee_table_final_psm <- gee_table_psm %>%
  mutate(
    Effect_measure = "OR",
    Estimate = sprintf("%.2f", estimate),
    CI = paste0(
      sprintf("%.2f", conf.low),
      "-",
      sprintf("%.2f", conf.high)
    ),
    P_value = p.value
  ) %>%
  dplyr::select(
    Outcome = outcome,
    Effect_measure,
    Estimate,
    CI,
    P_value
  )

# Cox model for POSTUTI
cox_uti_psm <- coxph(
  Surv(POSTUTIFUP, POSTUTI == 1) ~ TRT2 + strata(CENTER),
  data = mdata,
  weights = weights_psm, # weights have no effect
  robust = TRUE,
  cluster = subclass
)

cox_summarypsm <- summary(cox_uti_psm)

```

```

cox_row <- grep(
  "~TRT2",
  rownames(cox_summarypsm$coefficients)
)

POSTUTI_table_psm <- data.frame(
  Outcome = "POSTUTI",
  Effect_measure = "HR",
  Estimate = sprintf(
    "%.2f",
    cox_summarypsm$coefficients[
      cox_row,
      "exp(coef)"
    ]
  ),
  CI = paste0(
    sprintf("%.2f",
      cox_summarypsm$conf.int[cox_row, "lower .95"]),
    "-",
    sprintf("%.2f",
      cox_summarypsm$conf.int[cox_row, "upper .95"])
  ),
  P_value = cox_summarypsm$coefficients[
    cox_row,
    "Pr(>|z|)"
  ]
)

# Combine final results
final_resultspsm <- rbind(
  gee_table_final_psm,
  POSTUTI_table_psm
)

# Multiple correction

## Bonferroni correction
final_resultspsm$p_bonferroni <- p.adjust(
  final_resultspsm$P_value,
  method = "bonferroni"
)

## Holm-Bonferroni
final_resultspsm$p_holm <- p.adjust(
  final_resultspsm$P_value,
  method = "holm"
)

## Benjamini-Hochberg FDR
final_resultspsm$p_FDR <- p.adjust(
  final_resultspsm$P_value,
  method = "BH"
)

```

)

```
final_resultspsm <- final_resultspsm %>%
  mutate(
    across(
      c(P_value, p_bonferroni, p_holm, p_FDR),
      ~case_when(
        .x < 0.001 ~ "<0.001",
        TRUE ~ sprintf("%.3f", .x)
      )
    )
  )
```

final_resultspsm

A tibble: 9 x 8

	Outcome	Effect_measure	Estimate	CI	P_value	p_bonferroni	p_holm	p_FDR
	<chr>	<chr>	<chr>	<chr>	<chr>	<chr>	<chr>	<chr>
1	LUTD	OR	1.38	1.01-1.88	0.044	0.392	0.174	0.065
2	OBST	OR	3.76	2.02-7.00	<0.001	<0.001	<0.001	<0.001
3	URCOMPINJ	OR	0.40	0.28-0.56	<0.001	<0.001	<0.001	<0.001
4	SURG	OR	0.86	0.64-1.16	0.333	1.000	0.999	0.416
5	ILEUS	OR	0.04	0.02-0.10	<0.001	<0.001	<0.001	<0.001
6	GU30	OR	0.03	0.00-0.50	0.015	0.138	0.077	0.028
7	GU90	OR	0.22	0.09-0.51	<0.001	0.004	0.003	0.001
8	PROC365	OR	1.00	0.59-1.68	0.986	1.000	0.999	0.986
9	POSTUTI	HR	0.80	0.48-1.31	0.370	1.000	0.999	0.416

```
final_resultspsm %>%
  flextable() %>%
  theme_booktabs() %>%
  autofit() %>%
  align(
    j = c("Estimate", "CI", "P_value"),
    align = "center",
    part = "all"
  ) %>%
  save_as_docx(
    path = "output/treatment_unadj_m_norec.docx"
  )
```

Block 2 - without VUR recurrence

Block 2: exposure-effect analysis (stent) on postoperative outcomes = Multivariable analyses

```
outcomes <- c(
  "LUTD",
  "OBST",
  "URCOMPINJ",
  "SURG",
  "ILEUS",
```

```

"GU30",
"GU90",
"PROC365"
)

# convert outcomes to numeric
mdata[outcomes] <- lapply(
  mdata[outcomes],
  function(x) as.numeric(as.character(x))
)

# Predictors to include in the outcome models
predictors <- c("TRT2", "STENT")

# Apply GEE model after overlap weighting for all binary outcomes
gee_results_psm2 <- lapply(outcomes, function(y) {
  formula <- as.formula(
    paste(y, "~",
          paste(predictors, collapse = " + ")
    )
  )
  model <- geeglm(
    formula,
    data = mdata,
    id = CENTER,
    weights = weights_psm, # weights have no effect because weights are all equal
    family = binomial(link = "logit"),
    corstr = "exchangeable"
  )
  tidy(model, exponentiate = TRUE, conf.int = TRUE)
})
names(gee_results_psm2) <- outcomes

# Combine results into one table
gee_table_psm2 <- do.call(
  rbind,
  lapply(names(gee_results_psm2), function(x){
    tmp <- gee_results_psm2[[x]]
    tmp %>%
      mutate(outcome = x)
  })
)

# Create final manuscript table
gee_table_final_psm2 <- gee_table_psm2 %>%
  mutate(
    Effect_measure = "OR",
    Estimate = sprintf("%.2f", estimate),
    CI = paste0(
      sprintf("%.2f", conf.low),
      "-",
      sprintf("%.2f", conf.high)
    )
  )

```

```

    ),
    P_value = p.value
  ) %>%
  dplyr::select(
    Outcome = outcome,
    Predictor = term,
    Effect_measure,
    Estimate,
    CI,
    P_value
  )

# Cox model for POSTUTI

cox_uti_psm2 <- coxph(
  Surv(POSTUTIFUP, POSTUTI == 1) ~ TRT2 + STENT + strata(CENTER),
  data = mdata,
  weights = weights_psm, # weights have no effect
  robust = TRUE,
  cluster = subclass
)

cox_summarypsm2 <- summary(cox_uti_psm2)

cox_row <- rownames(cox_summarypsm2$coefficients)

POSTUTI_table_psm2 <- data.frame(
  Outcome = "POSTUTI",
  Predictor = rownames(cox_summarypsm2$coefficients),
  Effect_measure = "HR",
  Estimate = sprintf(
    "%.2f",
    cox_summarypsm2$coefficients[
      cox_row,
      "exp(coef)"
    ]
  ),
  CI = paste0(
    sprintf("%.2f",
      cox_summarypsm2$conf.int[cox_row, "lower .95"]),
    "-",
    sprintf("%.2f",
      cox_summarypsm2$conf.int[cox_row, "upper .95"])
  ),
  P_value = cox_summarypsm2$coefficients[
    cox_row,
    "Pr(>|z|)"
  ]
)

# Combine final results

```

```

final_results_psm2 <- rbind(
  gee_table_final_psm2,
  POSTUTI_table_psm2
)

final_results_psm2

# A tibble: 26 x 6
  Outcome Predictor Effect_measure Estimate CI P_value
* <chr> <chr> <chr> <chr> <chr> <dbl>
1 LUTD (Intercept) OR 0.35 0.29-0.42 0
2 LUTD TRT21 OR 1.15 0.91-1.46 0.245
3 LUTD STENT1 OR 0.23 0.06-0.92 0.0371
4 OBST (Intercept) OR 0.02 0.00-0.08 0.00000127
5 OBST TRT21 OR 3.90 2.49-6.09 0.0000000236
6 OBST STENT1 OR 1.84 0.91-3.73 0.0904
7 URCOMPINJ (Intercept) OR 0.02 0.01-0.03 0
8 URCOMPINJ TRT21 OR 0.41 0.28-0.60 0.00000394
9 URCOMPINJ STENT1 OR 2.94 0.37-23.06 0.305
10 SURG (Intercept) OR 0.84 0.71-0.99 0.0323
# i 16 more rows

final_results_psm2 <- final_results_psm2 %>%
  mutate(
    across(
      c(P_value),
      ~case_when(
        .x < 0.001 ~ "<0.001",
        TRUE ~ sprintf("%.3f", .x)
      )
    )
  )

final_results_psm2 %>%
  flextable() %>%
  theme_booktabs() %>%
  autofit() %>%
  align(
    j = c("Estimate", "CI", "P_value"),
    align = "center",
    part = "all"
  ) %>%
  save_as_docx(
    path = "output/treatment_adj_all_m_norec.docx"
  )

# Multiple correction

# Extract TRT2 rows only

```



```

gee_trt2_psm2 <- gee_table_final_psm2 %>%
  filter(grepl("^TRT2", Predictor))

POSTUTI_trt2_psm2 <- POSTUTI_table_psm2 %>%
  filter(grepl("^TRT2", Predictor))

# Combine only treatment effects
treatment_results_psm2 <- rbind(
  gee_trt2_psm2,
  POSTUTI_trt2_psm2
)

# Apply Correction

treatment_results_psm2$p_bonferroni <- p.adjust(
  as.numeric(treatment_results_psm2$P_value),
  method = "bonferroni"
) # Bonferroni correction

treatment_results_psm2$p_holm <- p.adjust(
  as.numeric(treatment_results_psm2$P_value),
  method = "holm"
) # Holm-Bonferroni

treatment_results_psm2$p_FDR <- p.adjust(
  as.numeric(treatment_results_psm2$P_value),
  method = "BH"
) # Benjamini-Hochberg FDR

treatment_results_psm2 <- treatment_results_psm2 %>%
  mutate(
    across(
      c(P_value, p_bonferroni, p_holm, p_FDR),
      ~case_when(
        .x < 0.001 ~ "<0.001",
        TRUE ~ sprintf("%.3f", .x)
      )
    )
  )

treatment_results_psm2

# A tibble: 9 x 9
  Outcome Predictor Effect_measure Estimate CI P_value p_bonferroni p_holm
  <chr> <chr> <chr> <chr> <chr> <chr> <chr>
1 LUTD TRT21 OR 1.15 0.91-- 0.245 1.000 0.734
2 OBST TRT21 OR 3.90 2.49-- <0.001 <0.001 <0.001
3 URCOMPINJ TRT21 OR 0.41 0.28-- <0.001 <0.001 <0.001
4 SURG TRT21 OR 0.74 0.60-- 0.006 0.052 0.023
5 ILEUS TRT21 OR 0.08 0.06-- <0.001 <0.001 <0.001
6 GU30 TRT21 OR 0.06 0.02-- <0.001 <0.001 <0.001
7 GU90 TRT21 OR 0.19 0.09-- <0.001 <0.001 <0.001

```

8	PROC365	TRT21	OR	0.95	0.62--	0.828	1.000	0.828
9	POSTUTI	TRT21	HR	0.78	0.48--	0.329	1.000	0.734

i 1 more variable: p_FDR <chr>

```

treatment_results_psm2 %>%
  flextable() %>%
  theme_booktabs() %>%
  autofit() %>%
  align(
    j = c("Estimate", "CI", "P_value"),
    align = "center",
    part = "all"
  ) %>%
  save_as_docx(
    path = "output/treatment_adj_m_norec.docx"
  )

# ADDITIONAL MULTIPLE COMPARISONS

# Extract STENT rows only

gee_st_psm2 <- gee_table_final_psm2 %>%
  filter(grepl("^STENT", Predictor))

POSTUTI_st_psm2 <- POSTUTI_table_psm2 %>%
  filter(grepl("^STENT", Predictor))

# Combine only stent effects
st_results_psm2 <- rbind(
  gee_st_psm2,
  POSTUTI_st_psm2
)

# Apply Correction

st_results_psm2$p_bonferroni <- p.adjust(
  as.numeric(st_results_psm2$P_value),
  method = "bonferroni"
) # Bonferroni correction

st_results_psm2$p_holm <- p.adjust(
  as.numeric(st_results_psm2$P_value),
  method = "holm"
) # Holm-Bonferroni

st_results_psm2$p_FDR <- p.adjust(
  as.numeric(st_results_psm2$P_value),
  method = "BH"
) # Benjamini-Hochberg FDR

st_results_psm2 <- st_results_psm2 %>%
  mutate(

```

```

    across(
      c(P_value, p_bonferroni, p_holm, p_FDR),
      ~case_when(
        .x < 0.001 ~ "<0.001",
        TRUE ~ sprintf("%.3f", .x)
      )
    )
  )
)

st_results_psm2

# A tibble: 9 x 9
  Outcome Predictor Effect_measure Estimate CI      P_value p_bonferroni p_holm
  <chr>      <chr>      <chr>      <chr> <chr> <chr>      <chr>      <chr>
1 LUTD      STENT1      OR          0.23  0.06-- 0.037    0.334    0.223
2 OBST      STENT1      OR          1.84  0.91-- 0.090    0.814    0.362
3 URCOMPINJ STENT1      OR          2.94  0.37-- 0.305    1.000    0.916
4 SURG      STENT1      OR          0.11  0.03-- 0.002    0.015    0.015
5 ILEUS     STENT1      OR          0.09  0.02-- 0.005    0.044    0.039
6 GU30      STENT1      OR          0.22  0.07-- 0.008    0.070    0.054
7 GU90      STENT1      OR          0.27  0.07-- 0.052    0.468    0.260
8 PROC365   STENT1      OR          0.65  0.09-- 0.677    1.000    1.000
9 POSTUTI   STENT1      HR          0.79  0.33-- 0.601    1.000    1.000
# i 1 more variable: p_FDR <chr>

st_results_psm2 %>%
  flextable() %>%
  theme_booktabs() %>%
  autofit() %>%
  align(
    j = c("Estimate", "CI", "P_value"),
    align = "center",
    part = "all"
  ) %>%
  save_as_docx(
    path = "output/stent_adj_m_norec.docx"
  )

```

Block 2i1 - without VUR recurrence

Block 2: exposure-effect analysis (stent) on postoperative outcomes = Multivariable analyses

```

outcomes <- c(
  "LUTD",
  "OBST",
  "URCOMPINJ",
  "SURG",
  "ILEUS",
  "GU30",
  "GU90",
  "PROC365"
)

```

```

# convert outcomes to numeric
mdata[outcomes] <- lapply(
  mdata[outcomes],
  function(x) as.numeric(as.character(x))
)

# Predictors to include in the outcome models
predictors <- "TRT2 * STENT"

# Apply GEE model after overlap weighting for all binary outcomes
gee_results_psm2 <- lapply(outcomes, function(y) {
  formula <- as.formula(
    paste(y, "~",
          paste(predictors, collapse = " + ")
    )
  )
  model <- geeglm(
    formula,
    data = mdata,
    id = CENTER,
    weights = weights_psm, # weights have no effect because weights are all equal
    family = binomial(link = "logit"),
    corstr = "exchangeable"
  )
  tidy(model, exponentiate = TRUE, conf.int = TRUE) # estimate = OR because exponentiate = TRUE
})
names(gee_results_psm2) <- outcomes

# Combine results into one table
gee_table_psm2 <- do.call(
  rbind,
  lapply(names(gee_results_psm2), function(x){
    tmp <- gee_results_psm2[[x]]
    tmp %>%
      mutate(outcome = x)
  })
)

# Create final manuscript table
gee_table_final_psm2 <- gee_table_psm2 %>%
  mutate(
    Effect_measure = "OR",
    Estimate = sprintf("%.2f", estimate),
    CI = paste0(
      sprintf("%.2f", conf.low),
      "-",
      sprintf("%.2f", conf.high)
    ),
    P_value = p.value
  ) %>%
  dplyr::select(
    Outcome = outcome,

```

```

    Predictor = term,
    Effect_measure,
    Estimate,
    CI,
    P_value
  )

# Cox model for POSTUTI

cox_uti_psm2 <- coxph(
  Surv(POSTUTIFUP, POSTUTI == 1) ~ TRT2*STENT + strata(CENTER),
  data = mdata,
  weights = weights_psm, # weights have no effect
  robust = TRUE,
  cluster = subclass
)

cox_summarypsm2 <- summary(cox_uti_psm2)

cox_row <- rownames(cox_summarypsm2$coefficients)

POSTUTI_table_psm2 <- data.frame(
  Outcome = "POSTUTI",
  Predictor = rownames(cox_summarypsm2$coefficients),
  Effect_measure = "HR",
  Estimate = sprintf(
    "%.2f",
    cox_summarypsm2$coefficients[
      cox_row,
      "exp(coef)"
    ]
  ),
  CI = paste0(
    sprintf("%.2f",
      cox_summarypsm2$conf.int[cox_row, "lower .95"]),
    "-",
    sprintf("%.2f",
      cox_summarypsm2$conf.int[cox_row, "upper .95"])
  ),
  P_value = cox_summarypsm2$coefficients[
    cox_row,
    "Pr(>|z|)"
  ]
)

# Combine final results
final_results_psm2 <- rbind(
  gee_table_final_psm2,
  POSTUTI_table_psm2
)

```

```
final_results_psm2
```

```
# A tibble: 35 x 6
```

	Outcome	Predictor	Effect_measure	Estimate	CI	P_value
*	<chr>	<chr>	<chr>	<chr>	<chr>	<dbl>
1	LUTD	(Intercept)	OR	0.34	0.26-0.45	1.53e-14
2	LUTD	TRT21	OR	1.22	0.86-1.73	2.71e- 1
3	LUTD	STENT1	OR	0.27	0.04-1.78	1.73e- 1
4	LUTD	TRT21:STENT1	OR	0.70	0.20-2.48	5.84e- 1
5	OBST	(Intercept)	OR	0.01	0.00-0.05	1.50e-12
6	OBST	TRT21	OR	4.16	2.19-7.91	1.32e- 5
7	OBST	STENT1	OR	2.77	1.10-6.93	2.99e- 2
8	OBST	TRT21:STENT1	OR	0.49	0.23-1.05	6.73e- 2
9	URCOMPINJ	(Intercept)	OR	0.01	0.01-0.02	0
10	URCOMPINJ	TRT21	OR	0.00	0.00-0.00	0

```
# i 25 more rows
```

```
final_results_psm2 <- final_results_psm2 %>%
```

```
  mutate(
    across(
      c(P_value),
      ~case_when(
        .x < 0.001 ~ "<0.001",
        TRUE ~ sprintf("%.3f", .x)
      )
    )
  )
```

```
final_results_psm2 %>%
```

```
  flextable() %>%
  theme_booktabs() %>%
  autofit() %>%
  align(
    j = c("Estimate", "CI", "P_value"),
    align = "center",
    part = "all"
  ) %>%
  save_as_docx(
    path = "output/treatment_adj_all_m_norec_i.docx"
  )
```

```
# Multiple correction
```

```
# Extract TRT2 rows only
```

```
gee_trt2_psm2 <- gee_table_final_psm2 %>%
```

```
  filter(grepl("^TRT2", Predictor))
```

```
POSTUTI_trt2_psm2 <- POSTUTI_table_psm2 %>%
```

```
  filter(grepl("^TRT2", Predictor))
```

```

# Combine only treatment effects
treatment_results_psm2 <- rbind(
  gee_trt2_psm2,
  POSTUTI_trt2_psm2
)

# Apply Correction

treatment_results_psm2$p_bonferroni <- p.adjust(
  as.numeric(treatment_results_psm2$P_value),
  method = "bonferroni"
) # Bonferroni correction

treatment_results_psm2$p_holm <- p.adjust(
  as.numeric(treatment_results_psm2$P_value),
  method = "holm"
) # Holm-Bonferroni

treatment_results_psm2$p_FDR <- p.adjust(
  as.numeric(treatment_results_psm2$P_value),
  method = "BH"
) # Benjamini-Hochberg FDR

treatment_results_psm2 <- treatment_results_psm2 %>%
  mutate(
    across(
      c(P_value, p_bonferroni, p_holm, p_FDR),
      ~case_when(
        .x < 0.001 ~ "<0.001",
        TRUE ~ sprintf("%.3f", .x)
      )
    )
  )

treatment_results_psm2

# A tibble: 18 x 9
  Outcome Predictor Effect_measure Estimate CI P_value p_bonferroni p_holm
  <chr> <chr> <chr> <chr> <chr> <chr> <chr>
1 LUTD TRT21 OR 1.22 0.86~ 0.271 1.000 1.000
2 LUTD TRT21:ST~ OR 0.70 0.20~ 0.584 1.000 1.000
3 OBST TRT21 OR 4.16 2.19~ <0.001 <0.001 <0.001
4 OBST TRT21:ST~ OR 0.49 0.23~ 0.067 1.000 0.605
5 URCOMPINJ TRT21 OR 0.00 0.00~ <0.001 <0.001 <0.001
6 URCOMPINJ TRT21:ST~ OR Inf Inf~ <0.001 <0.001 <0.001
7 SURG TRT21 OR 0.94 0.77~ 0.550 1.000 1.000
8 SURG TRT21:ST~ OR 0.00 0.00~ 0.420 1.000 1.000
9 ILEUS TRT21 OR 0.08 0.06~ <0.001 <0.001 <0.001
10 ILEUS TRT21:ST~ OR 0.00 0.00~ <0.001 <0.001 <0.001
11 GU30 TRT21 OR 0.02 0.00~ 0.096 1.000 0.772
12 GU30 TRT21:ST~ OR 0.00 0.00~ <0.001 <0.001 <0.001
13 GU90 TRT21 OR 0.17 0.08~ <0.001 <0.001 <0.001

```

14	GU90	TRT21:ST~	OR	2.53	1.35~	0.004	0.072	0.044
15	PROC365	TRT21	OR	0.65	0.33~	0.201	1.000	1.000
16	PROC365	TRT21:ST~	OR	4.22	1.40~	0.010	0.186	0.103
17	POSTUTI	TRT21	HR	0.85	0.50~	0.562	1.000	1.000
18	POSTUTI	TRT21:ST~	HR	0.35	0.04~	0.334	1.000	1.000

i 1 more variable: p_FDR <chr>

```
treatment_results_psm2 %>%
  flextable() %>%
  theme_booktabs() %>%
  autofit() %>%
  align(
    j = c("Estimate", "CI", "P_value"),
    align = "center",
    part = "all"
  ) %>%
  save_as_docx(
    path = "output/treatment_adj_m_norec_i.docx"
  )
```

ADDITIONAL MULTIPLE COMPARISONS

Extract STENT rows only

```
gee_st_psm2 <- gee_table_final_psm2 %>%
  filter(grepl("^STENT", Predictor))
```

```
POSTUTI_st_psm2 <- POSTUTI_table_psm2 %>%
  filter(grepl("^STENT", Predictor))
```

Combine only stent effects

```
st_results_psm2 <- rbind(
  gee_st_psm2,
  POSTUTI_st_psm2
)
```

Apply Correction

```
st_results_psm2$p_bonferroni <- p.adjust(
  as.numeric(st_results_psm2$P_value),
  method = "bonferroni"
) # Bonferroni correction
```

```
st_results_psm2$p_holm <- p.adjust(
  as.numeric(st_results_psm2$P_value),
  method = "holm"
) # Holm-Bonferroni
```

```
st_results_psm2$p_FDR <- p.adjust(
  as.numeric(st_results_psm2$P_value),
  method = "BH"
) # Benjamini-Hochberg FDR
```



```

st_results_psm2 <- st_results_psm2 %>%
  mutate(
    across(
      c(P_value, p_bonferroni, p_holm, p_FDR),
      ~case_when(
        .x < 0.001 ~ "<0.001",
        TRUE ~ sprintf("%.3f", .x)
      )
    )
  )

st_results_psm2

# A tibble: 9 x 9
  Outcome Predictor Effect_measure Estimate CI P_value p_bonferroni p_holm
  <chr> <chr> <chr> <chr> <chr> <chr> <chr> <chr>
1 LUTD STENT1 OR 0.27 0.04-- 0.173 1.000 0.503
2 OBST STENT1 OR 2.77 1.10-- 0.030 0.269 0.149
3 URCOMPINJ STENT1 OR 4.39 1.11-- 0.035 0.318 0.149
4 SURG STENT1 OR 0.22 0.06-- 0.021 0.191 0.148
5 ILEUS STENT1 OR 0.10 0.02-- 0.005 0.048 0.048
6 GU30 STENT1 OR 0.21 0.06-- 0.012 0.112 0.100
7 GU90 STENT1 OR 0.22 0.06-- 0.021 0.191 0.148
8 PROC365 STENT1 OR 0.32 0.06-- 0.168 1.000 0.503
9 POSTUTI STENT1 HR 1.03 0.41-- 0.947 1.000 0.947
# i 1 more variable: p_FDR <chr>

st_results_psm2 %>%
  flextable() %>%
  theme_booktabs() %>%
  autofit() %>%
  align(
    j = c("Estimate", "CI", "P_value"),
    align = "center",
    part = "all"
  ) %>%
  save_as_docx(
    path = "output/stent_adj_m_norec_i.docx"
  )

```

Block 2i2 - without VUR recurrence + with interaction (to calculate OR for two levels)
(secondary effect-modification analysis)

Block 2: exposure-effect analysis (stent) on postoperative outcomes = Multivariable analyses

```

outcomes <- c(
  "LUTD",
  "OBST",
  "URCOMPINJ",
  "SURG",
  "ILEUS",

```

```

"GU30",
"GU90",
"PROC365"
)

# convert outcomes to numeric
mdata[outcomes] <- lapply(
  mdata[outcomes],
  function(x) as.numeric(as.character(x))
)

# Apply GEE model after overlap weighting for all binary outcomes
get_gee_results_psm2 <- function(y) {

  formula <- as.formula(
    paste(y, "~ TRT2 * STENT")
  )

  model <- geeglm(
    formula,
    data = mdata,
    id = CENTER,
    weights = weights_psm, # weights have no effect because weights are all equal
    family = binomial(link = "logit"),
    corstr = "exchangeable"
  )

  # Coefficients
  beta <- coef(model)

  # Robust covariance matrix
  V <- vcov(model)

  # -----
  # LG vs Cohen among NO-STENT patients
  # -----

  b_no_stent <- beta["TRT21"]

  se_no_stent <- sqrt(
    V["TRT21", "TRT21"]
  )

  OR_no_stent <- exp(b_no_stent)

  CI_no_stent <- exp(
    b_no_stent + c(-1, 1) * 1.96 * se_no_stent
  )

  # -----
  # LG vs Cohen among STENTED patients
  # = TRT21 + TRT21:STENT1

```

```

# -----

b_stent <- beta["TRT21"] +
  beta["TRT21:STENT1"]

# Variance of the linear combination:
# Var(beta1 + beta3)

var_stent <-
  V["TRT21", "TRT21"] +
  V["TRT21:STENT1", "TRT21:STENT1"] +
  2 * V["TRT21", "TRT21:STENT1"]

se_stent <- sqrt(var_stent)

OR_stent <- exp(b_stent)

CI_stent <- exp(
  b_stent + c(-1, 1) * 1.96 * se_stent
)

# -----
# Interaction test
# -----

b_interaction <- beta["TRT21:STENT1"]

se_interaction <- sqrt(
  V["TRT21:STENT1", "TRT21:STENT1"]
)

z_interaction <- b_interaction / se_interaction

p_interaction <- 2 * pnorm(
  abs(z_interaction),
  lower.tail = FALSE
)

# -----
# Return manuscript-ready results
# -----

data.frame(
  Outcome = y,
  `LG vs Cohen, No Stent` =
    sprintf(
      "%.2f (%.2f-%.2f)",
      OR_no_stent,
      CI_no_stent[1],
      CI_no_stent[2]
    ),

```

```

`LG vs Cohen, Stent` =
  sprintf(
    "%.2f (%.2f-%.2f)",
    OR_stent,
    CI_stent[1],
    CI_stent[2]
  ),

  P_interaction = p_interaction,

  stringsAsFactors = FALSE
)
}

# Run for all 8 outcomes
interaction_results_m <- do.call(
  rbind,
  lapply(outcomes, get_gee_results_psm2)
)

```

Warning in sqrt(var_stent): NaNs produced

interaction_results_m

	Outcome	LG.vs.Cohen..No.Stent	LG.vs.Cohen..Stent
TRT21:STENT1	LUTD	1.22 (0.86-1.73)	0.86 (0.31-2.37)
TRT21:STENT11	OBST	4.16 (2.19-7.91)	2.03 (1.41-2.92)
TRT21:STENT12	URCOMPINJ	0.00 (0.00-0.00)	0.70 (NaN-NaN)
TRT21:STENT13	SURG	0.94 (0.77-1.15)	0.00 (0.00-50792788.77)
TRT21:STENT14	ILEUS	0.08 (0.06-0.10)	0.00 (0.00-0.00)
TRT21:STENT15	GU30	0.02 (0.00-2.03)	0.00 (0.00-0.00)
TRT21:STENT16	GU90	0.17 (0.08-0.35)	0.44 (0.36-0.53)
TRT21:STENT17	PROC365	0.65 (0.33-1.26)	2.72 (1.28-5.78)
	P_interaction		
TRT21:STENT1		5.843082e-01	
TRT21:STENT11		6.726598e-02	
TRT21:STENT12		0.000000e+00	
TRT21:STENT13		4.197991e-01	
TRT21:STENT14		5.005891e-07	
TRT21:STENT15		9.980972e-24	
TRT21:STENT16		3.983401e-03	
TRT21:STENT17		1.032062e-02	

Cox model for POSTUTI

```

cox_uti_psm2 <- coxph(
  Surv(POSTUTIFUP, POSTUTI == 1) ~ TRT2*STENT + strata(CENTER),
  data = mdata,
  weights = weights_psm, # weights have no effect
  robust = TRUE,
  cluster = subclass
)

```

```

)

# Coefficients
beta_cox <- coef(cox_uti_psm2)

# Covariance matrix
V_cox <- vcov(cox_uti_psm2)

# -----
# LG vs Cohen among NO-STENT patients
# -----

b_no_stent_cox <- beta_cox["TRT21"]

se_no_stent_cox <- sqrt(
  V_cox["TRT21", "TRT21"]
)

HR_no_stent <- exp(b_no_stent_cox)

CI_no_stent_cox <- exp(
  b_no_stent_cox +
  c(-1, 1) * 1.96 * se_no_stent_cox
)

# -----
# LG vs Cohen among STENTED patients
# = TRT21 + TRT21:STENT1
# -----

b_stent_cox <- beta_cox["TRT21"] +
  beta_cox["TRT21:STENT1"]

var_stent_cox <-
  V_cox["TRT21", "TRT21"] +
  V_cox["TRT21:STENT1", "TRT21:STENT1"] +
  2 * V_cox["TRT21", "TRT21:STENT1"]

se_stent_cox <- sqrt(var_stent_cox)

HR_stent <- exp(b_stent_cox)

CI_stent_cox <- exp(
  b_stent_cox +
  c(-1, 1) * 1.96 * se_stent_cox
)

# -----
# Interaction p-value
# -----

```

```

b_interaction_cox <- beta_cox["TRT21:STENT1"]

se_interaction_cox <- sqrt(
  V_cox["TRT21:STENT1", "TRT21:STENT1"]
)

z_interaction_cox <-
  b_interaction_cox / se_interaction_cox

p_interaction_cox <- 2 * pnorm(
  abs(z_interaction_cox),
  lower.tail = FALSE
)

# Create POSTUTI result
POSTUTI_interaction_m <- data.frame(
  Outcome = "POSTUTI",

  `LG vs Cohen, No Stent` =
    sprintf(
      "%.2f (%.2f-%.2f)",
      HR_no_stent,
      CI_no_stent_cox[1],
      CI_no_stent_cox[2]
    ),

  `LG vs Cohen, Stent` =
    sprintf(
      "%.2f (%.2f-%.2f)",
      HR_stent,
      CI_stent_cox[1],
      CI_stent_cox[2]
    ),

  P_interaction = p_interaction_cox
)

POSTUTI_interaction_m

      Outcome LG.vs.Cohen..No.Stent LG.vs.Cohen..Stent P_interaction
TRT21:STENT1 POSTUTI           0.85 (0.50-1.45)    0.30 (0.04-2.36)    0.3337667

```

4 Rosenbaum's sensitivity analyses

```

# Rosenbaum's primal sensitivity analysis

outcomes <- c(
  "LUTD",
  "OBST",
  "URCOMPINJ",

```

```

    "SURG",
    "ILEUS",
    "GU30",
    "GU90",
    "PROC365"
  )

# Convert outcomes to numeric
mdata[outcomes] <- lapply(
  mdata[outcomes],
  function(x) as.numeric(as.character(x))
)

# Function for Rosenbaum sensitivity analysis using rbounds
rosenbaum_binary <- function(outcome, data, match_object,
                             gamma_seq = seq(1, 11, by = 0.1),
                             alpha = 0.05) {

  # matched pairs
  treated_ids <- rownames(match_object$match.matrix)
  control_ids <- match_object$match.matrix[, 1]

  pairs <- data.frame(
    treated = data[treated_ids, outcome],
    control = data[control_ids, outcome]
  )

  # remove missing outcome pairs
  pairs <- na.omit(pairs)

  # discordant pairs
  # x = control has outcome, treated does not
  # y = treated has outcome, control does not
  x <- sum(pairs$control == 1 & pairs$treated == 0)
  y <- sum(pairs$treated == 1 & pairs$control == 0)

  # run Rosenbaum sensitivity
  if ((x + y) == 0) {
    return(data.frame(
      Outcome = outcome,
      Discordant_pairs = 0,
      x = x,
      y = y,
      p_Gamma1 = NA,
      gamma_max = NA,
      p_gamma_max = NA
    ))
  }

  # Run sensitivity over Gamma values
  gamma_results <- sapply(gamma_seq, function(g) {

```

```

res <- binarysens(
  x = x,
  y = y,
  Gamma = g,
  GammaInc = 0.1 # GammaInc defines the increment used to increase  $\Gamma$  values when the function eval
)

# extract upper bound p-value corresponding to Gamma g
res$bounds$`Upper bound`[
  res$bounds$Gamma == g
]
})

# p-value at Gamma = 1
p_gamma1 <- gamma_results[gamma_seq == 1]

# first Gamma where upper p-value >= alpha
idx <- which(gamma_results < alpha)

if(length(idx) == 0){

  if(p_gamma1 >= alpha){

    gamma_max <- "NS at Gamma=1"
    p_gamma_max <- NA

  } else {

    gamma_max <- paste0(">=", max(gamma_seq))
    p_gamma_max <- gamma_results[length(gamma_results)]

  }

} else {

  gamma_max_value <- gamma_seq[max(idx)]

  # If still significant at the maximum tested Gamma
  if(gamma_max_value == max(gamma_seq)){

    gamma_max <- paste0(">=", max(gamma_seq))

  } else {

    gamma_max <- gamma_max_value

  }

  p_gamma_max <- gamma_results[max(idx)]
}

# Extract p-values from rbounds object
data.frame(

```



```

    Outcome = outcome,
    Discordant_pairs = x + y,
    x = x,
    y = y,
    p_Gamma1 = round(p_gamma1, 3),
    gamma_max = gamma_max,
    p_gamma_max = round(p_gamma_max, 3)
  )
}

```

```

# Run all outcomes
rosenbaum_results <- do.call(
  rbind,
  lapply(
    outcomes,
    rosenbaum_binary,
    data = mdata,
    match_object = m.out
  )
)

```

```
rosenbaum_results
```

	Outcome	Discordant_pairs	x	y	p_Gamma1	gamma_max	p_gamma_max
1	LUTD	28 12 16	0.172	NS	at Gamma=1	NA	
2	OBST	18 5 13	0.015		1.2	0.038	
3	URCOMPINJ	4 3 1	0.062	NS	at Gamma=1	NA	
4	SURG	42 22 20	0.322	NS	at Gamma=1	NA	
5	ILEUS	12 12 0	0.000		>=11	0.000	
6	GU30	32 31 1	0.000		10.1	0.049	
7	GU90	33 27 6	0.000		2.4	0.048	
8	PROC365	10 5 5	0.377	NS	at Gamma=1	NA	

```

# Rename columns for publication
rosenbaum_results_pub <- rosenbaum_results %>%
  rename(
    Outcome = Outcome,
    `Discordant pairs` = Discordant_pairs,
    `Control only events (x)` = x,
    `Treated only events (y)` = y,
    `P-value (Γ=1)` = p_Gamma1,
    `Γmax` = gamma_max,
    `P-value at Γmax` = p_gamma_max
  )

```

```

# Create publication table
rosenbaum_table <- rosenbaum_results_pub %>%
  flextable() %>%

# Add title
add_header_lines(

```

```

    "Rosenbaum's primal sensitivity analysis for postoperative outcomes in the matched cohort."
  ) %>%

# Add footnote
add_footer_lines(
  "The Gamma value  $\Gamma$  is a measure of sensitivity to hidden bias, representing the magnitude of differ
) %>%

theme_booktabs() %>%
autofit() %>%

align(
  j = c(
    "Discordant pairs",
    "Control only events (x)",
    "Treated only events (y)",
    "P-value ( $\Gamma=1$ )",
    " $\Gamma_{max}$ ",
    "P-value at  $\Gamma_{max}$ "
  ),
  align = "center",
  part = "all"
)

# Save as Word document
save_as_docx(
  rosenbaum_table,
  path = "output/Rosenbaum_effect.docx"
)

```

5 PAIN - Non-epidural Cohort

5.1 Filter & match the new population

```

# Filter dataset to only those who did not receive EPIDURAL
Outp <- Out %>%
  filter(EPIDURAL == "0") %>%
  mutate(
    PAIN = factor(
      PAIN,
      levels = c("0", "1", "2", "3"),
      labels = c(
        "No pain",
        "Mild pain (score 1-3)",
        "Moderate pain (score 4-6)",
        "Severe pain (score 7-10)"
      ),
      ordered = TRUE
    )
  )

# Do matching

```

```

set.seed(123)

m.outp <- matchit(
  TRT2 ~ AGS + YEARS + GENDER + BMI + PREOPRNPLN + HISTOINJ +
    SURGIND2 + PREOPVURGRADE + VURSIDE + UDR7M + VURPHASE +
    PREOPEPISODE1 + PREOPTYPEU1 + PREOPRNSCR + SURGTECH2,
  data = Outp,
  method = "nearest",
  ratio = 1, # 1:1 matching (1 treated : 1 control)
  distance = "logit",
  caliper = 0.2
)

```

Warning: Fewer control units than treated units; not all treated units will get a match.

```

mdatap <- match.data(
  m.outp,
  weights = "weightsp_psm"
)
readr::write_csv(mdatap, "data/matched_nonepidural_data.csv")

```

```

# Table for balance
# This gives: SMD before and after matching, and flagged imbalance (>0.1)
balmp <- cobalt::bal.tab(m.outp, un = TRUE, ESS = TRUE, m.threshold = 0.1)
balmp

```

Balance Measures

	Type	Diff.Un	Diff.Adj	M.Threshold
distance	Distance	1.6163	0.1223	
AGS	Contin.	0.4157	-0.1366	Not Balanced, >0.1
YEARS	Contin.	-0.5177	-0.0743	Balanced, <0.1
GENDER	Binary	-0.0397	0.0000	Balanced, <0.1
BMI	Contin.	-0.9098	0.0991	Balanced, <0.1
PREOPRNPLN	Binary	-0.2680	-0.0222	Balanced, <0.1
HISTOINJ	Binary	0.0111	-0.0111	Balanced, <0.1
SURGIND2_1	Binary	0.0673	-0.0222	Balanced, <0.1
SURGIND2_2	Binary	0.0864	-0.0111	Balanced, <0.1
SURGIND2_3	Binary	-0.0123	0.0444	Balanced, <0.1
SURGIND2_4	Binary	-0.0520	0.0000	Balanced, <0.1
SURGIND2_5	Binary	-0.0894	-0.0111	Balanced, <0.1
PREOPVURGRADE_3	Binary	-0.0179	0.0111	Balanced, <0.1
PREOPVURGRADE_4	Binary	0.0999	0.0222	Balanced, <0.1
PREOPVURGRADE_5	Binary	-0.0820	-0.0333	Balanced, <0.1
VURSIDE	Binary	0.0233	-0.0111	Balanced, <0.1
UDR7M	Binary	-0.1485	-0.0556	Balanced, <0.1
VURPHASE_1	Binary	-0.0702	-0.0111	Balanced, <0.1
VURPHASE_2	Binary	0.3261	0.0000	Balanced, <0.1
VURPHASE_3	Binary	-0.2559	0.0111	Balanced, <0.1
PREOPEPISODE1_1	Binary	-0.0509	0.0222	Balanced, <0.1

PREOPEPISODE1_2	Binary	-0.1592	0.0222	Balanced, <0.1
PREOPEPISODE1_3	Binary	0.2101	-0.0444	Balanced, <0.1
PREOPTYPEU1	Binary	0.0120	-0.0111	Balanced, <0.1
PREOPRNSCR	Binary	-0.2746	-0.0222	Balanced, <0.1
SURGTECH2_1	Binary	0.2538	0.0111	Balanced, <0.1
SURGTECH2_2	Binary	-0.0625	0.0111	Balanced, <0.1
SURGTECH2_3	Binary	-0.1913	-0.0222	Balanced, <0.1

Balance tally for mean differences

	count
Balanced, <0.1	26
Not Balanced, >0.1	1

Variable with the greatest mean difference

Variable	Diff.Adj	M.Threshold
AGS	-0.1366	Not Balanced, >0.1

Sample sizes

	Control	Treated
All	178	205
Matched	90	90
Unmatched	88	115

```
balmp_df <- as.data.frame(balmp$Balance)
smd_tablemp <- balmp_df %>%
  mutate(
    Variable = rownames(balmp_df),
    SMD_Before = round(abs(Diff.Un), 3),
    SMD_After = round(abs(Diff.Adj), 3)
  ) %>%
  dplyr::select(Variable, SMD_Before, SMD_After)
```

```
smd_tablemp %>%
  flextable() %>%
  save_as_docx(path = "output/SMD_psm_p.docx")
```

Figure for balance - Love plot

```
plmp <- love.plot(
  m.outp,
  threshold = 0.1,
  abs = TRUE,
  stats = "mean.diffs",
  var.order = "unadjusted",
  drop.distance = TRUE,
  sample.names = c("Before matching", "After matching"),
  size = 4,
  var.names = c(
    AGS = "Age at surgery",
    YEARS = "Year of surgery",
    BMI = "Body mass index",
    GENDER = "Female gender",
    PREOPRNLN = "Preop pyelonephritis",
```

```

HISTOINJ = "History of injection",
VURSIDE = "Right-sided VUR",
UDR7M = "Ureteral dilation 7mm",
PREOPTYPEU1 = "Preop febrile UTIs",
PREOPRNSCR = "Preop renal scarring",
VURPHASE_1 = "Reflux: voiding phase",
VURPHASE_2 = "Reflux: filling phase",
VURPHASE_3 = "Reflux: filling + voiding",
SURGTECH2_1 = "Surgeon experience 100 operations",
SURGTECH2_2 = "Surgeon experience 50-100 operations",
SURGTECH2_3 = "Surgeon experience <50 operations",
PREOPEPISODE1_1 = "Preop UTI episodes: 1",
PREOPEPISODE1_2 = "Preop UTI episodes: 2",
PREOPEPISODE1_3 = "Preop UTI episodes: 3",
SURGIND2_1 = "Indication: breakthrough UTI",
SURGIND2_2 = "Indication: high-grade reflux (III-V)",
SURGIND2_3 = "Indication: poor compliance to medical treatment",
SURGIND2_4 = "Indication: ineffective/poorly tolerated CAP",
SURGIND2_5 = "Indication: other",
PREOPVURGRADE_3 = "Preop VUR grade: III",
PREOPVURGRADE_4 = "Preop VUR grade: IV",
PREOPVURGRADE_5 = "Preop VUR grade: V"
)
)

```

Warning: Standardized mean differences and raw mean differences are present in the same plot. Use the `stars` argument to distinguish between them and appropriately label the x-axis. See `love.plot()` for details.

```

plmp <- plmp +
  labs(
    x = "Absolute Standardized Differences",
    subtitle = "Matched population of children who did not receive epidural analgesia"
  )
  theme(
    plot.margin = margin(10, 10, 10, 10),
    axis.text = element_text(size = 16),
    axis.title = element_text(size = 16),
    plot.title = element_text(size = 16),
    legend.text = element_text(size = 16),
    legend.title = element_text(size = 16),
    axis.ticks = element_line(size = 0.5, color = "black"),
    axis.line = element_line(size = 0.5, color = "black"),
    panel.background = element_blank(),
    panel.grid.major = element_blank(),
    panel.grid.minor = element_blank(),
    panel.border = element_blank(),
    axis.ticks.length = unit(0.3, "cm"),
    panel.spacing = unit(2, "lines")
  )

```

```

<theme> List of 14
 $ axis.title      : <ggplot2::element_text>

```

```

..@ family      : NULL
..@ face        : NULL
..@ italic      : chr NA
..@ fontweight  : num NA
..@ fontwidth   : num NA
..@ colour      : NULL
..@ size        : num 16
..@ hjust       : NULL
..@ vjust       : NULL
..@ angle       : NULL
..@ lineheight  : NULL
..@ margin      : NULL
..@ debug       : NULL
..@ inherit.blank: logi FALSE
$ axis.text     : <ggplot2::element_text>
..@ family      : NULL
..@ face        : NULL
..@ italic      : chr NA
..@ fontweight  : num NA
..@ fontwidth   : num NA
..@ colour      : NULL
..@ size        : num 16
..@ hjust       : NULL
..@ vjust       : NULL
..@ angle       : NULL
..@ lineheight  : NULL
..@ margin      : NULL
..@ debug       : NULL
..@ inherit.blank: logi FALSE
$ axis.ticks    : <ggplot2::element_line>
..@ colour      : chr "black"
..@ linewidth   : num 0.5
..@ linetype    : NULL
..@ lineend     : NULL
..@ linejoin    : NULL
..@ arrow       : logi FALSE
..@ arrow.fill  : chr "black"
..@ inherit.blank: logi FALSE
$ axis.ticks.length: 'simpleUnit' num 0.3cm
..- attr(*, "unit")= int 1
$ axis.line     : <ggplot2::element_line>
..@ colour      : chr "black"
..@ linewidth   : num 0.5
..@ linetype    : NULL
..@ lineend     : NULL
..@ linejoin    : NULL
..@ arrow       : logi FALSE
..@ arrow.fill  : chr "black"
..@ inherit.blank: logi FALSE
$ legend.text   : <ggplot2::element_text>
..@ family      : NULL
..@ face        : NULL
..@ italic      : chr NA
..@ fontweight  : num NA

```

```

..@ fontwidth      : num NA
..@ colour         : NULL
..@ size           : num 16
..@ hjust          : NULL
..@ vjust          : NULL
..@ angle          : NULL
..@ lineheight     : NULL
..@ margin         : NULL
..@ debug          : NULL
..@ inherit.blank: logi FALSE
$ legend.title     : <ggplot2::element_text>
..@ family         : NULL
..@ face           : NULL
..@ italic         : chr NA
..@ fontweight     : num NA
..@ fontwidth      : num NA
..@ colour         : NULL
..@ size           : num 16
..@ hjust          : NULL
..@ vjust          : NULL
..@ angle          : NULL
..@ lineheight     : NULL
..@ margin         : NULL
..@ debug          : NULL
..@ inherit.blank: logi FALSE
$ panel.background : <ggplot2::element_blank>
$ panel.border      : <ggplot2::element_blank>
$ panel.spacing     : 'simpleUnit' num 2lines
..- attr(*, "unit")= int 3
$ panel.grid.major  : <ggplot2::element_blank>
$ panel.grid.minor  : <ggplot2::element_blank>
$ plot.title        : <ggplot2::element_text>
..@ family         : NULL
..@ face           : NULL
..@ italic         : chr NA
..@ fontweight     : num NA
..@ fontwidth      : num NA
..@ colour         : NULL
..@ size           : num 16
..@ hjust          : NULL
..@ vjust          : NULL
..@ angle          : NULL
..@ lineheight     : NULL
..@ margin         : NULL
..@ debug          : NULL
..@ inherit.blank: logi FALSE
$ plot.margin       : <ggplot2::margin> num [1:4] 10 10 10 10
@ complete: logi FALSE
@ validate: logi TRUE

ggsave ("figures/love_plot_pm.png", plot = plmp, width = 12, height = 10, dpi = 300)

# Compute mean SMD
smdmp_df <- balmp$Balance

```

```

mean_smdmp_before <- mean(abs(smdmp_df$Diff.Un), na.rm = TRUE)
mean_smdmp_after <- mean(abs(smdmp_df$Diff.Adj), na.rm = TRUE)
mean_smdmp <- data.frame(
  Mean_SMDmp_Before = mean_smdmp_before,
  Mean_SMDmp_After = mean_smdmp_after
)

mean_smdmp

  Mean_SMDmp_Before Mean_SMDmp_After
1           0.225855           0.03131203

# how much effective sample size remains (overall ESS: calculated manually)?
summary(m.outp$weights)

  Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
  0.00   0.00   0.00   0.47   1.00   1.00

bal.tab(m.outp)$Effective.Sample.Size

NULL

sum(m.outp$weights)^2 / sum(m.outp$weights^2)

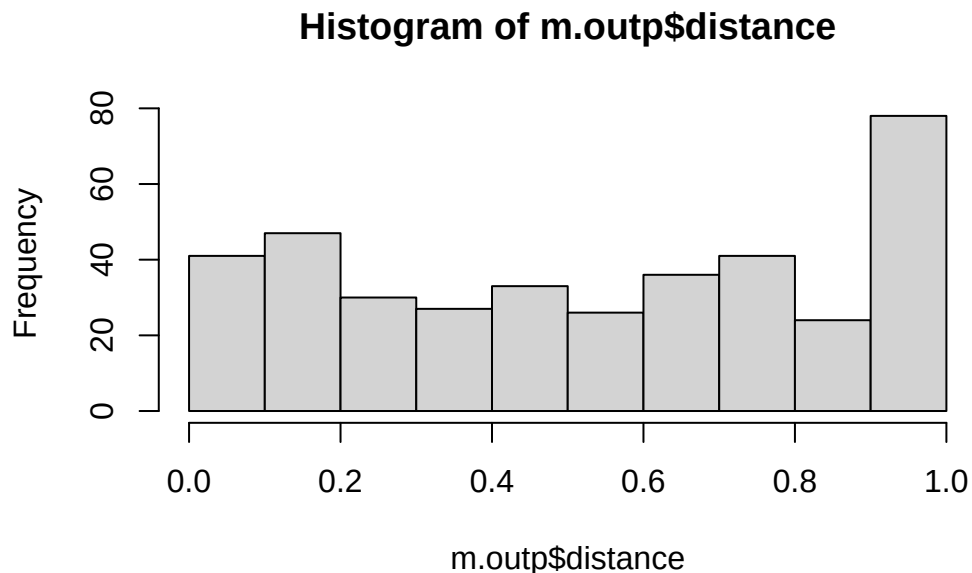
[1] 180

# propensity score distribution
summary(m.outp$distance)

  Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.006964 0.240110 0.543399 0.535248 0.819747 0.999273

hist(m.outp$distance)

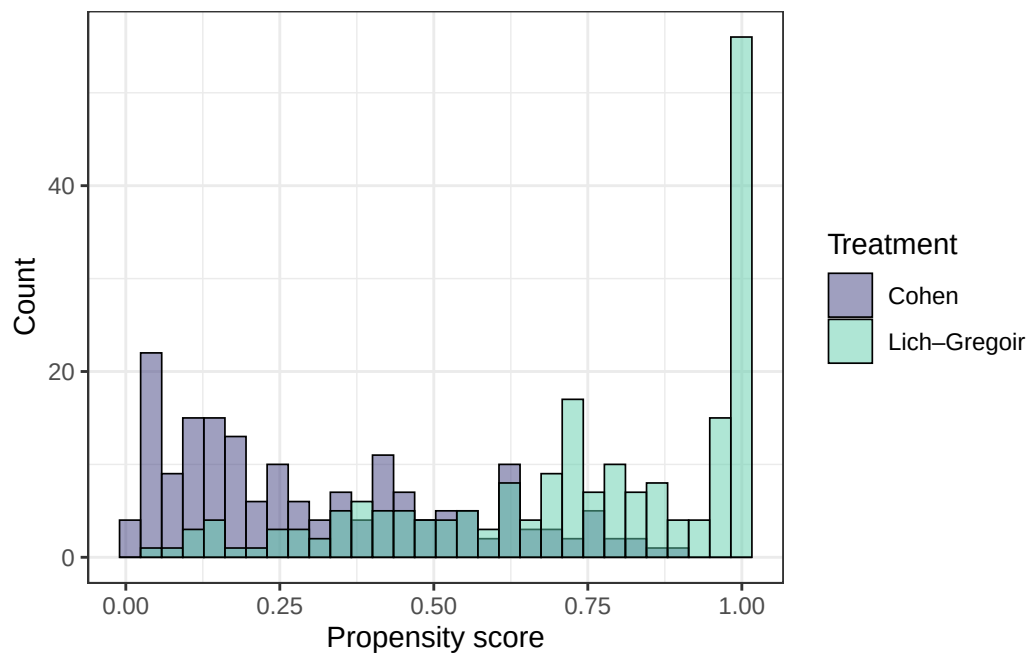
```




```

ggplot(
  data.frame(
    ps = m.outp$distance,
    Treatment = factor(Outp$Treatment)
  ),
  aes(ps, fill = Treatment)
) +
  geom_histogram(alpha = 0.5, bins = 30, position = "identity", color = "black", linewidth = 0.3) +
  scale_fill_viridis_d(
    option = "mako",
    begin = 0.3,
    end = 0.8,
    labels = c("Cohen", "Lich-Gregoir")
  ) +
  labs(
    x = "Propensity score",
    y = "Count",
    fill = "Treatment"
  ) +
  theme_bw()

```



```

ggsave("figures/propensity_psm_p.png",
  width = 12,
  height = 10,
  dpi = 300)

# Distributional balance for propensity score

# Before-After weighting
p_psm_pain <- bal.plot(
  m.outp,

```

```

var.name = "distance",
type = "density",
which = "both"
)

p_psm_pain <- p_psm_pain +
  scale_color_viridis_d(
    option = "mako",
    begin = 0.3,
    end = 0.8,
    labels = c("Cohen group", "Lich-Gregoir group")
  ) +
  scale_fill_viridis_d(
    option = "mako",
    begin = 0.3,
    end = 0.8,
    labels = c("Cohen group", "Lich-Gregoir group")
  )

```

Scale for fill is already present.
 Adding another scale for fill, which will replace the existing scale.

```

ggsave(
  "figures/PS_density_before_after_pm.png",
  p_psm_pain,
  width = 10,
  height = 7,
  dpi = 300
)

```

5.2 Descriptive analysis

```

# mdatap is the dataset with added weightsp_psm

# Convert variables to appropriate data types for analysis
mdatap <- mdatap %>%
  mutate(
    across(c(PID, QCODE), as.character),

    across(c(AGS, YEARS, BLDLOSS, BMI, CATHPERIOD, FUPM, HOSPTIME, OPRTIME,
              POSTUTIFUP, POSTUTI_T), as.numeric),

    across(c(ANASTOMO, BLDCOMPLEAK, BLDCOMPRETN,
              BLDCOMPVOID, BLDSPASM, CATH, CENTER, CLAVIENCLASS, EPIDURAL, GENDER,
              GU30, GU90N, GUER30, GUER90, HEMAT,
              HISTOINJ, ILEUS, INCCOMPBLEED, INCCOMPINF, INCCOMPSCAR, NARC,
              NONARC, PAIN, PREOPEPISODE1, PREOPRNPLN, PREOPRNSCR, PREOPTYPEU1,
              PREOPVURGRADE, PROC365, STENT, SURGIND, SURGTECH2, TRT2, UDR7M,
              URCOMPVUR, URCOMPINJ, URCOMPOBST, URCOMPSTEN, POSTUTI, VURPHASE,
              VURSIDE, RECGRADE_MAX, SURGIND2, LUTD, OBST, SURG, GU90,
              CATHPERIOD_G, ANY, MINOR, MAJOR), as.factor)
  )

```

```
str(mdatap)
```

```
Classes 'matchdata' and 'data.frame': 180 obs. of 69 variables:
```

```
$ PID      : chr  "P1" "P4" "P5" "P6" ...
$ QCODE    : chr  "102" "4" "7" "8" ...
$ REC011   : int   0 1 0 1 0 0 0 0 1 0 ...
$ AGS      : num   5 10 7 9 5 8 8 7 10 6 ...
$ YEARS    : num   2011 2014 2011 2018 2010 ...
$ ANASTOMO : Factor w/ 2 levels "0","1": 1 1 1 1 1 1 1 1 1 1 ...
$ BLDCOMPLEAK : Factor w/ 2 levels "0","1": 1 1 1 1 1 1 1 1 1 1 ...
$ BLDCOMPRETN : Factor w/ 2 levels "0","1": 1 1 1 2 1 2 1 1 2 2 ...
$ BLDCOMPVOID : Factor w/ 2 levels "0","1": 2 1 1 1 1 1 1 1 1 1 ...
$ BLDLOSS   : num   70 55 55 190 50 80 80 50 45 40 ...
$ BLDSPASM  : Factor w/ 2 levels "0","1": 1 1 1 2 2 1 2 1 2 2 ...
$ BMI       : num   13.5 13.3 15 15.8 13.7 15 15 14.4 15.3 15.4 ...
$ CATH      : Factor w/ 2 levels "0","1": 2 2 2 2 2 2 2 2 2 2 ...
$ CATHPERIOD : num   2 5 4 7 1 5 5 4 5 5 ...
$ CENTER    : Factor w/ 4 levels "1","2","3","5": 1 1 1 1 1 1 1 1 1 1 ...
$ CLAVIENCLASS : Factor w/ 4 levels "1","2","3","4": 2 2 1 2 1 2 1 2 3 2 ...
$ EPIDURAL  : Factor w/ 2 levels "0","1": 1 1 1 1 1 1 1 1 1 1 ...
$ FUPM      : num   24 36 120 36 24 36 24 24 36 24 ...
$ GENDER    : Factor w/ 2 levels "0","1": 1 2 2 1 2 2 2 1 2 2 ...
$ GU30      : Factor w/ 2 levels "0","1": 1 1 1 1 1 1 1 1 1 1 ...
$ GU90N     : Factor w/ 3 levels "0","1","2": 1 2 1 2 1 1 1 1 2 1 ...
$ GUER30    : Factor w/ 2 levels "0","1": 1 1 1 1 1 1 1 1 2 1 ...
$ GUER90    : Factor w/ 2 levels "0","1": 2 2 1 1 1 1 1 1 2 1 ...
$ HEMAT     : Factor w/ 2 levels "0","1": 1 2 1 2 1 1 1 1 1 1 ...
$ HISTOINJ  : Factor w/ 2 levels "0","1": 2 1 1 1 1 1 1 1 2 1 ...
$ HOSPTIME  : num   2 5 4 5 2 5 5 5 6 5 ...
$ ILEUS     : Factor w/ 2 levels "0","1": 1 1 1 1 1 1 1 1 1 1 ...
$ INCCOMPBLEED : Factor w/ 2 levels "0","1": 2 1 1 1 1 2 1 1 1 2 ...
$ INCCOMPINF : Factor w/ 2 levels "0","1": 1 1 1 2 1 1 1 1 2 1 ...
$ INCCOMPSCAR : Factor w/ 2 levels "0","1": 1 1 1 1 1 1 1 1 1 1 ...
$ NARC      : Factor w/ 2 levels "0","1": 1 1 1 2 1 1 1 1 2 1 ...
$ NONARC    : Factor w/ 2 levels "0","1": 2 2 2 2 2 2 2 2 2 2 ...
$ OPRTIME   : num   76 83 90 75 68 75 78 76 80 85 ...
$ PAIN      : Ord.factor w/ 4 levels "No pain"<"Mild pain (score 1-3)"<.: 2 2 2 3 2 2 2 2 4 2 ...
$ PREOPEPISODE1 : Factor w/ 3 levels "1","2","3": 2 3 3 3 3 3 3 3 3 3 ...
$ PREOPRNPLN : Factor w/ 2 levels "0","1": 1 1 1 1 1 1 1 1 2 1 ...
$ PREOPRNSCR : Factor w/ 2 levels "0","1": 1 2 1 1 1 1 1 1 1 1 ...
$ PREOPTYPEU1 : Factor w/ 2 levels "0","1": 2 2 2 2 2 2 2 2 2 2 ...
$ PREOPVURGRADE : Factor w/ 3 levels "3","4","5": 3 3 2 3 3 3 2 3 3 3 ...
$ PROC365   : Factor w/ 2 levels "0","1": 1 1 1 1 1 1 1 1 2 1 ...
$ STENT     : Factor w/ 2 levels "0","1": 1 1 1 1 2 1 1 1 1 1 ...
$ SURGIND   : Factor w/ 11 levels "1","2","3","4",...: 3 3 4 4 4 3 3 4 3 4 ...
$ SURGTECH2 : Factor w/ 3 levels "1","2","3": 3 2 2 2 2 2 2 2 2 2 ...
$ TRT2      : Factor w/ 2 levels "0","1": 2 2 2 2 2 2 2 2 2 2 ...
$ UDR7M     : Factor w/ 2 levels "0","1": 2 2 1 2 2 2 2 2 2 2 ...
$ URCOMPVCVUR : Factor w/ 2 levels "0","1": 1 1 1 1 1 1 1 1 1 1 ...
$ URCOMPINJ  : Factor w/ 2 levels "0","1": 1 1 1 1 1 1 1 1 1 1 ...
$ URCOMPOBST : Factor w/ 2 levels "0","1": 1 1 1 1 1 1 1 1 1 1 ...
$ URCOMPSTEN : Factor w/ 2 levels "0","1": 1 1 1 1 1 1 1 2 1 1 ...
$ POSTUTI   : Factor w/ 2 levels "0","1": 2 1 1 1 1 1 1 1 2 2 ...
```

```

$ POSTUTIFUP : num 1 36 120 36 24 36 24 24 1 1 ...
$ POSTUTI_T : num 1 0 0 0 0 0 0 0 1 1 ...
$ VURPHASE : Factor w/ 3 levels "1","2","3": 1 3 1 1 1 1 1 1 1 1 ...
$ VURSIDE : Factor w/ 2 levels "0","1": 1 2 1 2 1 2 1 1 2 1 ...
$ RECGRADE_MAX : Factor w/ 6 levels "0","1","2","3",...: 2 3 1 3 2 2 1 2 4 2 ...
$ SURGIND2 : Factor w/ 5 levels "1","2","3","4",...: 2 2 1 1 1 2 2 1 2 1 ...
$ LUTD : Factor w/ 2 levels "0","1": 2 1 1 2 1 2 1 1 2 2 ...
$ OBST : Factor w/ 2 levels "0","1": 1 1 1 1 1 1 1 2 1 1 ...
$ SURG : Factor w/ 2 levels "0","1": 2 1 1 2 1 2 1 1 2 2 ...
$ GU90 : Factor w/ 2 levels "0","1": 1 2 1 2 1 1 1 1 2 1 ...
$ CATHPERIOD_G : Factor w/ 3 levels "Early removal 3 days",...: 1 2 2 2 1 2 2 2 2 2 ...
$ ANY : Factor w/ 2 levels "0","1": 2 1 1 2 1 2 1 2 2 2 ...
$ MINOR : Factor w/ 2 levels "0","1": 2 1 1 2 1 2 1 1 2 2 ...
$ MAJOR : Factor w/ 2 levels "0","1": 1 1 1 1 1 1 1 1 2 1 ...
$ Treatment : Factor w/ 2 levels "Cohen","Lich-Gregoir": 2 2 2 2 2 2 2 2 2 2 ...
$ weights : num 0.578 0.254 0.103 0.577 0.209 ...
$ distance : num 0.466 0.819 0.883 0.338 0.762 ...
$ weightsp_psm : num 1 1 1 1 1 1 1 1 1 1 ...
$ subclass : Factor w/ 90 levels "1","2","3","4",...: 1 2 3 4 5 6 7 8 9 10 ...

# Matched baseline characteristics
table1mp<- mdatap %>%
  select(YEARS, AGS, GENDER, BMI, HISTOINJ, SURGIND2, PREOPVURGRADE, VURSIDE, UDR7M, VURPHASE, PREOPTYPE,
    SURGTECH2, CENTER, TRT2
  ) %>%
  tbl_summary(
    by = TRT2, # for stratification
    statistic = list(
      all_continuous() ~ "{median} ({p25}-{p75})",
      all_categorical() ~ "{n} ({p}%)",
    ),
    digits = all_categorical() ~ c(0, 1), # n = 0 decimals, p = 1 decimal
    missing = "ifany", # shows missing only when present
    missing_text = "Missing"
  ) %>%
  add_overall() %>%
  add_p(pvalue_fun = ~style_pvalue(.x, digits = 3)) %>%
  bold_labels()

table1mp <- table1mp %>%
  modify_caption(
    "Baseline characteristics after 1:1 propensity score matching among children who did not receive ep
  )

table1mp %>%
  as_flex_table() %>%
  flextable::save_as_docx(path = "output/baseline_m_pain.docx")

# Matched procedural factors
table2mp<- mdatap %>%
  select(TRT2, OPRTIME, STENT, CATH, CATHPERIOD, CATHPERIOD_G, HOSPTIME) %>%
  tbl_summary(
    by = TRT2, # for stratification

```

Characteristic	Overall N = 180 ¹	0 N = 90 ¹	1 N = 90 ¹	p-value ²
OPRTIME	85 (70-140)	140 (95-160)	70 (68-80)	<0.001
STENT				0.149
0	123 (68.3%)	57 (63.3%)	66 (73.3%)	
1	57 (31.7%)	33 (36.7%)	24 (26.7%)	
CATH				>0.999
0	4 (2.2%)	2 (2.2%)	2 (2.2%)	
1	176 (97.8%)	88 (97.8%)	88 (97.8%)	
CATHPERIOD	4 (2-5)	4 (4-5)	2 (1-5)	<0.001
CATHPERIOD_G				<0.001
Early removal 3 days	65 (36.1%)	8 (8.9%)	57 (63.3%)	
Standard removal 4-7 days	88 (48.9%)	68 (75.6%)	20 (22.2%)	
Delayed removal >7 days	27 (15.0%)	14 (15.6%)	13 (14.4%)	
HOSPTIME	5.00 (3.00-5.00)	5.00 (5.00-6.00)	3.00 (2.00-5.00)	<0.001

¹Median (Q1-Q3); n (%)

²Wilcoxon rank sum test; Pearson's Chi-squared test; Fisher's exact test

```

statistic = list(
  all_continuous() ~ "{median} ({p25}-{p75})", # or ~ "{mean} ± {sd}",
  all_categorical() ~ "{n} ({p}%)"
),
digits = all_categorical() ~ c(0, 1), # n = 0 decimals, p = 1 decimal
missing = "ifany",
missing_text = "Missing"
) %>%
add_overall() %>%
add_p(pvalue_fun = ~style_pvalue(.x, digits = 3)) %>% # 3 decimals
bold_labels()

table2mp

table2mp %>%
as_flex_table() %>%
flextable::save_as_docx(path = "output/procedural_m_pain.docx")

# Matched pain

# Before matching

table4 <- Outp %>%
dplyr::select(PAIN, NARC, NONARC, Treatment) %>%
tbl_summary(
  by = Treatment,
  statistic = list(
    all_categorical() ~ "{n} ({p}%)"
  ),
  label = list(

```

```

    PAIN ~ "Pain score",
    NARC ~ "Narcotic use",
    NONARC ~ "Non-narcotic use"
  ),
  digits = all_categorical() ~ c(0, 1),
  missing = "ifany",
  missing_text = "Missing"
) %>%
add_overall() %>%
add_p(pvalue_fun = ~style_pvalue(.x, digits = 3)) %>%
bold_labels()

# After matching

# Matched pain characteristics

table4m<- mdatap %>%
  select(PAIN, NARC, NONARC, TRT2
) %>%
tbl_summary(
  by = TRT2, # for stratification
  statistic = list(
    all_continuous() ~ "{median} ({p25}-{p75})",
    all_categorical() ~ "{n} ({p}%"
  ),
  label = list(
    PAIN ~ "Pain score",
    NARC ~ "Narcotic use",
    NONARC ~ "Non-narcotic use"
  ),
  digits = all_categorical() ~ c(0, 1), # n = 0 decimals, p = 1 decimal
  missing = "ifany", # shows missing only when present
  missing_text = "Missing"
) %>%
add_overall() %>%
add_p(pvalue_fun = ~style_pvalue(.x, digits = 3)) %>%
bold_labels()

table4m <- table4m %>%
  modify_caption(
    "Pain characteristics after 1:1 propensity score matching among children who did not receive epidural
  )

# MERGE TABLES
table_combined_m <- tbl_merge(
  tbls = list(table4, table4m),
  tab_spanner = c("**Before matching**", "**After matching**")
) %>%
  modify_caption(
    "Table 4. Comparison of pain intensity and analgesic use between treatment groups before and after

```

Characteristic	Before matching			p-value
	Overall N = 383 ¹	Cohen N = 178 ¹	Lich-Gregoir N = 205 ¹	
Pain score				<0.001
No pain	3 (0.8%)	0 (0.0%)	3 (1.5%)	
Mild pain (score 1-3)	127 (33.2%)	45 (25.3%)	82 (40.0%)	
Moderate pain (score 4-6)	123 (32.1%)	53 (29.8%)	70 (34.1%)	
Severe pain (score 7-10)	130 (33.9%)	80 (44.9%)	50 (24.4%)	
Narcotic use				0.2
0	215 (56.1%)	105 (59.0%)	110 (53.7%)	
1	168 (43.9%)	73 (41.0%)	95 (46.3%)	
Non-narcotic use				0.2
0	2 (0.5%)	2 (1.1%)	0 (0.0%)	
1	381 (99.5%)	176 (98.9%)	205 (100.0%)	

¹n (%)

²Fisher's exact test; Pearson's Chi-squared test

Analysis restricted to patients not receiving epidural analgesia.

```
) %>%
  modify_source_note(
    "Analysis restricted to patients not receiving epidural analgesia."
  )
```

table_combined_m

```
table_combined_m %>%
  as_flex_table() %>%
  flextable::save_as_docx(path = "output/pain_m.docx")
```

5.3 Plot: BEFORE-AFTER matching

```
# =====
# Pain distribution before and after 1:1 propensity-score matching
# Non-epidural subgroup
# =====

# -----
# 1. Prepare datasets
# -----

# Before matching: all children without epidural analgesia
before <- Outp %>%
  mutate(
    plot_weight = 1,
    Status = "Before matching (N=383), p<0.001"
  ) %>%
  dplyr::select(PAIN, TRT2, Treatment, plot_weight, Status)
```

```

# After matching: matched non-epidural population
after <- mdatap %>%
  mutate(
    plot_weight = weightsp_psm,
    Status = "After matching (N=180), p<0.001"
  ) %>%
  dplyr::select(PAIN, TRT2, Treatment, plot_weight, Status)

# -----
# 2. Combine datasets
# -----

plot_data <- bind_rows(before, after)

# -----
# 3. Pain labels
# -----

plot_data <- plot_data %>%
  mutate(
    PAIN_plot = factor(
      PAIN,
      levels = levels(PAIN),
      labels = c(
        "No pain",
        "Mild pain",
        "Moderate pain",
        "Severe pain"
      )
    )
  )

# -----
# 4. Define order of panels
# -----

plot_data <- plot_data %>%
  mutate(
    Status = factor(
      Status,
      levels = c(
        "Before matching (N=383), p<0.001",
        "After matching (N=180), p<0.001"
      )
    )
  )

# =====
# Figure : 100% stacked bar plot
# =====

```



```

plot_summary1 <- plot_data %>%
  group_by(Status, Treatment, PAIN_plot) %>%
  summarise(
    weighted_n = sum(plot_weight, na.rm = TRUE),
    .groups = "drop"
  ) %>%
  group_by(Status, Treatment) %>%
  mutate(
    Percent = 100 * weighted_n / sum(weighted_n)
  )

s1 <- ggplot(
  plot_summary1,
  aes(
    x = Treatment,
    y = Percent,
    fill = PAIN_plot
  )
) +
  geom_bar(
    stat = "identity",
    position = "fill",
    width = 0.7
  ) +
  facet_wrap(~Status) +
  labs(
    x = "Treatment groups",
    y = "Percentage of patients not receiving epidural analgesia (%)",
    fill = "1-Day postoperative pain score"
  ) +
  scale_fill_manual(
    values = c(
      "No pain" = "#f5f5f5",
      "Mild pain" = "#fee8c8",
      "Moderate pain" = "#fdbb84",
      "Severe pain" = "#e34a33"
    )
  ) +
  scale_y_continuous(
    labels = scales::percent_format(scale = 100)
  ) +
  theme_minimal() +
  theme(
    panel.grid = element_blank(),
    axis.title = element_text(size = 12),
    axis.text = element_text(size = 11),
    panel.border = element_rect(
      color = "black",
      fill = NA
    ),
    strip.text = element_text(
      face = "bold"
    )
  )

```

```

    ),
    panel.spacing = unit(
      0,
      "lines"
    )
  )
)

ggsave(
  filename = "figures/pain_stacked_m.png",
  plot = s1,
  width = 9,
  height = 6,
  dpi = 300
)

```

5.4 Regression

```

# -----
### Moderate-severe pain vs. Other levels
# -----

mdatap <- mdatap %>%
  mutate(
    PAIN_ms = factor(
      ifelse(
        PAIN %in% c(
          "Moderate pain (score 4-6)",
          "Severe pain (score 7-10)"
        ),
        1,
        0
      ),
      levels = c(0, 1),
      labels = c(
        "No/Mild pain (score 0-3)",
        "Moderate-Severe pain (score 4-10)"
      )
    )
  )

mdatap$PAIN_ms <- ifelse(
  mdatap$PAIN_ms == "Moderate-Severe pain (score 4-10)",
  1,
  0
)

# -----
### Exploratory analysis: Severe pain vs. Other levels
# -----

mdatap <- mdatap %>%

```

```

mutate(
  PAIN_s = factor(
    ifelse(
      PAIN == "Severe pain (score 7-10)",
      1,
      0
    ),
    levels = c(0, 1),
    labels = c(
      "No severe pain (score 0-6)",
      "Severe pain (score 7-10)"
    )
  )
)

# a numeric binary outcome:
mdatap$PAIN_s <- ifelse(
  mdatap$PAIN_s == "Severe pain (score 7-10)",
  1,
  0
)

#-----
# Block 1
#-----

outcomes <- c(
  "PAIN_ms",
  "PAIN_s"
)

# convert outcomes to numeric
mdatap[outcomes] <- lapply(
  mdatap[outcomes],
  function(x) as.numeric(as.character(x))
)

# Apply GEE model after matching for all binary outcomes
gee_results_psm <- lapply(outcomes, function(y) {
  formula <- as.formula(
    paste(y, "~ TRT2")
  )
  model <- geeglm(
    formula,
    data = mdatap,
    id = CENTER,
    weights = weightsp_psm,
    family = binomial(link = "logit"),
    corstr = "exchangeable"
  )
  tidy(model, exponentiate = TRUE, conf.int = TRUE) # estimate = OR because exponentiate = TRUE
})
names(gee_results_psm) <- outcomes

```

```

# Combine results into one table
gee_table_psmpp <- do.call(
  rbind,
  lapply(names(gee_results_psmpp), function(x){
    tmp <- gee_results_psmpp[[x]]
    tmp <- tmp %>%
      filter(grepl("^TRT2", term)) %>%
      mutate(outcome = x)
    tmp
  })
)

# Create final manuscript table
gee_table_final_psmpp <- gee_table_psmpp %>%
  mutate(
    Effect_measure = "OR",
    Estimate = sprintf("%.2f", estimate),
    CI = paste0(
      sprintf("%.2f", conf.low),
      "-",
      sprintf("%.2f", conf.high)
    ),
    P_value = ifelse(p.value < 0.001, "<0.001", sprintf("%.3f", p.value))
  ) %>%
  dplyr::select(
    Outcome = outcome,
    Effect_measure,
    Estimate,
    CI,
    P_value
  )

gee_table_final_psmpp

# A tibble: 2 x 5
  Outcome Effect_measure Estimate CI          P_value
  <chr>    <chr>          <chr> <chr>      <chr>
1 PAIN_ms OR            0.31    0.11-0.86 0.025
2 PAIN_s  OR            0.15    0.11-0.20 <0.001

gee_table_final_psmpp %>%
  flextable() %>%
  theme_booktabs() %>%
  autofit() %>%
  align(
    j = c("Estimate", "CI", "P_value"),
    align = "center",
    part = "all"
  ) %>%
  save_as_docx(
    path = "output/pain_unadj_m.docx"
  )

```

```

)

#-----
# Block 2
#-----

outcomes <- c(
  "PAIN_ms",
  "PAIN_s"
)

# convert outcomes to numeric
mdatap[outcomes] <- lapply(
  mdatap[outcomes],
  function(x) as.numeric(as.character(x))
)

# Predictors to include in the outcome models
predictors <- c("TRT2", "STENT")

# Apply GEE model after overlap weighting for all binary outcomes
gee_results_psm <- lapply(outcomes, function(y) {
  formula <- as.formula(
    paste(y, "~",
          paste(predictors, collapse = " + ")
    )
  )
  model <- geeglm(
    formula,
    data = mdatap,
    id = CENTER,
    weights = weightsp_psm,
    family = binomial(link = "logit"),
    corstr = "exchangeable"
  )
  tidy(model, exponentiate = TRUE, conf.int = TRUE) # estimate = OR because exponentiate = TRUE
})
names(gee_results_psm) <- outcomes

# Combine results into one table
gee_table_psm <- do.call(
  rbind,
  lapply(names(gee_results_psm), function(x){
    tmp <- gee_results_psm[[x]]
    tmp %>%
      mutate(outcome = x)
  })
)

# Create final manuscript table

```

```
gee_table_final_psm <- gee_table_psm %>%
  mutate(
    Effect_measure = "OR",
    Estimate = sprintf("%.2f", estimate),
    CI = paste0(
      sprintf("%.2f", conf.low),
      "-",
      sprintf("%.2f", conf.high)
    ),
    P_value = ifelse(p.value < 0.001, "<0.001", sprintf("%.3f", p.value))
  ) %>%
  dplyr::select(
    Outcome = outcome,
    Predictor = term,
    Effect_measure,
    Estimate,
    CI,
    P_value
  )
```

```
gee_table_final_psm
```

```
# A tibble: 6 x 6
  Outcome Predictor Effect_measure Estimate CI P_value
<chr> <chr> <chr> <chr> <chr> <chr>
1 PAIN_ms (Intercept) OR 2.21 0.43-11.28 0.339
2 PAIN_ms TRT21 OR 0.34 0.12-0.92 0.033
3 PAIN_ms STENT1 OR 1.76 1.04-2.97 0.035
4 PAIN_s (Intercept) OR 1.01 0.56-1.83 0.965
5 PAIN_s TRT21 OR 0.12 0.10-0.15 <0.001
6 PAIN_s STENT1 OR 0.30 0.16-0.58 <0.001
```

```
gee_table_final_psm %>%
  flextable() %>%
  theme_booktabs() %>%
  autofit() %>%
  align(
    j = c("Estimate", "CI", "P_value"),
    align = "center",
    part = "all"
  ) %>%
  save_as_docx(
    path = "output/pain_adj_m.docx"
  )
```

```
#-----
# Block 2ai- Interaction
#-----
```

```
outcomes <- c(
  "PAIN_ms",
  "PAIN_s"
```

```

)

# convert outcomes to numeric
mdatap[outcomes] <- lapply(
  mdatap[outcomes],
  function(x) as.numeric(as.character(x))
)

# Function to extract treatment effects within each stent stratum
get_gee_interaction_results <- function(y) {

  formula <- as.formula(
    paste(y, "~ TRT2 * STENT")
  )

  model <- geeglm(
    formula,
    data = mdatap,
    id = CENTER,
    weights = weightsp_psm,
    family = binomial(link = "logit"),
    corstr = "exchangeable"
  )

  # Coefficients
  beta <- coef(model)

  # Robust covariance matrix
  V <- vcov(model)

  # -----
  # LG vs Cohen among NO-STENT patients
  # -----

  b_no_stent <- beta["TRT21"]

  se_no_stent <- sqrt(
    V["TRT21", "TRT21"]
  )

  OR_no_stent <- exp(b_no_stent)

  CI_no_stent <- exp(
    b_no_stent + c(-1, 1) * 1.96 * se_no_stent
  )

  # -----
  # LG vs Cohen among STENTED patients
  # = TRT21 + TRT21:STENT1
  # -----

  b_stent <- beta["TRT21"] +

```

```

    beta["TRT21:STENT1"]

# Variance of the linear combination:
# Var(beta1 + beta3)

var_stent <-
  V["TRT21", "TRT21"] +
  V["TRT21:STENT1", "TRT21:STENT1"] +
  2 * V["TRT21", "TRT21:STENT1"]

se_stent <- sqrt(var_stent)

OR_stent <- exp(b_stent)

CI_stent <- exp(
  b_stent + c(-1, 1) * 1.96 * se_stent
)

# -----
# Interaction test
# -----

b_interaction <- beta["TRT21:STENT1"]

se_interaction <- sqrt(
  V["TRT21:STENT1", "TRT21:STENT1"]
)

z_interaction <- b_interaction / se_interaction

p_interaction <- 2 * pnorm(
  abs(z_interaction),
  lower.tail = FALSE
)

# -----
# Return manuscript-ready results
# -----

data.frame(
  Outcome = y,

  `LG vs Cohen, No Stent` =
    sprintf(
      "%.2f (%.2f-%.2f)",
      OR_no_stent,
      CI_no_stent[1],
      CI_no_stent[2]
    ),

  `LG vs Cohen, Stent` =
    sprintf(
      "%.2f (%.2f-%.2f)",

```



```

      OR_stent,
      CI_stent[1],
      CI_stent[2]
    ),

    P_interaction = p_interaction,

    stringsAsFactors = FALSE
  )
}

# Run for all 2 outcomes
interaction_results <- do.call(
  rbind,
  lapply(outcomes, get_gee_interaction_results)
)

```

```
interaction_results
```

	Outcome	LG.vs.Cohen..No.Stent	LG.vs.Cohen..Stent	P_interaction
TRT21:STENT1	PAIN_ms	0.19 (0.15-0.23)	1.03 (0.46-2.34)	2.414117e-05
TRT21:STENT11	PAIN_s	0.11 (0.09-0.14)	0.21 (0.13-0.36)	6.159767e-02

5.5 Rosenbaum's sensitivity analyses

```

# Rosenbaum's primal sensitivity analysis

outcomes <- c(
  "PAIN_ms",
  "PAIN_s"
)

# Convert outcomes to numeric
mdatap[outcomes] <- lapply(
  mdatap[outcomes],
  function(x) as.numeric(as.character(x))
)

# Function for Rosenbaum sensitivity analysis using rbounds
rosenbaum_binary <- function(outcome, data, match_object,
                             gamma_seq = seq(1, 11, by = 0.1),
                             alpha = 0.05) {

  # matched pairs
  treated_ids <- rownames(match_object$match.matrix)
  control_ids <- match_object$match.matrix[, 1]

  pairs <- data.frame(
    treated = data[treated_ids, outcome],

```

```

    control = data[control_ids, outcome]
  )

# remove missing outcome pairs
pairs <- na.omit(pairs)

# discordant pairs
# x = control has outcome, treated does not
# y = treated has outcome, control does not
x <- sum(pairs$control == 1 & pairs$treated == 0)
y <- sum(pairs$treated == 1 & pairs$control == 0)

# run Rosenbaum sensitivity
if ((x + y) == 0) {
  return(data.frame(
    Outcome = outcome,
    Discordant_pairs = 0,
    x = x,
    y = y,
    p_Gamma1 = NA,
    gamma_max = NA,
    p_gamma_max = NA
  ))
}

# Run sensitivity over Gamma values

gamma_results <- sapply(gamma_seq, function(g) {

  res <- binarysens(
    x = x,
    y = y,
    Gamma = g,
    GammaInc = 0.1 # GammaInc defines the increment used to increase  $\Gamma$  values when the function eval
  )

  # extract upper bound p-value corresponding to Gamma g
  res$bounds$`Upper bound`[
    res$bounds$Gamma == g
  ]
})

# p-value at Gamma = 1
p_gamma1 <- gamma_results[gamma_seq == 1]

# first Gamma where upper p-value >= alpha
idx <- which(gamma_results < alpha)

if(length(idx) == 0){

  if(p_gamma1 >= alpha){

    gamma_max <- "NS at Gamma=1"
  }
}

```

```

    p_gamma_max <- NA
  } else {

    gamma_max <- paste0(">=", max(gamma_seq))
    p_gamma_max <- gamma_results[length(gamma_results)]

  }
} else {

  gamma_max_value <- gamma_seq[max(idx)]

  # If still significant at the maximum tested Gamma
  if(gamma_max_value == max(gamma_seq)){

    gamma_max <- paste0(">=", max(gamma_seq))

  } else {

    gamma_max <- gamma_max_value

  }

  p_gamma_max <- gamma_results[max(idx)]
}

# Extract p-values from rbounds object
data.frame(
  Outcome = outcome,
  Discordant_pairs = x + y,
  x = x,
  y = y,
  p_Gamma1 = round(p_gamma1, 3),
  gamma_max = gamma_max,
  p_gamma_max = round(p_gamma_max, 3)
)
}

# Run all outcomes
rosenbaum_results <- do.call(
  rbind,
  lapply(
    outcomes,
    rosenbaum_binary,
    data = mdatap,
    match_object = m.outp
  )
)

rosenbaum_results

```

	Outcome	Discordant_pairs	x	y	p_Gamma1	gamma_max	p_gamma_max
1	PAIN_ms	40	32	8	0	2.2	0.039
2	PAIN_s	46	40	6	0	3.6	0.046

```
# Rename columns for publication
```

```
rosenbaum_results_pub <- rosenbaum_results %>%
```

```
  rename(
    Outcome = Outcome,
    `Discordant pairs` = Discordant_pairs,
    `Control only events (x)` = x,
    `Treated only events (y)` = y,
    `P-value ( $\Gamma=1$ )` = p_Gamma1,
    ` $\Gamma$ max` = gamma_max,
    `P-value at  $\Gamma$ max` = p_gamma_max
  )
```

```
# Create publication table
```

```
rosenbaum_tablep <- rosenbaum_results_pub %>%
  flextable() %>%
```

```
# Add title
```

```
add_header_lines(
  "Rosenbaum's primal sensitivity analysis for pain in the matched cohort."
) %>%
```

```
# Add footnote
```

```
add_footer_lines(
  "The Gamma value  $\Gamma$  is a measure of sensitivity to hidden bias, representing the magnitude of differ"
) %>%
```

```
theme_booktabs() %>%
```

```
autofit() %>%
```

```
align(
```

```
  j = c(
    "Discordant pairs",
    "Control only events (x)",
    "Treated only events (y)",
    "P-value ( $\Gamma=1$ )",
    " $\Gamma$ max",
    "P-value at  $\Gamma$ max"
  ),
```

```
  align = "center",
  part = "all"
```

```
)
```

```
# Save as Word document
```

```
save_as_docx(
  rosenbaum_tablep,
  path = "output/Rosenbaum_effect_pain.docx"
```

```
)
```

TEMPORAL CHANGES

Number of procedures per year

Plots

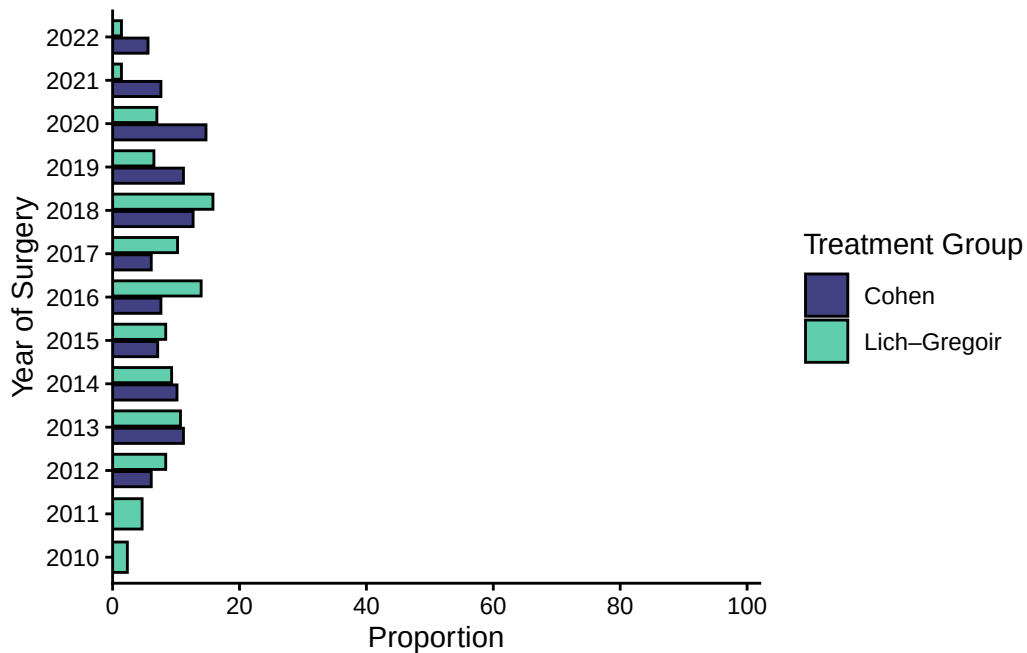
```
#-----
# Horizontal bar for each year
#-----

# -----
# The whole cohort
# -----

plot_data1 <- Out %>%
  count(YEARS, Treatment) %>%
  group_by(Treatment) %>%
  mutate(
    Percent = 100 * n / sum(n)
  ) %>%
  ungroup()

p_y_m1 <- ggplot(
  plot_data1,
  aes(
    x = Percent,
    y = factor(YEARS),
    fill = Treatment
  )
) +
  geom_col(
    position = position_dodge(width = 0.8),
    width = 0.7,
    colour = "black"
  ) +
  scale_x_continuous(
    limits = c(0, 100),
    breaks = seq(0, 100, 20),
    expand = expansion(mult = c(0, 0.02))
  ) +
  scale_fill_viridis_d(option = "mako", begin = 0.3, end = 0.8)
  ) +
  labs(x = "Proportion", y = "Year of Surgery", fill = "Treatment Group") +
  theme_classic()

p_y_m1
```



```
# -----
# After ATO weighting
# -----

plot_data_ato <- Out %>%
  group_by(YEARS, Treatment) %>%
  summarise(
    weighted_n = sum(weights),
    .groups = "drop"
  ) %>%
  group_by(Treatment) %>%
  mutate(
    Percent = 100 * weighted_n / sum(weighted_n)
  ) %>%
  ungroup()

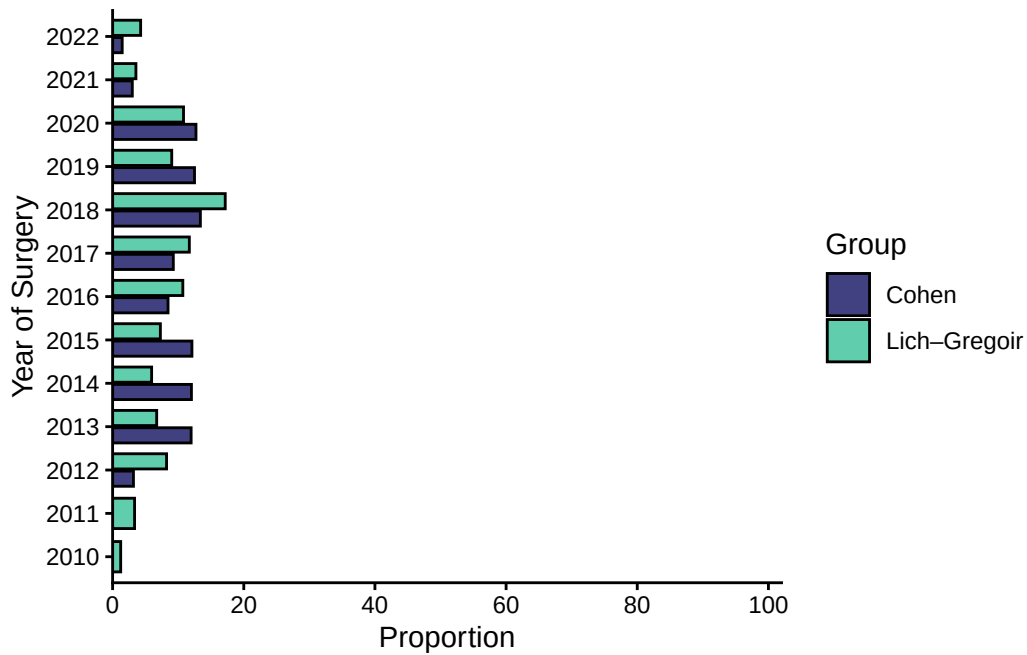
p_y_m_ato <- ggplot(
  plot_data_ato,
  aes(
    x = Percent,
    y = factor(YEARS),
    fill = Treatment
  )
) +
  geom_col(
    position = position_dodge(width = 0.8),
    width = 0.7,
    colour = "black"
  ) +
  scale_x_continuous(
    limits = c(0, 100),
    breaks = seq(0, 100, 20),
  )
```

```

    expand = expansion(mult = c(0, 0.02))
  ) +
  scale_fill_viridis_d(
    option = "mako",
    begin = 0.3,
    end = 0.8
  ) +
  labs(
    x = "Proportion",
    y = "Year of Surgery",
    fill = "Group"
  ) +
  theme_classic()

```

p_y_m_ato



```

# -----
# After matching
# -----

plot_data <- mdata %>%
  count(YEARS, Treatment) %>%
  group_by(Treatment) %>%
  mutate(
    Percent = 100 * n / sum(n)
  ) %>%
  ungroup()

p_y_m <- ggplot(
  plot_data,
  aes(
    x = Percent,

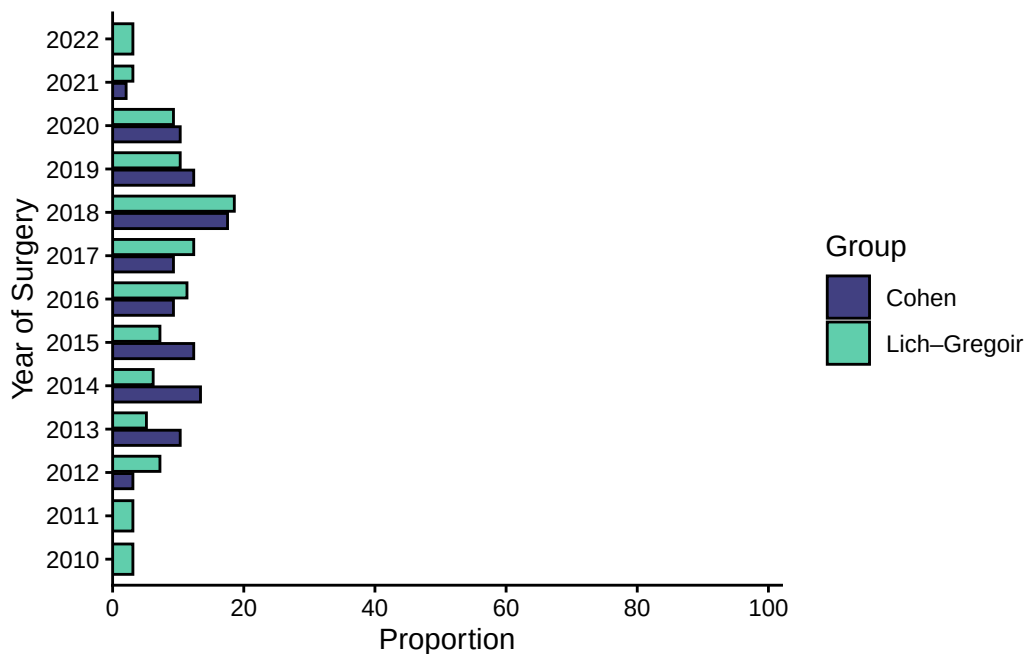
```

```

    y = factor(YEARS),
    fill = Treatment
  )
) +
  geom_col(
    position = position_dodge(width = 0.8),
    width = 0.7,
    colour = "black"
  ) +
  scale_x_continuous(
    limits = c(0, 100),
    breaks = seq(0, 100, 20),
    expand = expansion(mult = c(0, 0.02))
  ) +
  scale_fill_viridis_d(option = "mako", begin = 0.3, end = 0.8)
) +
  labs(x = "Proportion", y = "Year of Surgery", fill = "Group") +
  theme_classic()

```

p_y_m



```

#-----
# Add subtitle (we can keep or remove them from the final figure)
#-----

```

```

p_y_m1 <- p_y_m1 +
  labs(subtitle = "Whole cohort")

```

```

p_y_m_ato <- p_y_m_ato +
  labs(subtitle = "Overlap weighting population")

```



```

p_y_m <- p_y_m +
  labs(subtitle = "Matched cohort")

#-----
# Combine plot
#-----

treatment_plot <-
  p_y_m1 /
  p_y_m_ato /
  p_y_m +
  plot_layout(heights = c(2,2,2)) +
  plot_annotation(
    tag_levels = "A"
  )

#-----
# Save the figure
#-----

ggsave(filename = "figures/treatment_across_years.png",
  plot = treatment_plot,
  width = 8, # width in inches
  height = 10, # height in inches
  dpi = 300) # resolution

```

Table

```

# -----
# The whole cohort
# -----

year_table <- Out %>%
  count(YEARS, Treatment) %>%
  group_by(YEARS) %>%
  mutate(
    Percent = 100 * n / sum(n),
    n_percent = paste0(n, " (", round(Percent, 1), "%)")
  ) %>%
  select(YEARS, Treatment, n_percent) %>%
  pivot_wider(
    names_from = Treatment,
    values_from = n_percent
  ) %>%
  left_join(
    Out %>%
      count(YEARS, name = "Total_n") %>%
      mutate(
        Total_percent = 100 * Total_n / nrow(Out),
        Total = paste0(Total_n, " (", round(Total_percent, 1), "%)")
      ) %>%
      select(YEARS, Total),
    by = "YEARS"
  ) %>%

```

```

  arrange(YEARS)

year_table <- year_table %>%
  rename(
    `Year of Surgery` = YEARS,
    `Cohen n (%)` = Cohen,
    `Lich-Gregoir n (%)` = `Lich-Gregoir`,
    `Total surgeries` = Total
  )

year_table %>%
  gt()

year_table

# A tibble: 13 x 4
# Groups:   Year of Surgery [13]
  `Year of Surgery` `Lich-Gregoir n (%)` `Cohen n (%)` `Total surgeries`
      <dbl> <chr>                <chr>          <chr>
1         2010 5 (100%)          <NA>          5 (1.2%)
2         2011 10 (100%)         <NA>          10 (2.4%)
3         2012 18 (60%)          12 (40%)      30 (7.3%)
4         2013 23 (51.1%)        22 (48.9%)    45 (10.9%)
5         2014 20 (50%)          20 (50%)      40 (9.7%)
6         2015 18 (56.2%)        14 (43.8%)    32 (7.8%)
7         2016 30 (66.7%)        15 (33.3%)    45 (10.9%)
8         2017 22 (64.7%)        12 (35.3%)    34 (8.3%)
9         2018 34 (57.6%)        25 (42.4%)    59 (14.3%)
10        2019 14 (38.9%)        22 (61.1%)    36 (8.7%)
11        2020 15 (34.1%)        29 (65.9%)    44 (10.7%)
12        2021 3 (16.7%)         15 (83.3%)    18 (4.4%)
13        2022 3 (21.4%)         11 (78.6%)    14 (3.4%)

year_table <- year_table %>%
  mutate(
    `Year of Surgery` = as.character(`Year of Surgery`)
  )

year_table %>%
  flextable() %>%
  theme_booktabs() %>%
  autofit() %>%
  align(
    align = "center",
    part = "all"
  ) %>%
  save_as_docx(
    path = "output/treatment_year_whole_cohort.docx"
  )

```

Lich-Gregoir n (%)	Cohen n (%)	Total surgeries
2010		
5 (100%)	NA	5 (1.2%)
2011		
10 (100%)	NA	10 (2.4%)
2012		
18 (60%)	12 (40%)	30 (7.3%)
2013		
23 (51.1%)	22 (48.9%)	45 (10.9%)
2014		
20 (50%)	20 (50%)	40 (9.7%)
2015		
18 (56.2%)	14 (43.8%)	32 (7.8%)
2016		
30 (66.7%)	15 (33.3%)	45 (10.9%)
2017		
22 (64.7%)	12 (35.3%)	34 (8.3%)
2018		
34 (57.6%)	25 (42.4%)	59 (14.3%)
2019		
14 (38.9%)	22 (61.1%)	36 (8.7%)
2020		
15 (34.1%)	29 (65.9%)	44 (10.7%)
2021		
3 (16.7%)	15 (83.3%)	18 (4.4%)
2022		
3 (21.4%)	11 (78.6%)	14 (3.4%)

```

# -----
# After matching
# -----

year_tablem <- mdata %>%
  count(YEARS, Treatment) %>%
  group_by(YEARS) %>%
  mutate(
    Percent = 100 * n / sum(n),
    n_percent = paste0(n, " (", round(Percent, 1), "%)")
  ) %>%
  select(YEARS, Treatment, n_percent) %>%
  pivot_wider(
    names_from = Treatment,
    values_from = n_percent
  ) %>%
  left_join(
    Out %>%
      count(YEARS, name = "Total_n") %>%
      mutate(
        Total_percent = 100 * Total_n / nrow(Out),
        Total = paste0(Total_n, " (", round(Total_percent, 1), "%)")
      ) %>%
      select(YEARS, Total),
    by = "YEARS"
  ) %>%
  arrange(YEARS)

year_tablem <- year_tablem %>%
  rename(
    `Year of Surgery` = YEARS,
    `Cohen n (%)` = Cohen,
    `Lich-Gregoir n (%)` = `Lich-Gregoir`,
    `Total surgeries` = Total
  )

year_tablem %>%
  gt()

year_tablem

# A tibble: 13 x 4
# Groups:   Year of Surgery [13]
  `Year of Surgery` `Lich-Gregoir n (%)` `Cohen n (%)` `Total surgeries`
      <dbl> <chr>                <chr>         <chr>
1         2010 3 (100%)          <NA>          5 (1.2%)
2         2011 3 (100%)          <NA>          10 (2.4%)
3         2012 7 (70%)           3 (30%)       30 (7.3%)
4         2013 5 (33.3%)         10 (66.7%)    45 (10.9%)
5         2014 6 (31.6%)         13 (68.4%)    40 (9.7%)
6         2015 7 (36.8%)         12 (63.2%)    32 (7.8%)
7         2016 11 (55%)          9 (45%)       45 (10.9%)

```

Lich-Gregoir n (%)	Cohen n (%)	Total surgeries
2010		
3 (100%)	NA	5 (1.2%)
2011		
3 (100%)	NA	10 (2.4%)
2012		
7 (70%)	3 (30%)	30 (7.3%)
2013		
5 (33.3%)	10 (66.7%)	45 (10.9%)
2014		
6 (31.6%)	13 (68.4%)	40 (9.7%)
2015		
7 (36.8%)	12 (63.2%)	32 (7.8%)
2016		
11 (55%)	9 (45%)	45 (10.9%)
2017		
12 (57.1%)	9 (42.9%)	34 (8.3%)
2018		
18 (51.4%)	17 (48.6%)	59 (14.3%)
2019		
10 (45.5%)	12 (54.5%)	36 (8.7%)
2020		
9 (47.4%)	10 (52.6%)	44 (10.7%)
2021		
3 (60%)	2 (40%)	18 (4.4%)
2022		
3 (100%)	NA	14 (3.4%)

8	2017	12 (57.1%)	9 (42.9%)	34 (8.3%)
9	2018	18 (51.4%)	17 (48.6%)	59 (14.3%)
10	2019	10 (45.5%)	12 (54.5%)	36 (8.7%)
11	2020	9 (47.4%)	10 (52.6%)	44 (10.7%)
12	2021	3 (60%)	2 (40%)	18 (4.4%)
13	2022	3 (100%)	<NA>	14 (3.4%)

```
year_tablem <- year_tablem %>%
  mutate(
    `Year of Surgery` = as.character(`Year of Surgery`)
  )
```

```
year_tablem %>%
  flextable() %>%
  theme_booktabs() %>%
  autofit() %>%
  align(
    align = "center",
    part = "all"
  ) %>%
  save_as_docx(
    path = "output/treatment_year_matched_cohort.docx"
  )
```

```
# -----
# After ATO weighting
# -----
```

```
year_table_ato <- Out %>%
  group_by(YEARS, Treatment) %>%
  summarise(
    weighted_n = sum(weights),
    .groups = "drop"
  ) %>%
  group_by(YEARS) %>%
  mutate(
    Percent = 100 * weighted_n / sum(weighted_n),
    n_percent = paste0(
      round(weighted_n, 1),
      " (",
      round(Percent, 1),
      "%)"
    )
  ) %>%
  select(YEARS, Treatment, n_percent) %>%
  pivot_wider(
    names_from = Treatment,
    values_from = n_percent
  ) %>%
  left_join(
    Out %>%
      group_by(YEARS) %>%
      summarise(
```

```

      weighted_total = sum(weights),
      .groups = "drop"
    ) %>%
    mutate(
      Total_percent = 100 * weighted_total / sum(weighted_total),
      Total = paste0(
        round(weighted_total, 1),
        " (",
        round(Total_percent, 1),
        "%)"
      )
    ),
    by = "YEARS"
  ) %>%
  arrange(YEARS)

year_table_ato <- year_table_ato %>%
  rename(
    `Year of Surgery` = YEARS,
    `Cohen n (%)` = Cohen,
    `Lich-Gregoir n (%)` = `Lich-Gregoir`,
    `Total surgeries` = Total
  )

year_table_ato %>%
  gt()

year_table_ato

# A tibble: 13 x 6
# Groups:   Year of Surgery [13]
  `Year of Surgery` `Lich-Gregoir n (%)` `Cohen n (%)` weighted_total
    <dbl> <chr>                <chr>          <dbl>
1      2010 0.7 (100%)          <NA>           0.740
2      2011 2 (100%)            <NA>           2.01
3      2012 4.9 (72.1%)         1.9 (27.9%)     6.83
4      2013 4 (36%)              7.2 (64%)       11.2
5      2014 3.6 (33.1%)         7.2 (66.9%)     10.8
6      2015 4.4 (37.6%)         7.2 (62.4%)     11.6
7      2016 6.4 (55.9%)         5.1 (44.1%)     11.5
8      2017 7 (55.8%)           5.5 (44.2%)     12.6
9      2018 10.3 (56.2%)        8 (43.8%)       18.3
10     2019 5.4 (41.9%)         7.5 (58.1%)     12.9
11     2020 6.5 (46%)           7.6 (54%)       14.1
12     2021 2.1 (54.2%)         1.8 (45.8%)     3.93
13     2022 2.6 (74.5%)         0.9 (25.5%)     3.44
# i 2 more variables: Total_percent <dbl>, `Total surgeries` <chr>

year_table_ato <- year_table_ato %>%
  mutate(
    `Year of Surgery` = as.character(`Year of Surgery`)
  )

```

Lich-Gregoir n (%)	Cohen n (%)	weighted_total	Total_percent	Total surgeries
2010				
0.7 (100%)	NA	0.7395953	0.6169953	0.7 (0.6%)
2011				
2 (100%)	NA	2.0084437	1.6755112	2 (1.7%)
2012				
4.9 (72.1%)	1.9 (27.9%)	6.8321610	5.6996181	6.8 (5.7%)
2013				
4 (36%)	7.2 (64%)	11.2030486	9.3459593	11.2 (9.3%)
2014				
3.6 (33.1%)	7.2 (66.9%)	10.7679299	8.9829687	10.8 (9%)
2015				
4.4 (37.6%)	7.2 (62.4%)	11.6140830	9.6888580	11.6 (9.7%)
2016				
6.4 (55.9%)	5.1 (44.1%)	11.4809209	9.5777697	11.5 (9.6%)
2017				
7 (55.8%)	5.5 (44.2%)	12.5586325	10.4768329	12.6 (10.5%)
2018				
10.3 (56.2%)	8 (43.8%)	18.3131712	15.2774623	18.3 (15.3%)
2019				
5.4 (41.9%)	7.5 (58.1%)	12.8868221	10.7506197	12.9 (10.8%)
2020				
6.5 (46%)	7.6 (54%)	14.0985203	11.7614591	14.1 (11.8%)
2021				
2.1 (54.2%)	1.8 (45.8%)	3.9291750	3.2778497	3.9 (3.3%)
2022				
2.6 (74.5%)	0.9 (25.5%)	3.4380010	2.8680959	3.4 (2.9%)


```

year_table_ato %>%
  flextable() %>%
  theme_booktabs() %>%
  autofit() %>%
  align(
    align = "center",
    part = "all"
  ) %>%
save_as_docx(
  path = "output/treatment_year_overlap_cohort.docx"
)

```

Transition in practice: Observed transition in practice

```

# -----
# The whole cohort
# -----

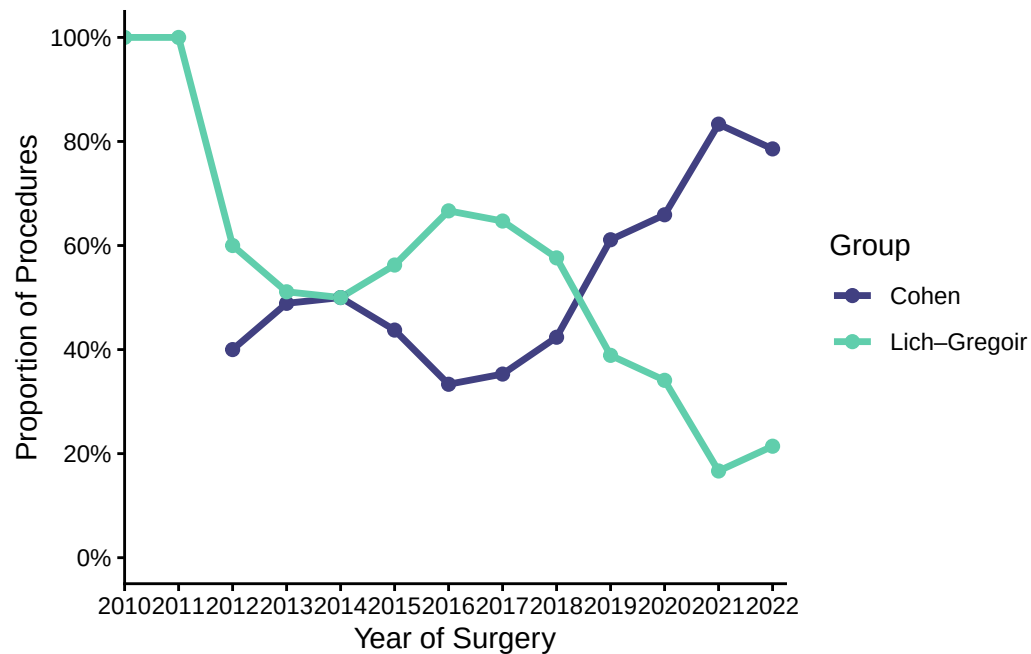
plot_data1 <- Out %>%
  count(YEARS, Treatment) %>%
  group_by(YEARS) %>%
  mutate(
    Percent = 100 * n / sum(n)
  ) %>%
  ungroup()

p_transition1 <- ggplot(
  plot_data1,
  aes(
    x = YEARS,
    y = Percent,
    color = Treatment,
    group = Treatment
  )
) +
  geom_line(linewidth = 1.2) +
  geom_point(size = 2) +
  scale_y_continuous(
    limits = c(0, 100),
    breaks = seq(0, 100, 20),
    labels = function(x) paste0(x, "%")
  ) +
  scale_x_continuous(
    limits = c(2010, 2022),
    breaks = seq(2010, 2022, 1),
    expand = expansion(mult = c(0, 0.02))
  ) +
  scale_color_viridis_d(
    option = "mako",
    begin = 0.3,
    end = 0.8
  ) +

```

```
labs(
  x = "Year of Surgery",
  y = "Proportion of Procedures",
  color = "Group"
) +
theme_classic()
```

p_transition1



```
# -----
# After matching
# -----

plot_data <- mdata %>%
  count(YEARS, Treatment) %>%
  group_by(YEARS) %>%
  mutate(
    Percent = 100 * n / sum(n)
  ) %>%
  ungroup()

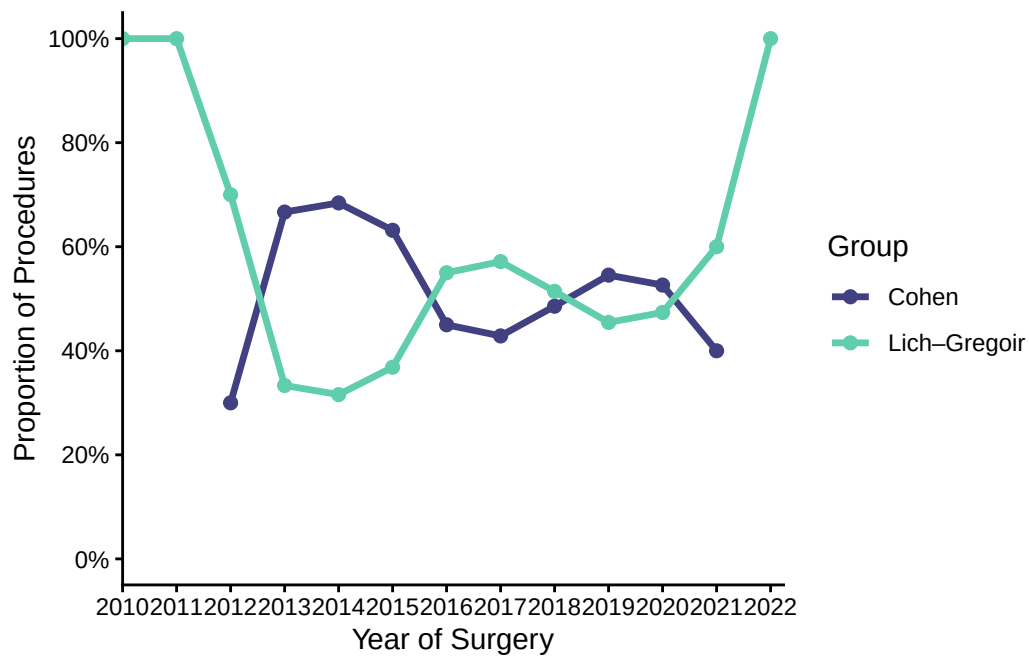
p_transition <- ggplot(
  plot_data,
  aes(
    x = YEARS,
    y = Percent,
    color = Treatment,
    group = Treatment
  )
) +
  geom_line(linewidth = 1.2) +
  geom_point(size = 2) +
```

```

scale_y_continuous(
  limits = c(0, 100),
  breaks = seq(0, 100, 20),
  labels = function(x) paste0(x, "%")
) +
scale_x_continuous(
  limits = c(2010, 2022),
  breaks = seq(2010, 2022, 1),
  expand = expansion(mult = c(0, 0.02))
) +
scale_color_viridis_d(
  option = "mako",
  begin = 0.3,
  end = 0.8
) +
labs(
  x = "Year of Surgery",
  y = "Proportion of Procedures",
  color = "Group"
) +
theme_classic()

```

p_transition



```

# -----
# After ATO weighting
# -----

```

```

plot_data_ato <- Out %>%
  group_by(YEARS, Treatment) %>%
  summarise(
    weighted_n = sum(weights),

```

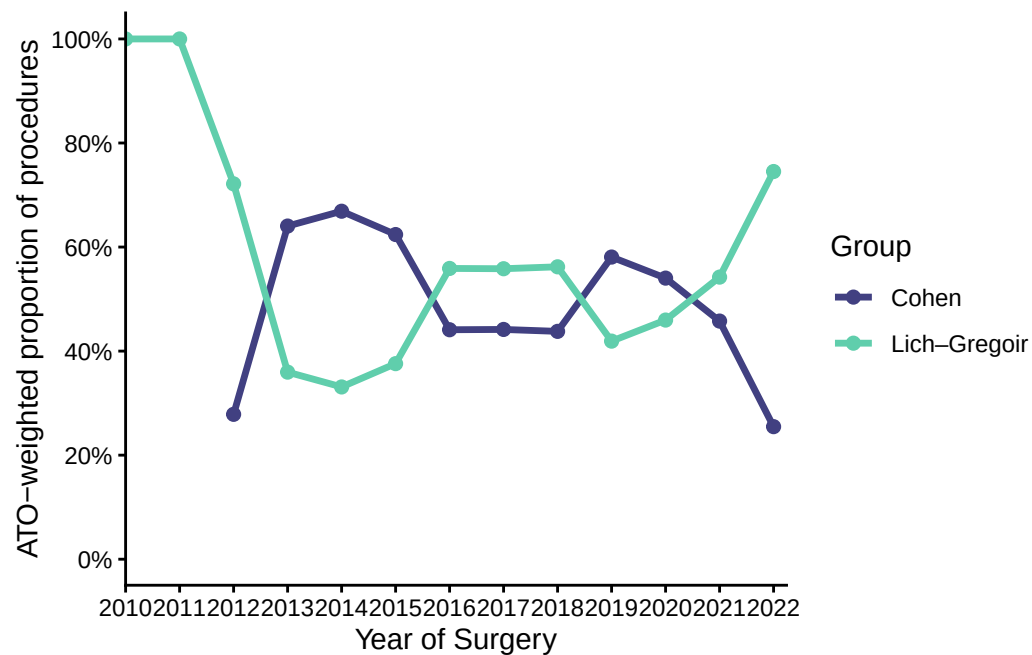
```

    .groups = "drop"
  ) %>%
  group_by(YEARS) %>%
  mutate(
    Percent = 100 * weighted_n / sum(weighted_n)
  ) %>%
  ungroup()

p_transition_ato <- ggplot(
  plot_data_ato,
  aes(
    x = YEARS,
    y = Percent,
    color = Treatment,
    group = Treatment
  )
) +
  geom_line(linewidth = 1.2) +
  geom_point(size = 2) +
  scale_y_continuous(
    limits = c(0,100),
    breaks = seq(0,100,20),
    labels = function(x) paste0(x,"%")
  ) +
  scale_x_continuous(
    limits = c(2010, 2022),
    breaks = seq(2010, 2022, 1),
    expand = expansion(mult = c(0, 0.02))
  ) +
  scale_color_viridis_d(
    option = "mako",
    begin = 0.3,
    end = 0.8
  ) +
  labs(
    x = "Year of Surgery",
    y = "ATO-weighted proportion of procedures",
    color = "Group"
  ) +
  theme_classic()

p_transition_ato

```



```

#-----
# Add subtitle (we can keep or remove them from the final figure)
#-----

p_transition1 <- p_transition1 +
  labs(subtitle = "Whole cohort")

p_transition <- p_transition +
  labs(subtitle = "Matched cohort")

p_transition_ato <- p_transition_ato +
  labs(subtitle = "Overlap weighting population")

#-----
# Combine plot
#-----

transition_plot <-
  p_transition1 /
  p_transition /
  p_transition_ato +
  plot_layout(heights = c(2,2,2)) +
  plot_annotation(
    tag_levels = "A"
  )

#-----
# Save the figure
#-----

ggsave(filename = "figures/transition_in practice.png",
  plot = transition_plot,
  width = 8, # width in inches

```

```
height = 10, # height in inches
dpi = 300) # resolution
```

FLOWCHART

```
f <- grViz("
digraph CONSORT {

graph [
  layout = dot,
  rankdir = TB,
  splines = ortho,
  nodesep = 0.4,
  ranksep = 0.6
]

node [
  shape = box,
  style = 'rounded,filled',
  fillcolor = white
]

# Main pathway
A [label =
'Children undergoing unilateral ureteral reimplantation
June 2010 - September 2022
Assessed for eligibility
N = 441']

C [label = 'Original cohort\\nN = 412 (93.4%)

Cohen group n= 197\\nLich-Gregoir group n= 215']

D [label = 'Propensity score overlap weighting (ATO) using R software

Propensity score model covariates:
Calendar year
Indications for surgery
Prior Deflux injection
Age at surgery
Gender
BMI
VUR grade and laterality
Ureteral dilation
VUR phase on VCUG
Type and number of preoperative UTIs
Preoperative acute pyelonephritis and renal scarring on DMSA
Surgeon experience

Covariate balance: Absolute standardized difference (ASD) <0.1']
```

```

E [label = 'Weighted overlap pseudo-population\\nN = 412

No exclusions after weighting, all patients contributed to analysis with weights']

F [label = 'Weighted outcome analyses']

# Side exclusion
B [label = 'Excluded n = 29 (6.6%)

Secondary VUR n = 0\\nRepeat or tapering reimplantation procedures n = 0\\nBladder and bowel dysfunction

# Invisible branching point
X [shape = point, width = 0.01, label = '']

# Main flow
A -> X [arrowhead = none]
X -> C
C -> D
D -> E
E -> F

# Side branch
X -> B [style = dashed]

# Keep exclusion box beside the main flow
{ rank = same; X; B }

}
")

svg <- export_svg(f)
writeLines(svg, "figures/flowchart.svg")

rsvg_png(
  "figures/flowchart.svg",
  "figures/flowchart.png",
  width = 10000
)

saveWidget(
  f,
  file = "figures/flowchart.html",
  selfcontained = TRUE
)

webshot(
  "figures/flowchart.html",

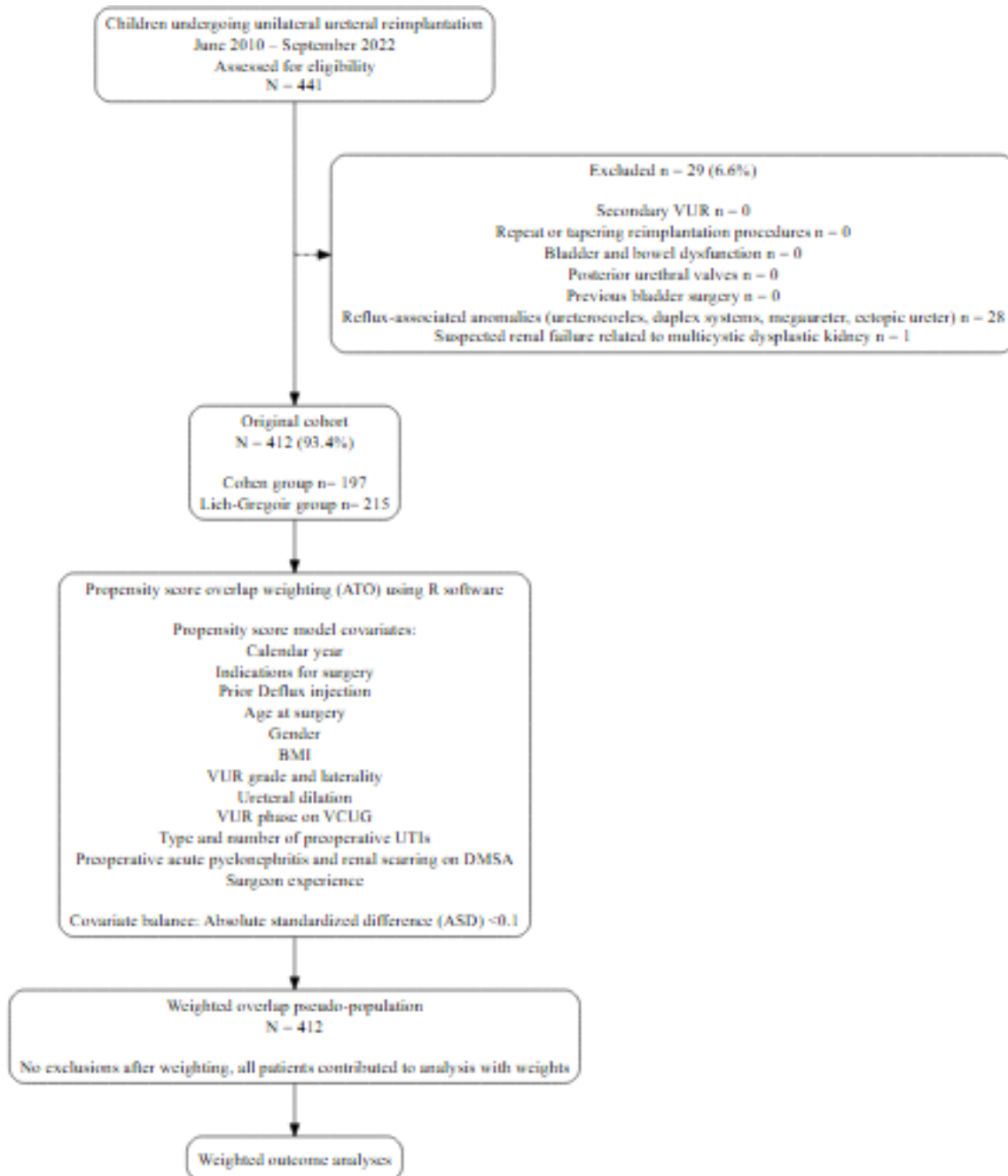
```

```

file = "figures/flowchart.png",
vwidth = 800
)

```

file:///C:/Users/USER/OneDrive/Documents/KI_PhD Materials/2024-2025/- Article_Outcomes/Outcome_Project/



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